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Editorial

Dear valued readers,

The Faculty of Education is delighted to announce the release of the second issue of the XIX volume of the National Journal of Education. The journal provides a forum for scholars, researchers, and practitioners to showcase their work in the field of education, ultimately contributing to the advancement of education on a national and international scale. We believe this journal will serve as a valuable resource for the education community, promoting a culture of innovation and excellence in the field. We cordially invite all education stakeholders to actively participate in this endeavour and contribute to the growth and development of the National Journal of Education.

Peer review is a critical process that ensures the quality and credibility of academic publications, and we acknowledge the significant contribution of the reviewers in upholding the high standards of the journal. We extend our sincere gratitude to the reviewers who dedicated their time and effort to thoroughly evaluating the papers and providing invaluable insights and recommendations.

The current volume of the journal features a diverse range of papers on educational issues that reflect the current trends in education research and practice. We are confident that these papers will be of great interest and value to the education community, and we look forward to their engagement and feedback. We are committed to the continuous improvement of the National Journal of Education and strive to make it a leading platform for the dissemination of knowledge in the field of education.

This issue of the journal includes papers that address significant concerns in the education sector that have emerged due to the Covid-19 pandemic. The pandemic has caused a significant shift in traditional teaching methods, learning strategies, and assessment systems. These papers offer insights into the impact of Covid-19 on education, the changing dynamics of teaching and learning, and the potential of technology in shaping the future of education. For example, Nutan Pandey and Amita Bajpai discuss the effects of Covid-19 on early intervention programs for children with Autism Spectrum Disorder, while Rukshar and M. Altaf Khan analyze factors motivating students' behavior on online education from the perspective of business schools.

Jeemina Baglari presents the challenges, opportunities, and experiences of open-book examinations from the students' perspective, while Sujata Pal and R.N. Sharma review ICT initiatives targeting inclusive development in India. Sangeeta Suman explores the use of ICT and Edmodo as a learning tool, and M Nandan discusses the role of artificial intelligence in fostering mastery learning.

This issue also contains papers that emphasize the intersection of gender and education, exploring the challenges and opportunities for inclusive education. Sumedha Mukherjee and Umakant Prasad discuss the need for gender-inclusive education and the steps necessary to achieve it, while Anthea Patrao and Triveni S. examine the role of gender in the development of social competence in young children. A Jayashree and N. Murugeswari focus on the specific challenges faced by rural girls in accessing education and propose a creative solution to bridge this gap. Together, these papers emphasize the need for education to be gender-inclusive and highlight the unique challenges faced by different genders in accessing education.

Additionally, this issue includes papers that underscore the importance of effective teaching strategies in enhancing mathematical skills and conceptual understanding among students. Yogeshwari S and H Indu highlight the effectiveness of Vedic Mathematics-based instruction in achieving better results in mathematics, while also examining the impact of gender on academic achievement. Priyanka Singh and Ajay Singh emphasize the effectiveness of instructional modules based on Van-Hiele's theory in enhancing students' geometric conceptual understanding. Together, these papers highlight the significance of innovative teaching methods in facilitating effective learning outcomes in mathematics.

In their paper titled 'A Study on Prosociality and Attitude towards Teachers of Higher Secondary School Students Studying in Schools Managed by Different Religious Communities' Mahejabin and Md. Akhtar Raza investigate the correlation between prosociality and attitude towards teachers among higher secondary school students from schools managed by different religious communities, finding significant correlations and differences in prosociality and attitude towards teachers among different religious community-managed schools.

We would like to draw attention to the paper by Vimal Singh that explores how the digital age has caused various insecurities and fears in society, and argues that J. Krishnamurti's philosophy of education, which emphasizes freedom from fear, can help children overcome these insecurities and live fearlessly. [The review article by Arabinda Kumar Sahu and Poonam](#)

Kharwar, focuses on the challenges and issues related to academic adjustment in students of later adolescence, their effects on overall development, and coping strategies to address these issues.

Lastly, the paper by Raj Ballav Panda et al., analyses the implementation of the Curriculum Framework for Teachers Education, 2009 in four institutes offering the 3 Years integrated B.Ed.-M.Ed. Program, identifying major challenges such as overload of papers and time restrictions, lack of logical sequencing of topics, and overlapping of contents, and suggests solutions to address them.

In conclusion, the release of the second issue of the XIX volume of the National Journal of Education represents an important milestone in the promotion of education research and practice. We extend our gratitude to all contributors, reviewers, and editorial staff for their dedicated efforts in making this issue a success. We hope that this journal will continue to serve as a valuable platform for the dissemination of knowledge and the exchange of ideas and contribute to the advancement of education in India and beyond.

With best wishes!

Editorial Board

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**Teachers' Perception Regarding English- Medium Instruction to Rural Students in
Uttar Pradesh**

Archana Singh* Meenakshi Singh R. P. Shukla****

Abstract

Alarmed by the continuous decrease in enrolment at government Parishadiya primary schools (DISE report 2011) and responding to its peoples' aspirations, Uttar Pradesh government decided to introduce English Medium education on experimental level from 2014-15 in a few hand- picked primary schools of rural area. The teachers selected for these schools were those, who had been teaching in Parishadiya primary schools and had English as a subject in the intermediate level. No training had been given to such teachers. The paper explores whether the teachers feel competent, confident, and skilled enough for imparting education in English (a third language here) medium to Bhojpuri speaking rural children, while they themselves have been throughout Hindi medium students? This researcher took all the teachers of four Model English Medium Primary schools from two districts of Uttar Pradesh, namely, Varanasi and Mau as sample. The qualitative data obtained through a questionnaire and semi structured interview, was analyzed, and interpreted on the aspiration parameters of language policies, various documents and NCF-2005. Thereby, the paper brings forth some major paradoxical praxis or issues about the policies and practices, demand, and supply of English medium education in rural areas.

Keywords- *Teachers' perception and awareness, English medium instruction, contextual teaching to rural students, self-esteem, identity*

Introduction

‘Medium of instruction is the most direct agent of maintaining and revitalizing a culture as well as a power (re)distribution and social reconstruction’ (Tsui, 2004) and in India English language, which was a key part of the mechanism of exclusion and native culture destroyer, (Crystal, 2000) is now seen as a means of inclusion and prosperity (Graddol, 2010) and thereby desired as a medium of education countrywide. EMI (English Medium Instruction) is advocated for

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English language proficiency as it is believed that English can be improved by repeating the targeted structures - by connecting vocabulary with contents, a series of repeated expressions can be easily recalled (Kashiwagi and Tomecsek, 2015). Dearden (2014) defines English Medium Instruction as, “The use of the English language to teach academic subjects in countries or jurisdictions where the first language (L1) of the majority of the populations is not English.” (English as a Medium of Instruction-a growing global phenomenon, p.3)

Background of the study

The State Government of Uttar Pradesh, in the year 2014 introduced English Medium Education in government Parishadiya primary schools. ‘The idea was to bring government primary schools at par with private schools to attract students towards government primary education’ according to Shri Ram Govind Choudhary, the then education minister of U.P. (TOI, September 14, 2014). Such competitive fear stemmed out of survey reports which estimated that private school’s enrolment is increasing at the rate of 10% per year. It was feared that if the trend continued private schools would take over government schools as the provider of elementary education by 2020 (DISE, 2011-12).

The Basic Education department of the state introduced English Medium education from 2015-16 in some hand-picked primary schools. This was a very first for ‘Sarkari’ schools. Such schools were named **Model English Medium Primary Schools** (under the notification of Basic Shiksha Parishad/12883-13128/2014-15). Two such primary schools were established at experimental level in each district without much preparation in terms of planning or teacher preparation. No extra pain was taken regarding syllabus, curriculum, and teaching methodology. The teachers were selected based on a nominal departmental exam from among the teachers already teaching in the government primary schools. These teachers were to have had English as a subject at least up to their intermediate level. The students in these schools were majorly poor schedule caste backward children who spoke only Bhojpuri, which was their mother tongue. They could understand Hindi but didn’t use it. They had minimal exposure of English in their surroundings both in school and in their neighborhood. In such a pathetic condition it seemed like a daunting task to not just teach English as a language but to teach all subjects using the English medium.

Justification and objectives of the study

If any change is brought abruptly to the medium of instruction without any proper plan for its introduction, it may produce cultural shock and affect teaching-learning adversely. As Gilbert (1982) opines, ‘students’ experience and knowledge can be astonishingly resistant to change and tenacious. The main stake holders for EMI at primary education level were teachers, students, and parents. Nevertheless, it is the teacher who faces the actual problems, plans accordingly, and executes the policy recommendations in the classrooms, thus he is the main agent for bringing any change. What teachers know and can do affects all the core tasks of teaching (Pandian, 2002).

It becomes vital to study teachers’ perception, attitude, awareness, and readiness before executing any change, as perception decides preparation and readiness decides execution. If teachers don’t find themselves confident enough for teaching through EMI or if they have an anti EMI attitude, they cannot be expected to implement EMI successfully. The issue becomes even more complex because the students to be taught in these schools are from a rural background and are solely dependent on their teachers for any kind of English exposure. Therefore, the present study was conducted with a view to study teachers’ perception over EMI related issues in Model English Medium Primary schools. The research questions addressed in the study are:

1. What are the perceptions of the teachers regarding English Medium education in general and English Medium Instruction at Primary level to Bhojpuri Speaking rural students?
2. What are their perceptions regarding the training program, textbooks?
3. What are the perceptions of the teachers regarding the impact of English medium instruction on cognitive development and language acquisition of primary students?
4. Are the teachers aware and ready for creating an atmosphere of English Medium education to rural background students?
5. Do the teachers find themselves skilled enough to teach through English medium and what skills they apply in classrooms for EMI?

Methodology

This is a qualitative exploratory study conducted on a small scale, adopting an Interpretive approach to know the perception of government primary teachers towards English Medium Instruction. According to Agrawal and Sharma (2011) an interpretive approach provides a deep

insight into “the complex world of lived experience from the point of view of those who live it”. The Interpretive researcher’s ontological assumption is that social reality is locally and specifically constructed “by humans through their action and interaction.”

Sample of teachers included all the 20 teachers from four Model English Medium Primary Schools of Varanasi and Mau District in the Purvanchal region of Uttar Pradesh where Bhojpuri is the mother tongue and home language.

A questionnaire followed by open-ended written response was prepared. It had two parts. First part enquired the general information about teachers’ age, sex, languages known, teaching experience, teachers’ own medium of instruction, their qualifications etc. The second part included 10 questions with choices of agree/ disagree. It had space after each question to write their response related to that question. Teachers were quite reluctant to write their responses and gave guarded and cautious responses which did not align with observations made by the researcher. The underlying reasons being a fear of breach of confidentiality and of them being leaked to their officials Another probable reason was that the questionnaire was in English. Although the teachers were given the option to respond in either Hindi or English, they felt pressured to write in English due to being teachers in a Model English Medium Primary School. Consequently, few of them (despite having poor vocabulary and using incorrect grammatical structures) chose to write brief comments in English, while most of them only marked the appropriate option and left the writing space for comments blank instead of responding in Hindi.

To address this issue, a semi-structured interview was created with the same questions but phrased differently. The interview was conducted with utmost freedom given to the teachers, and probing questions were asked in between to elicit more detailed responses.

Response analysis and interpretation

The responses to items of the questionnaire of each teacher were matched and triangulated with their oral responses as coded and decoded from the interview. Here is an analysis of teachers’ perception on various EMI related issues in the area-

1. Issue of Social Prestige and Benefits of EMI: 14 out of total 20 teachers agreed that ‘attending an English Medium education improves social prestige’ of a teacher as well as of students and of his/her family too. Those 6 teachers, who disagreed to this statement to be their personal belief, accepted in the interview that it is the societal truth that cannot be neglected.

Headmasters acknowledged the fact that as soon as their schools converted to an English medium school, the enrolment increased automatically, and it had nearly doubled by the time of the study. Table1 shows the increase in enrolment.

Table 1: Increase in enrolment after 2015.

Name of the school	Total enrolment		
	2015	2016	2017
Model English Medium primary school, Badagaon, Mau (near town)	110	180	245
Model English Medium primary school, Kopagunj, Mau (in rural area)	278	390	476
Model English Medium primary school, Harahua, Varanasi (in rural area)	105	145	181
Model English Medium primary school, Nawabgunj, Varanasi (in main city)	60	79	94

It is quite apparent that even poor, backward, and illiterate parents from a rural area desire to send their child to an English-medium school (Chan, Hoare & Johnson, 1997). A teacher expressed that "speaking in English portrays a person as more confident and held in high esteem among their peers". Dearden (2014) reported through a survey conducted by the British Council that in India, "the demand for English-medium instruction is increasing, particularly at the school level, to access opportunities in tertiary education and employment." When asked about the relative usefulness of English Medium Instruction (EMI) compared to Hindi Medium Instruction (HMI), eight teachers disagreed while twelve agreed. Their concerns and priorities were different. Teachers who disagreed with the usefulness of EMI over HMI viewed its drawbacks from the perspective of child learning psychology and found it to be tedious, troublesome, and an obstacle for young children, destroying their enjoyment of learning. On the other hand, teachers who found EMI useful at the primary level considered it to be early preparation for the later challenges of higher education, employment, and cultured socialization of the child. They believed that despite the difficulties encountered in learning English at present, it would prove beneficial in the long run.

2. Teachers' Perception of Their Teaching Proficiency for EMI: All 20 of the respondent teachers believed that they were skilled enough to teach in an English medium school, despite lacking an understanding of what it entails to be an English medium teacher to non-native speakers of the language. Dearden (2011) similarly noted that most teachers who were expected to teach in EMI were not native English speakers, and there existed uncertainty surrounding the necessary English language competency requirements.

A teacher wrote proudly,

"I am agreed because I am from convent background, and I am able to teach the pupils in English."

This teacher holds a B.Sc., M.B.A., and B.T.C. degree and is T.E.T. qualified, so he believes he is qualified enough for the job. However, his incorrect use of verb forms makes him unsuitable for teaching in an English Medium school.

When asked about the skills they apply to teach a subject in English, teachers gave varied and confused responses. Most of them said, "I make students note down and learn the Hindi meaning of important words from each lesson of each subject." It is noteworthy that at the primary level, most Bhojpuri-speaking students cannot read and write Hindi either. Sercu (2004) acknowledges the problems with this type of confused EMI, such as a decrease in the quality of instruction and learning, accompanied by an increase in the teaching and study load.

3. Teachers' perception of the Training for EMI: Although the teachers expressed confidence in their English proficiency, they expressed dissatisfaction with the quality of training they received prior to joining Model English Medium Primary schools. According to them, the training by SCERT was inadequate and lasted only 4 to 6 days, with the sole purpose of meeting the requirement on paper.. They stated that the training did not provide them with any specific information related to English Medium education.

4. Teachers' Perception over Text Books and Teacher Guidelines: Fifteen out of the twenty teachers disagreed on the usefulness of textbooks and stated that there was no prescribed teacher guideline (shikshakSamdershika). The remaining five teachers were uncertain about the textbooks. The teachers claimed that the books were merely translated versions of earlier Hindi medium books. They believed that some practice workbooks with drills and books with large

attractive pictures, like those used in private English Medium Schools, should have been introduced.

5. Teachers Perception of creating an English Environment: All the teachers concurred that an English Medium education is not only about teaching in English, but also about fostering an environment for the English language. However, the only concept they had of an English atmosphere was one that was orderly, well-organized, and tidy, where students imitate the English mannerisms of private English Medium schools, rather than an environment that fosters a love and curiosity for the language.

6. Teachers perception of EMI to Rural Students: All the teachers agreed unanimously that ‘it is good to introduce English Medium at primary level to rural students’. Level of the introduction of EMI or issues like identity crisis, cultural death (Crystal, 2000: 78) and language endangerment (Skutnabb-Kangas, 2000) are not their concern. They demand EMI to rural backward students on social inclusion and equality ground, as early as possible (Pennycook, 1994).

All the teachers unanimously agreed that introducing English Medium at the primary level to rural students is beneficial. However, they did not seem to be concerned about the level of introduction of EMI or issues related to identity crisis, cultural erosion/death (Crystal, 2000: 78), and language endangerment (Skutnabb-Kangas, 2000). Their focus is on providing social inclusion and equality to rural students as early as possible. They demanded EMI to rural backward students as soon as possible on grounds of social inclusion and equality. (Pennycook, 1994).

As a teacher stated,

“inhe bhi shahri ameer bachhon ki tarah angrezi madhyam wali achhi shiksha ka poora huq hai.” (These kids are also entitled to good English medium education like those rich urban kids.)

Even though these students may not understand anything taught in English and may not receive support at home due to their different home surroundings and their guardians' indifference (Saeed, Iqbal & Azam, 2015)., the teachers believe that EMI has no negative impact on their learning of other subjects or academic achievement. They argue that since everything is translated by the teacher and exams are not too rigorous, there is no issue. However, Hong

(2008) disagrees with this practice, stating that limited use of the mother tongue and over-reliance on translation can hinder language development and discourage learners from using the second language. (cf. Saeed, Iqbal & Azam, 2015).

Conclusion and Recommendation:

The study indicates that teachers in Model English Medium Primary Schools are enthusiastic and positive about the English medium teaching-learning environment. When asked why they chose to work in such schools without any salary increment or other monetary benefits, they replied that, despite better classroom infrastructure and toilet facilities, shorter distances from home, etc., they found it more prestigious and challenging to teach in an English medium school and felt motivated.

However, they reported major problems, which are summarized as follows:

1. The teachers lack knowledge of the key issues, norms, and expectations of teaching in an English medium primary school to Bhojpuri speaking children.
2. Textbooks, curriculum, in-service training, and infrastructure provide little insight, proper plans, or appropriate help to the teachers.
3. The biggest concern is how to integrate English language in content teaching without disturbing the understanding of academic subjects. The uncertain teaching methodology in an English medium school is also a problem (Dearden, 2014).

The paper recommends that policy makers and researchers pay special attention to sensitizing and raising teachers' consciousness to develop the desired English Medium Education system. Their enthusiasm for EMI should be met with appropriate in-service training to develop their language proficiency and create an appropriate English Medium environment. As Markee (2001) says, teachers can help bring about deeper changes if they understand why there is a need to change.

The Position Paper of the National Focus group on the Teaching of Indian Languages, 1.3 (2006) suggests that teachers can be the best agents to hear the real voices of children and their parents. Training teachers to be sensitive observers at all linguistic levels could produce more effective and challenging textbooks and teaching aids.

It is time to actualize plans on a grassroots level beyond any confusion and duality, lest it should remain merely a pseudo 'political response to the people's aspiration.' (National Focus Group, Position Paper on Teaching of English 2006: 1). As National Knowledge Commission believes, “ time has come for us to teach our people, ordinary people, English as a language in schools. Early action in this sphere would help us build an inclusive society and transform India into a knowledge society.”(GOI, 2007)

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**Effectiveness of the Instructional Module based on Van-Hiele's theory on the Geometric
Conceptual Understanding of 9th Grade Students**

Priyanka Singh* Ajay Kumar Singh**

Abstract

Mathematics as a school subject is considered important for developing a child in all aspects i.e., mentally, socially, emotionally etc. An instructional module based on constructivist strategy was developed by following Van Hiele's model-based instruction process. The study aimed to find out the effectiveness of the module on the conceptual understanding of geometry among students of 9th grade. Quasi experimental design with pre- test post-test non- equivalent control group was used in the study. Sample consists of hundred and twenty 9th grade students in the academic year 2022-2023. Self-made "Geometric Conceptual Understanding Tool (GCUT)" was used for assessing conceptual understanding of students prior to and at the end of the intervention. Data was analyzed using a t-test. Decisively, there exists significant difference between GCUT scores in pretest and posttest of Experimental Group ($t\text{-score}=7.4575 > t\text{-critical two tail value}=2.0210$) and Control Group ($t\text{-score}=2.1697 > t\text{-critical two tail value}=2.030$), but also a significant difference was found in posttest scores of both the groups ($t\text{-score}=3.3983 > t\text{-critical two tail value}=1.993$). Result's showed that instructional module based on Van Hiele's found to be more effective than traditional method for geometry teaching.

Key Words: Geometry, Van Hiele's Model, Geometric Conceptual Understanding

Introduction

Geometry is an off shoot of mathematics that associates the idea of points, lines, angles, and two dimensional and three-dimensional figures (Salim k Tiawa, 2015). It helps students to understand the world they live in as geometry structurally consists of shapes and objects (Goo, Spencer 2003; Pesen, 2003).

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With the changing world new technologies influence the field of education, the changes in field of mathematics, particularly in geometry are revealed enormously. Based on the National Council of Teacher of Mathematics (NCTM) standards, new innovations, and new perspectives in the field of mathematics are taking place. As per guidelines of NCTM “The Van Hiele’s Model” was taken as a building block to design instruction. (Choi, Koh 1999; NCTM, 2000).

In India for consolidating these new patterns of technologies and for fulfilling the new objective of child centered education drastic changes in the field of Mathematics, particularly in Geometry, were done by the stakeholders in this field. The syllabus of Geometry after NCF 2005 changed desperately as Geometry has changed from remembering definitions and proofs to understanding space and spatial relations. Theorems and proofs are a part of understanding shapes and their properties, but Geometry is not synonyms to theorems and proofs it is about visualization, symmetry representation of objects, projecting, and mapping, plotting functional relationships along with ability to use mathematical logic. In this perspective, entire system of teaching and learning becomes student centric and aims to develop high level thinking skills and conceptual understanding of students become core of whole system.

The Van Hiele’s Model and Conceptual Understanding of Geometry

The Van Hiele’s Model is a model which promises development of Geometric Understanding (E. Tolga, 2006). Mostly researchers took Van Hiele’s theory as foundation to develop geometric thinking (Erdoğan, 2006). Classroom activities are taken as core of this model. To attain instructional objectives students’ participation in pre-set learning situation is an essential fragment of this model (Gutierrez, 1992).

The remarkable feature of the Van Hiele Model is that it provides an explanation of five interconnected levels for the development of geometric conceptual understanding. These five levels progress through a hierarchy of concept formation processes to enhance geometric understanding. This model explains the type of thinking pattern students have at a specific stage and the pattern of geometric schemas they are engaged with, instead of the knowledge they possess at that level (Van de Walle, 2004). The five levels of geometric thinking and understanding, as elaborated by the Van Hiele Model, are as follows:

Visual period (Level 0): In the initial stage, people based on observations in the setting, can make comment about the geometric structure. Based on appearances of their surrounding

students at this level can determine names and compare shapes. (Baykul, 2002; Crowley, 1987; Hoffer, 1983; Kiliç 2003; Olkun&Toluk,2003; Van Hiele, 1986)

Analysis period (Level I): This stage is not confined only to observation. At this stage, students begin to adapt the characteristics of shapes by means of experimentation along with observation. At this period, the investigation of geometric concepts takes shape. At this level, the students check out and demonstrate characteristics and laws about the shape through activities as observation and folding. (Baykul, 2002; Hoffer, 1983; Mason, & Schell, 2001)

Inferences related to experience (Level II): This is the level of categorization. At this level categories are developed by seeing pattern of similarity and dissimilarity among different shapes. Students gain an understanding of ordering and making group of structures in accord with their attributes. By employing casual statements, students figure out new associations from the earlier known relations (Crowely, 1987; Olkun &Toluk, 2003; Van Hiele, 1986).

Inference Resolutions (Level III): Students at this level through induction make association between the concepts and they can differentiate the different logical rationales associated to the common theorem. (Altun, 2002; Baykul, 2002; Crowley, 1987; Hoffer, 1983; Maso and Schell, 2001)

Advanced Period (Level IV): An individual at this last level can grasp the distinct ideas included in non- identical axiomatic systems. They declare theorems and postulates indifferent axiomatic frameworks by comparative analysis of these systems (; Hoffer, 1983; Van Hiele, 1986).

Besides describing the evolution of student's geometric thinking, these levels provide a framework to geometry teaching and classroom activities related to geometry. This model consists of an instructional plan consisting of five steps. It helps in passage from one level to another too to intensify geometric thinking; a student must go through these five steps. (Crowley, 1987; Erdoğan, Durmus & Bekci, 2007; Kiliç, 2003; Olkun & Toluk, 2003)

(i) Interview (Information): This is the first step of instruction where geometric thinking level of students are assessed. The core purpose of this step is to determine the student's geometric level of thinking. Direct communication is crucial at this level to conduct task.

(ii) Direct orientation (Guided Orientation): At this level, the teacher takes command into his hands and gives instructions and assignment related to the studies considering the answers given

by the students. The rationale of the task given to the students is to open out the configuration of the topic, where student involved actively.

(iii) Making clear (Explication): In this step teacher presents the topic in front of students and students combine their experiences related to the topic in their own words. This is a major step to involve students into the teaching-learning process and arouse interest of students.

(iv) Free Performance/Orientation (Activities): In this step students work on multisided solutions of the problem in hand. They invent the linkage among the various elements of the compositions in the current topic. The teacher plays the role of guide to direct students thinking to discover different solutions.

(v) Integration: In this step student assembles the entire concept they learnt. Students internalize and make equilibration between all the concepts what they learned as a new thinking structure.

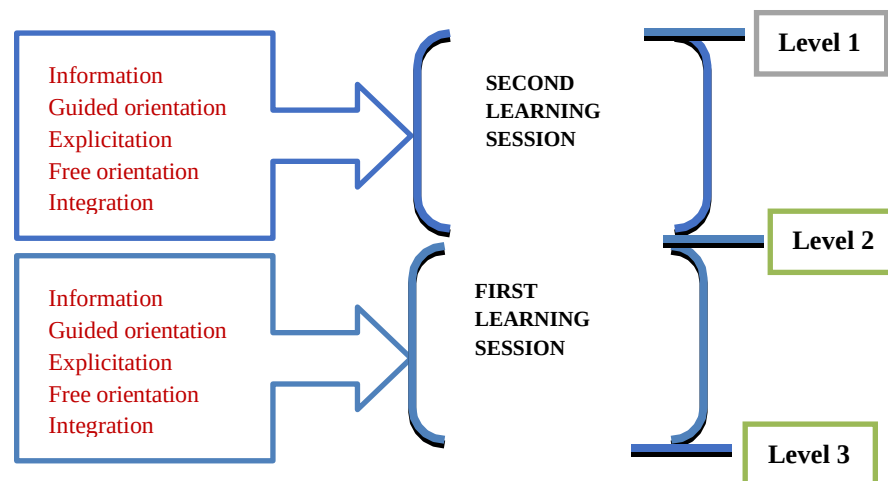


Fig 1: The Van Hiele's phases for Geometry Learning

The educational environment created according to the Van Hiele's Model focuses on developing geometric concepts and the relations among these concepts to enhance Conceptual Understanding. According to Yadil (2009)- " Van Hiele's model improve students conceptual knowledge , because it is based on stages of thinking, so it able to know student's need, thus the content conveyed to students according to thinking stage they belong to." Alex & Mammen, 2015 has also supported the fact as they stated that Van Hiele's theory provides a geometry conceptual Understanding.

Based on the explanation above, the researcher designed an instructional module based on Van Hiele's Model as a tool for student in learning and helped students in understanding Geometry.

The Aim of the study

To determine the Effectiveness of the Instructional Module based on Van-Hiele's theory on the Geometric Conceptual Understanding of 9th Grade Students.

Problem Statement

Are the students who are taught according to instructional module based on the Van Hiele's model and traditional methods in the 9th grade differ significantly in Geometric Conceptual Understanding?

Hypotheses

Following are the hypotheses subjected to test at 0.05 level of significance:

- (i) There is no significant difference between the pre-test and post-test GCUT score of experimental group students.
- (ii) There is no significant difference between the pre-test and post-test GCUT score of control group students.
- (iii) There is no significant difference between the post-test GCUT scores of experimental group and Control group.

Method

Research Design

Non- equivalent pre- test post-test quasi experimental design with control group was used in this study. There were two groups involved in data collection; one group was the experimental group and the other was the control group. Instruction by module based on Van Hiele's model was carried in the experimental group, while traditional method of teaching was used in the control group. Instruction in both groups was carried out by the researcher. Statistical analyses were done on quantitative data.

The Study Sample

Sample comprised of 120 students from standard 9 from a school in Varanasi district named "*Sant Atulanand Residential Academy*". Two sections, section C and section D were selected out of nine sections consisting of 60 students in each section. Section C was randomly selected

as the experimental group and section D was selected as the control group. Data for pre-test and post-test was obtained from 46 students in the experimental group and 38 students in control group.

Data Collection Tool

In the study, Geometric Conceptual Understanding Tool (GCUT) was applied to determine the conceptual understanding of students to obtain pre-test post-test scores of both groups.

Procedure

Thirty days instruction about “*Lines and Angles*” was carried out in both groups. Discussion, group work, cooperate learning, learning by doing were used to carry out session in experimental group, and the traditional method was used in the control group which is teacher centered approach. The researcher guides the students according to instructional module provided in the experimental group. Tools, such as smart board, geometry box, and computer were used in experimental group. Software app Geo Gebrawas also used for instruction in experimental group.

Analysis of data

In the study, geometric conceptual understanding of students of experimental group and control group before and after the instruction were determined and compared. Also post test scores of experimental and control groups compared. Obtained data were examined using Microsoft Excel 2010 data analysis tool to determine significance of difference between the Geometric Conceptual Understanding of students among the groups and before and after the interventions given to both the groups.

Result

Finding Related to First Hypothesis

There is a significant difference between pretest and posttest GCUT scores of experimental groups was found during the analysis. The value of t-score was found 7.4575, which was significantly greater than t table value (t- critical two tail value=2.021075) at degree of freedom 40. Hence the null hypothesis was rejected.

Table 1: t-Test: Paired Two Sample for Means

	<i>Pre test</i>	<i>post test</i>
Mean	13	18.97

Variance	41.95	30.62
Observations	41	41
Df	40	
t Stat	7.45	
t Critical two-tail	2.02	

Discussion: The above result shows that Van Hiele's Model based Instructional Module is effective on the Geometric Conceptual Understanding of 9th Grade students. Many studies demonstrated that Van Hiele theory can help in improving geometric understanding. (Crowley, 1987; Mason, 2002).

Finding Related to Second Hypothesis

After analysis it was depicted that pretest and posttest GCUT scores of control group differ significantly. The value of t-score was found 2.169, which was significantly greater than t table value (t- critical two tail value=2.03010) at degree of freedom 35. Hence the null hypothesis was rejected.

Table 2: t-Test: Paired Two Sample for Means

	<i>pre test</i>	<i>post test</i>
Mean	12.5	14.44
Variance	45.97	37.11
Observations	36	36
Df	35	
t Stat	2.169	
t Critical two-tail	2.03	

Discussion: The above result showed that as null hypothesis was rejected means traditional method of teaching geometry had effect on student's geometric understanding, but effect size is quite low.

Finding Related to third Hypothesis

Post test scores of experimental group and control group in GCUT test were found significantly different and null hypotheses were rejected. t-value was found 3.398 which is greater than t critical value 1.993 at degree of freedom 71.

Table 3: t-Test: Two-Sample Assuming Unequal Variances

	<i>post test</i>	<i>post test</i>
Mean	18.97	14.44
Variance	30.62	37.11
Observations	41	36
Df	71	
t Stat	3.39	
t Critical two-tail	1.99	

Discussion: post test scores of both the groups in GCUT differ significantly. experimental group mean score is higher than that of control group. t-value was found 3.398 which is greater than t critical value 1.993 at degree of freedom 71. Result shows that Van Hiele's Model based Instructional Module is more effective in enhancing conceptual understanding of geometry.

Conclusion

The present investigation aimed to see the effectiveness of the Instructional Module based on Van-Hiele's theory on the Geometric Conceptual Understanding of 9th Grade Students. Results conveyed that instructional module has significant effect on Geometric Conceptual Understanding as GCUT score of pretest and posttest of experimental group was significantly different and also there was significant difference found in post test scores of experimental and control ground, pretest and posttest geometric conceptual understanding score of control group was also significant different but effect size was found less in this case, hence instructional module based on Van Hiele's model of instruction was found to be quite effective in enhancing conceptual understanding of geometry among students.

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Facing Fears Challenges and Realities: Understanding Krishnamurti's Philosophy

Vimal Singh*

Abstract

The members of present society are occupied with various types of known and unknown fears, insecurities, threats etc. and due to that they are incapable of utilising their potentials to fullest. The digital technology has helped people connect with others living across the world but then the question arises 'Are we really connected with others in its true sense?' We have become more skeptical while communicating and interacting with others as technology has not only made our life easier on one hand but on the other it has made our life uncertain and insecure. This paper throws light on various insecurities faced by people of the global world with an example of real life. It also discusses the philosophy of J. Krishnamurti which says that education should provide us freedom from fear. The present paper analyses how Krishnamurti's educational philosophy can play a role of saviour for children of new age who are facing threats, fears, and insecurities since birth. The philosophy 'freedom from fear' helps children in living fearlessly and freely.

Keywords: *Krishnamurti's Philosophy, Fear, Challenges and Realities, Educational Philosophy, Freedom from Fear*

Introduction

The digital technology has opened all the boundaries created in the name of country, religion, caste, race etc. and made everybody a member of one global society. New vistas are being unfolded for the members for fulfilling their dreams according to their aptitude, attitude, interest & capability. Technology has made all kinds of information easily accessible & approachable for all and sundry sans any discrimination. The citizens of the world relate to each other through technology for sharing innovative & creative ideas, thoughts, viewpoints, observations, perceptions etc. freely & frankly. It can be easily said that the new world is having a positive impact on the lives of individuals. But the question is, "Is this a complete picture? Or are we missing another side of it which is gloomy & depressing side of it? No doubt, the world

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has given us freedom from various obstacles for our betterment but what about the freedom from various fears like war, genocide, pollution, infanticide, sexual abuse, mental harassment, cut-throat competition, corruption, cheating, discrimination etc. These fears pose a great threat for us. Today we live in a free & open society but are we really free from the threats & fears, which consume our positive energies & leave us with negative emotions like frustration, jealousy & hatred. We need freedom from negative forces which are controlling our minds to make us their slaves. What kind of freedom is this which is making us slaves? Here we can easily use an oxymoron and that is-Free Slaves.

There are many reasons in society for people to fear. Psychologists say that fear is only manipulated perception of danger, but not actual danger. It is said that our imagination places us in frightening situation that become real threat in our minds. Modern society encompasses us in the castle of fear and keeps us in virtual freedom. In the modern risky society security and safety become one of the major concerns in the life of its citizens.

According to Beck (1990), fear is a social phenomenon with its emergent practices, processes, and trends. We can relate it to uncertainty that becomes the part of peoples' life." Fear is "an unpleasant experience tied to a specific object or situation, which creates a concern regarding peril in the individual. Thus, it is a reaction to known dangers and has a signaling and defensive function. As an experience, the emotion encompasses tension, disquiet, tightness and even paralysis." (Vymětal, 2004, p. 12).

The individual who considers himself as a free citizen is a slave of various threats & fears (concrete & manipulated both) and thus, has very less control over his mind. This paper analyses the perceived threats of contemporary society with the help of an example and then tries to provide solutions through J. Krishnamurti's educational ideology on freedom from fear.

Background of the theme

Giving birth to a new life is always considered as an incredibly beautiful & exciting feeling. When a woman comes to know that she has conceived & would soon deliver a baby, she & her well wishers start praying for the healthy child. Since a child is a future of the nation it also becomes a duty of the government to provide facilities for the proper mental & physical growth of the child. Child birth is celebrated all over the world as a festival. Let's take an

example of say, Ragini, a lady. Ragini is expecting a baby. Her mind is filled with varied emotions like happiness, excitement, anxiety, curiosity etc. She has started planning for the new member of the family. She is also taking proper diet and medicines. But the picture is not so beautiful & bright as it seems to be. There are many fears attached to it. Her orthodox family wants a son only, and then what would happen if she delivered a baby girl? And, even if she has no such restriction then the fear of having unhealthy child due to pollution & adulterated food intake cannot be ruled out. Suppose she delivers a healthy baby. Birth of a baby brings yet another set of fears. She cannot trust even the nearest of the family members as there are various incidents of sexual harassment in the society. As the child grows older he must take admission in school but again the same insecurity of ‘whom to trust?’. There are so many infections & communicable diseases prevail in society due to neglect of the government & the citizens as well. Somehow that child is saved from the diseases, but the social insecurity still prevails. Schools are supposed to make students feel safe and secure as school is a second home for them. Schools try to take security measure or at least they show that they are doing so but ‘Can we say in affirmation that the reality is this only?’ The woman who was excited at the birth of her child is now captivated in the cage of fears like mental, physical & sexual harassment of her child apart from other fears like failing in exam, not getting top position in the class, not becoming master of all activities, not looking smart, not winning too many awards & trophies to show off and many more. As the child grows older there are other types of threats like suicide attempt, drug addiction, depression, anxiety, accident etc. which start taking a fairly good position in the mother’s mind. A mother wanted to give birth to the ‘Abhimanyu’ who can break into the ‘Chakravyuh of life’ is now captured in that Chakravyuh herself. She is surrounded by insecurity & mistrust, and this leads to loneliness, depression & frustration. This vicious cycle of mistrust, jealousy & hatred keeps on growing with the growth of the child.

A child who comes into the world with enthusiasm & zeal starts wasting his positive energy in combating the perceived fears given by society as a return gift.

We need to prepare Abhimanyus with divine armor to break into the Chakravyuh of life, but we are providing them with the armors of fears, insecurity, jealousy, hatred, competitiveness, cheating etc. and expecting them to win over the world with love. We give different input to get different output with yet another process. It is making the entire system a failure. There is no

coherence in what we teach & what we expect. There is a dire need of that education which takes us to the world of freedom, love, trust and faith, the path of freedom from fear.

Since time immemorial education has played its role in revolutionizing society and removing the social evils whereas philosophy supports education in planning the education system and then implementing them. In the above-mentioned scenario it is being talked about that the society is full of various kinds of fears and we need to tackle them effectively. For this a proper education system is required. When we start looking for that proper educational philosophy we find that the great Philosopher Sri J. Krishnamurti has talked a lot about freedom from all kinds of fear. There are many educational institutes, which are working on his philosophy.

Educational Philosophy of Krishnamoorthi

Krishnamurti is regarded globally as one of the greatest thinkers and educationist in modern society. He talked about the issues that concern all of us in our everyday lives, of the problems of living in modern society with its intolerance, violence, and corruption, of the individual's need for security and happiness, and the need for mankind to free himself from inner burdens of fear, anger, hatred, and sorrow. He explained the subtle working of the human mind and pointed to the need for making it free from any kind of control.

Krishnamurti described good society as one 'without violence, without the contradictions of various beliefs, dogmas, rituals, gods, without national and economic divisions.' One of the fundamental changes on the way to a good human being involves shedding special attachments that Krishnamurti considered divisive, and so facing life 'as a human being without a label.' Such a person would be free of anchorage, and thereby be well positioned to have 'right' values and 'right' relationships.

Krishnamurti once expressed, "I want to free man from chains and fear, I am not finding a new religion. My only purpose is to free man completely." He considered that belief comes from fear and is the most destructive thing. Fear & belief in the mind curb man from understanding the truth. Fear is one of the greatest problems & it destroys people when they think about the future. For him freedom means freedom from fear & any form of resistance.

Krishnamurti realized that education is making life dependent upon machines & its aim is to produce scholars, technicians & job hunters not the integrated human beings who are free from any kind of fear & slavery.

The function of education,' he said, 'is not to help the young conform to this rotten society, but to be free from its influences so that they may create a new society, a different world.'

Education is not about preparation for only a part of life (like work) but is about preparation for the whole of life and the deepest aspects of living. A teacher plays a key role in guiding the students towards achieving the aims of education. True teacher or a philosopher 'directs' the student towards true living, at the very beginning of life. He meets the children at an early age and teaches them the art of living life unconditionally by keeping their minds free and fresh. Krishnamurti devoted his life to the task of keeping the young minds uncluttered by thought.

According to Krishnamurti, the flowering of goodness is possible only in freedom and in the choice less awareness of our daily existence and activity. It is the total unfolding and cultivation of our minds, hearts, and our physical well-being. It is living in complete harmony in which there is clear, objective, non-personal perception unburdened by any kind of conditioning. It is the release of our total energy and its total freedom. Awareness is the will to understand and therefore a golden key to freedom.

Keeping in mind the educational ideas of Krishnamurti, The Krishnamurti Foundations set up many schools in India, America, and the U.K. and the teachings have been the inspiration for many others over the decades. The Krishnamurti entrées are intended for the study of the teachings of J. Krishnamurti. They are for people who would like to be in an environment of quietness and great natural beauty, where they can give full attention to these teachings and their implications in their own lives.

The threats and fears of modern society can be overcome with the educational ideas of J. Krishnamurti. The school children are also the member of society. If their minds are free from any kind of fear & insecurity they can create a better environment for themselves and for others as well.

If we learn from the teachings of Krishnamurti we can lead our paths towards knowing our own true self in an environment of freedom from any kind of fear.

The above-mentioned problem is faced by one or the other in different forms in various places, the form and cause of the fear may differ, but fears & threats remain. If we want to provide a peaceful life to mothers like Ragini then we should first create an environment where society is free from fears.

Conclusion

It is already discussed that modern society is facing various kinds of threats & fears. Apart from natural hazards there are several social fears & insecurities which prevail everywhere and affect its members. An individual is always having one or the other kind of fear in his mind. But for the proper development of an individual as well as society it is especially important to make them free from any kind of fear as fear restricts proper development.

Whenever we face any threatening situation we try to take solace in the hands of education. Education tries to fulfil the needs of society by providing conducive environment to the children. When we talk about preparing fearless young minds, we must take support of education.

J. Krishnamurti's education philosophy talks a lot about freedom of mind from any kind of bondage. The schools run by Krishnamurti foundation are taking care of this need as they are following the teachings of Krishnamurti.

According to him freedom never comes through discipline. Only individual attention and assistance enable the children to keep themselves disciplined. Obedience has no place in the right kind of education. Freedom enables the child to grow free & fearless. It enables the child to understand him and life.

Krishnamurti rightly says, "Freedom must begin with the individual. If the individual transforms himself, his life, then for him there is freedom and because he is the result of a total process, when he liberates himself from nationalism, from greed, from exploitation, he has direct action on the whole."

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**Effects of Covid-19 Outbreak on Early Intervention Programs for Children with Autism
Spectrum Disorder**

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Abstract

Autism is neurological developmental disorder, where the child appears physically normal, but their social behavior remains rudimentary, they never connect emotionally, and communication skills remain near to zero or zero. But early intervention may bring changes in social behavior which in return may develop expressive language in children with autism. This is a case study of a special school, where early intervention program is being run for the children with Autism. Before the outbreak of Covid-19, children with Autism Spectrum disorder were regularly came to school, teachers were giving early interventions and children were showing positive changes in their behavior. During the period of lockdown children with Autism could not attend the special school and the results of this pause were visible in the behavior of autistic children; through this paper researcher is trying to compare the effect of intervention program on autistic children, also the implications of long pause on the results of interventions.

Keyword: Autism, Intervention, Covid-19, Special School.

Introduction

The outbreak of covid-19 impacted the world bleakly. If we talk about the education system, offline classes were ruthlessly derailed for a stretched period of two years, studies were shaken (as online mode had many glitches and affording devices were not everyone's cup of tea), assessment of students' improvement were unmanageable. Many parents and teacher were upset as this online mode has engulfed all the seriousness required. The list of dilemmas is long, but when we investigate the ground level situation, at least those students who were studying in the normal schools, teachers had some sort of check via online mode. Students were in touch with

the syllabus, teachers, and friends and with other things, though there were many issues but the possibility of their draining into level zero was negligible, but just think about a special school teacher who face faced this trauma, as many of their students slipped back to the level zero. The need and requirement of especially abled students are very different from normal students and those were not possible to make them learn through online mode. In many houses parents didn't even bother to make them revise things that were taught in the school. The result was that such special students reached back to the level where they started and in many cases lots and lots of tantrums were added with this level. Through this paper a case study of a special school where early interventions was being provided to the Autistic student is presented. To the knowledge of teachers all these 13 autistic students were showing remarkable changes in their behavior. But with the outbreak of covid-19 special schools were also closed, autistic children along with other children were compelled to remain at their houses with no interventions, no therapy, and no sessions. When the schools are opened after a long pause of two years, teachers are so upset that they do not understand how to work to get them back on track as many of them have returned with prominent symptoms of autism spectrum disorder.

Autism spectrum disorder is condition where child appears indifferent. Despite being amid people they seem to be in some other world. They lack communicative language, are socially unresponsive and show repetitive pattern in their behavior. Autistic children lack any kind of emotions, don't understand how to react, or respond when present in a social setting. As a matter of fact, you will never be able to recognize an autistic child just in a single glance. Proper observation for a long period of time is required to label a child as an autistic one. There is no medical test or examination which can diagnose autism spectrum disorder. Leo Kanner in 1943 first time discovered a very peculiar characteristic in 11 children 1) coldness towards social circle 2) repetitiveness for certain things. Later Asperger discovered this syndrome in four children. Though Autism has come into light since 1943, but cause is unknown, so is the cure. Only early interventions are helpful in bringing changes in the behavior of children with autism. According to DSM-V (2013) Autism is a spectrum of disorder characterized by insufficiency in societal connections and social dealings. RPWD act 2016 defines autism as a neuro-developmental state typically appearing in the first three years of lifecycle that meaningfully affects a person's ability to communicate, understand association and relates to others and is

frequently linked with unusual or stereotypical rituals or behaviors. As per Hourigan, R., and Hourigan(2009) Autism is a complex developmental disability that characteristically appears during the first three years of life and is the result of neurological disorder that affects the usual working of the brain. It is said by cognitive psychologist that autistic people lack 'theory of mind' any straight observation or idea of other minds or other situations of mind- and that this lies at the heart of their different. Charles Hart, the author of "without reason"- a book about his autistic son and brother, sums up his son's thinking in one sentence: "Ted's statements aren't logical, they are associational". It has been found that autistic brain thinks in foremost three forms- Either they will be visual thinker or Music and Math thinkers or verbal logic thinkers.

What are the symptoms of Autism Spectrum Disorder?

Spectrum is a beautiful word which accepts a range of things. Autism also has a continuum of symptoms; there are various symptoms of autism, with varied intensity and range. It is possible that few symptoms which are present in one child with autism are not at all present in the other child with autism spectrum disorder. In Some autistic children the same symptom may remain present at the severe level whereas in some other child the same symptom may present at mild level.

1. Do not get socially gel with others, do not initiate talks, replies very rarely or never, behave as if nobody else is present in front of them, seems they don't understand anything, appears to be intensely thinking.
2. Don't have any best friend, don't go on outings with group of friends on their own, they never become part of any music or dance band. They don't have any foe as well because their social circle never develops. If anyone slaps them they will just cry, but never slap back.
3. Lacks expressive as well as receptive languages, they do not ask things verbally, either they will cry or show tantrum to get the thing. Self-harm is another way of asking for something.
4. No or very nominal eye contact, keeps their head down, flapping of hands, keeping hand on ear are the body languages being shown by children with ASD.
5. Some children with Autism also have Savant syndrome so those children show extra talent generally in the field of music, mathematics, thinking in pictures.
6. They show great fixation towards certain things, they will not do it easily, remain fixed there for long period of time.

7. They will show repetitive behavior, follow the same sequence daily, arrange things in a sequence.

8. Co-morbid condition is very normal among the children with autism. Co-morbidity is nothing but presence of other medical

Time table of early intervention program for autistic children being followed by a special school before lockdown -

School has maintained a case profile of every autistic child, and therapies, classes and sessions were provided according to it.

	8:00 am to 9:00 am	9:00am to 10:30am	10:30am to 12:30 pm	12:30 pm to 1:15 pm	1:15pm-to 2:15pm	2:15pm to 3:00pm
Monday	Yoga	Reading and writing	Speech therapy	Lunch	Sensory training	Craft training
Tuesday	Yoga	Reading and writing	Speech therapy	Lunch	Sensory training	Craft training
Wednesday	Cycling	Reading and writing	Speech therapy	lunch	Music	Group activity
Thursday	cycling	Reading and writing	Psychological sessions	Lunch	Music	Group activity
Friday	Yoga	Reading and writing	Psychological sessions	Lunch	Dance	Group activity
Saturday	Yoga	Reading and writing	Psychological session	Lunch	Dance	Craft training

Students came to the school sharp at 8 am and stayed till 3 pm in the evening. This special school for autistic starts their day with yoga for autistic students. A case profile of every student has been maintained, and according to the profile, students perform yoga in the presence of experts.

After this session, students were supposed to complete reading and writing works with teachers. They were meant to sit on a semi-circular table; only 3 students under the supervision of a teacher. Students were given simple mathematical problems like addition and subtractions, which they will need in future. These were not simple paper pen activities, instead teacher was making use of different items like pebbles, toys, fruits, or any other favorite thing . Besides this Mobile no. their name, parents and sibling's name, Address, and very simple things were made to read and write.

Autistic children do not communicate; they lack expressive as well as receptive language. So, speech therapy to improve communication, speech, articulation, to strengthen weakened vocal cord and to improve overall quality of life of Autistic students was being provided in three days of a week. Along with this in remaining three days psychological sessions were organized to control the moods, tantrum, and behavior of the autistic child.

Every day a 45 min long Lunch break was given, where they all were made to sit together so that they could learn to share. Parents were directed to send healthy food, no chips, or oily eateries. A caretaker was employed specially to make every child wash hands, eat their food, keep their lunch box on place and drink plenty of water after lunch.

In a week two days were also devoted for sensory training as these children have some sort of sensory issues. Craft, music, dance, and group activity for developing social behavior, as they were made to sit, sing and dance together; so that more and more situations arise to initiate and continue the talk.

Children with autism are hard to train, they will not follow the above-mentioned time table as a normal student will follow it. Say for example, autistic student may take entire one period in writing one word and to be true making them write a single word is also a big achievement. So, before the covid-19 outbreak students of rainbow school were showing steady growth. Let's understand this with the example of two autistic children, what they were before Covid 19, and what are they after Covid 19.

06 years old Satvik is a low performing autistic child, full of tantrums. He keeps on walking in the hall of school as he has fixation issue with place, seems restless at times and very poor at

expressive as well as receptive languages. But joining and getting early interventions, Satvik was showing great change. He started sitting on the chair, followed a few of the directions being given, started recognizing letters, was doing match the following exercise very aptly. Earlier he was not allowing the therapists to even touch him but with time he was okay with attending therapy sessions. Speech therapy was also bringing changes in his speech. Yoga and exercise helped in lowering tantrum and aggression. Overall changes were being seen and intervention had a positive effect on him.

Another child with autism, Sukhena, 05-year-old, when joined the school was most difficult to handle. She had lots of sensory issues. She used to bite her everywhere, pulling of hair, spitting, crying was her daily routine. But with the intervention program she settled down, her spitting behavior was under control, occasionally she was biting her and on teachers intervention she was coming into control also. Work was done on her sensory issues; therapies were showing changes.

But with the spread of Covid 19, schools were closed and as a results intervention programs were interrupted for a long period of time. Statuses of students were showing nobody in family paid attention towards their time table. Though few things were not possible to provide at the home during the lockdown period, small efforts would have helped them to not to slip at level zero.

Satvik after two years has started showing fixation as his prominent behavior, has forgot all the new words learnt and tantrum is at peak same is the case of Sukhena, stopping her biting herself has become difficult, keeps and crying and not ready to read and write anything. They both have gained a lot of weight also.

Effects of Long Pause on Early Intervention Program:

- 1) Many children with autism lose their communicative, expressive, receptive language. Effect of speech therapy either became zero or only negligible things were remembered by the students.
- 2) Fixation and tantrum became their most prominent characteristic.
- 3) Weight gain made them lethargic, which affected their physiological profile. Sugar level and cholesterol level were on peak.

4) They forgot dance, music, painting, and craft. Teacher must pay more and more extra efforts to revive them.

5) Repetitive behavior was at swing, breaking it has become problem.

6) They forgot to share things which earlier helped in maintaining their social behavior.

7) Many of them have started self-harm. Earlier none of them were not on medicine for aggression, now many of them are being put on dosage to stop their aggression.

8) They have stopped recognizing many of their teachers and friends, which is acting as a hindrance in their social behavior. Now they do not get ready to play, paint or dance with others. So, magnitude of the impact of interruption is so large that many of the students reached at the same standard, where they have started from, and it was way difficult to bring back them to the last level where they were before the lockdown. But this affect could have been minimized just by keeping few points in mind by the parents and teacher

1) Parents and caretakers should pay attention in intervention program being given to the child at the school. They can maintain the detailed description explanation, and changes being observed in the child.

2) Parents and caretakers should take interest in keeping the knowledge of talent (if any) present in the child. They should organize activities to hone the talent and do the needful to promote it.

3) Teacher should organize time-to-time meeting with parents and make them acquaint with the latest evidence-based intervention programs. Parents must be given the copy of time-timely ever week, so that they also should get information regarding the activities being planned at school

4) Parents and teachers should sign a pact of sharing of responsibility, everything should not be imparted at the school, and instead things could be shared and revised at the home.

5) Parents should keep on trying new interventions and note down the changes in the child.

6) At the home, it should always keep in mind that child should not be left isolated; he/she should be treated as the other members of the family. It will give a chance of interaction to and with the autistic child.

Conclusion

It has been observed that when a special child takes birth, he/she is considered as a burden. Those parents who can afford education generally send them to special school on the other hand poor people lack resources so they in turn are bound to keep the child at home. Generally, efforts

from parents (be it financially sound or poor) remain negligible at personal level. In the beginning years, parents do have some hope; that one day their child will be like the children of his/her age. But with the passage of time this hope diminishes, they accept their child as “nothing can be done”. Same is the case of autistic children either they are placed at the school because for a while he/she will remain at school and trouble created by him/her will be less at the home, or parents send them to special school in the hope of a miracle . Being a parent or as sibling or a community member everyone has to share their share and understand that it is true that “nothing can be done” to make them alike the other children, but lots of things could be done in order to make them “what they can be” by simply understanding the level of Autism, talent of the child in case of both “autistic savant” and “autistic only”; presence of co-morbid condition (if any), and giving the child need and evidence based intervention program since from the very beginning and continuously at the school and home as well; otherwise interruption caused by Covid 19 is in the front of everyone.

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**Challenges in Implementation of NCFTE – 2009 with Reference to Three Year
Integrated B.Ed. – M.Ed. Self-Financing Program in Odisha**

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Abstract

Transaction of the curriculum is one of the most critical aspects of the proposed teacher education program. The Curriculum Framework for Teachers Education, 2009 developed by the National Council for Teacher Education, deals with preparing and re-orienting teachers for enabling the child to learn in a child-friendly and child-centered manner through activities, discovery and exploration of his environment and surroundings. In Odisha, Fakir Mohan University (2016-17), Maharaja Sriram Chandra Bhanja Deo University (erstwhile North Odisha University) (2016-17), Rajendra University (Erstwhile Rajendra Autonomous College, Balangir) (2017-18) and Sambalpur University (2018-19) have started 3 Years integrated B.Ed.-M.Ed. Program on self financing mode in view of the NCFTE, 2009 to improve quality teachers education program. This paper analyses the implementation of curriculum of all four institutes particularly overload of papers and time restrictions, Lack of logical sequencing of topics in classroom transaction, overlapping of contents, unavailability of resource material, absence of collaboration with local schools, etc. are major challenges in implementation and suggestions to address these challenges are also described.

Keywords: Curriculum, NCFTE, Integrated B.Ed.-M.Ed., NEP

Introduction

Learning among children can be enhanced with the help of quality teachers in schools. The quality of teachers depends on the teacher educators who are prepared through various Teacher Education Programs. The National curriculum framework for school Education NCF 2005 by NCERT recommends that teacher Education must fulfill emerging needs and demands of school education in the country. The National Education Commission (1964-1966) viewed, “The

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destiny of India is now being shaped in her classrooms.” So, the Draft National Policy on Education 1986 emphasized that the socio-cultural ethos of the society is reflected by the status of the teacher, and it is said that no people can rise above the level of its teachers. Further, the Yashpal Committee Report (1993), Learning without Burden, noted: “The emphasis in these programs should be on enabling trainees to acquire the ability for self-learning and independent thinking.” In the light of Right to Education Act-2009, National Curriculum Framework 2005 and National Knowledge Commission Report 2007, the horizon of education has completely changed and the field of teacher education in the country is confronting a paradigm shift. The National Curriculum Framework for Teacher Education 2009 is a vital document in this regard, towards Preparing Professional and Humane Teacher. This article tries to critically analyze the challenges in implementation of NCFTE –2009 with reference to 3 years integrated B.Ed.-M.Ed. Program on Self-Financing Program in Odisha and suggests measures which should be implemented in the teacher education scenario.

3 Years Integrated Teachers Education Program in Odisha on Self Financing Mode: Govt. of Odisha in Higher Education Department allowed 8 State Universities and 17 Reputed Government Colleges to run Self Financing Course Teachers Training Program on Self Financing Mode from 2016-17. In accordance with the approval of Government and subsequent reorganization of NCTE, New Delhi, Fakir Mohan University (2016-17), Maharaja Sriram Chandra Bhanja Deo University (erstwhile North Odisha University) (2016-17) and Rajendra University (erstwhile Rajendra College (Autonomous)) (2017-18) and Sambalpur University (2018-19) have started 3 Years integrated B.Ed.-M.Ed. Program on self-financing mode. The curriculum of all four institutes has been analyzed and found that all these universities/institutions adopted followed the same curriculum as per the recommendation of NCFTE, 2009.

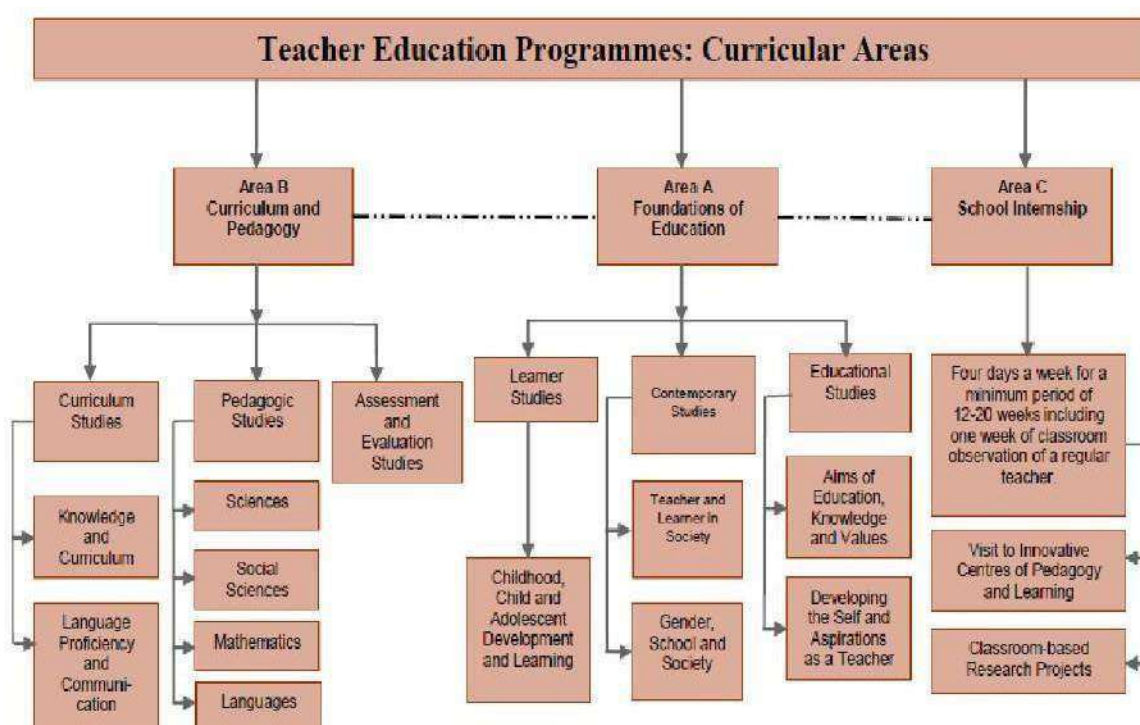
Highlights of NCFTE, 2009

According to the National Curriculum Framework for Teacher Education (NCFTE, 2009), ‘A teacher needs to be prepared in relation to the needs and demands arising in the school context, to engage with questions of school knowledge, the learner, and the learning process. The expectations of the school system from a teacher change from time to time, responding to the broader social, economic, and political changes taking place in the society.’ Moreover, The

NCFTE, 2009 has given a systematic and comprehending framework of curriculum for teacher education and highlights the strategies to implement it. Every aspect of teacher's education got preference in this framework. If we look at the framework some important points will come out such as follows.

- NCFTE, 2009 touches all spheres of education e.g., Theoretical, Practical, Psychological, Philosophical, and Socio-Economic aspects.

Fig. I. Flow Chart of main curricular areas along with potential courses



Source: NCFTE, 2009, P.27

- In this framework Area-C is a very crucial and essential part of the teacher education program. The practice teaching and innovative programs enriching the programs of teacher education are focused in this area. The other two areas i.e., Area-A and Area-B are common and already known areas and it is extracted from the earlier frameworks e.g., NCF- 2005. But the significant point is that NCFTE, 2009 gave a single paradigm and caught all the features of the curriculum to bring a desirable change in teacher education. The Flow Chart-I shows the core curricular areas and potential courses.

- NCFTE, 2009 has made an impact in the scenario of the educational system. The view of NCFTE – 2009 helped to provide Open and Distance learning (ODL) program for teachers in the state. NCFTE, 2009 along with pre-service teacher education programs also carefully and practically emphasized In-service and CPD programs and its various perspectives.
- NCFTE, 2009 streamlines the way for implementation of curricular areas by giving practical and reasonable strategies.
- This framework is much reliable for making professional and humane teachers.

Major Challenges in Implementation of NCFTE, 2009

- **Lack of multidisciplinary approach**

Preparation of quality teachers must be an activity of multidisciplinary perspectives. Teacher Education Programs require multidisciplinary inputs, and education in high-quality content along with pedagogical perspectives. NPE, 2020 aims at developing educationally sound, multidisciplinary, and integrated teacher education programs by 2030. Though the teacher education institution/ teacher education department has been brought under a single umbrella, these institutions are not intellectually and academically interconnected with other departments of the university and remain isolated.

- **Overload of papers and time restrictions**

The four-year program was clubbed into the 3 years program by increasing more numbers of papers in each semester. The number of papers is more in each semester, which consists of core papers as well as practical papers. It has increased the workload instead of reducing it as all of them have to be covered in a particular semester. In each semester some days are also to be allotted to Preliminary School Engagement which further reduces the tuition time in the institution. The burden of curriculum encourages rote learning, which will not be able to make learning a joyful, participatory, and meaningful activity.

- **Overlapping and outdated content**

The contents of various core papers overlap in different terms. Many of the topics like approaches to teaching and learning, gender issues, Use of TLM, inclusive education etc. are repetitive in various papers. Syllabi are not updated regularly with the changing policy and government guidelines in the field of education and as a result syllabus got outdated

with time. Besides, important aspects such as foundational literacy and numeracy, knowledge of India and its values/ethos/art/traditions have not been technically integrated with the syllabi.

- **Lack of Logical sequencing of topics in classroom transaction**

Few of the units of core papers are not in proper sequence which restricts the proper flow of content like application constructive approach has been taught under Pedagogy without teaching theories of learning. There was no coherence in classroom transaction of two core papers.

- **Practical**

NCFTE, 2009 emphasizes contextual learning and promotes constructive approach in teaching learning for real life experience. But pupil teachers find very little time to implement practical classes on Constructive driven Methods such as 5E and ICON Model in teaching various subjects. Further, the institutions are not well equipped with TLMs to explain various difficult/abstract concepts.

- **Field works**

With the vast content, each core paper has fieldwork which needs to be covered along with the theoretical content. The number of field works is more, and it is difficult to cover these field works in the given or available time.

- **Use of ICT**

Technology in education has made inroads into teacher education. Under SSA and RMSA schools have also been provided with technologies. Curriculum of teacher education program needs integration of technologies into the learning process of children. The changing time demands teachers to be well equipped with modern technologies so that they can use them effectively in the classroom for a better teaching learning process. Keeping this requirement in mind ICT has been made an important part of teacher education. In the three-year integrated course, this aspect has been included but challenges related to it are that it is included as one of the practical in the second semester; it's not a part of core theory papers. Another problem related to it is that all practical require trainees to prepare e-portfolios and digital profiles of various activities to be conducted during practical, so they

need to be trained with computers and ICT, which highlights the importance of making ICT as an important part of core papers that is missing in presently available curriculum.

- **Unavailability of Books**

The course has introduced many new subjects like language across the curriculum, understanding discipline and subjects but the reference material and books related to them are not easily available. Books of foreign authors are included in suggested readings mentioned in the curriculum, most of which are difficult to access.

- **Teacher's Content Knowledge**

Classroom instruction and the richness of learners' experiences are influenced by the content knowledge of the teachers. Many new subjects and new topics such as SDGs, educational Diaspora, SPSS, QDA for data analysis, Web 2.0 Technology and integration of mobile device, e-reader, laptop, interactive whiteboard have been introduced in three-year curriculum. Because of the complex nature of the topics and teachers' unfamiliarity with the content, the introduction of new topics is going to place a greater burden on these teachers. Curricular reforms are not part of the curriculum development process; thus, teachers lack clarity on it, and they also lack orientation which is needed for properly completing these courses.

- **Availability of Schools**

Due to the increase in the number of teacher education institutes, the schools are not available in sufficient manners for practicing teaching skills. School environment is dynamic in nature and hence changing day by day. It is not positive to practice the skills. The approach of student's teachers & admin education has been changed. Negation of school concerns results in poor preparation of student-teachers. Universities with a Teacher Education Program should have a network of government and private schools to work closely with local schools and teachers trainees can participate in school activities like community services, adult education, and vocational education, etc. But universities are not keen on building networks with the local schools resulting in overlapping of school examinations/events with the practice teaching of teacher education programs.

- **Lack of Institutional arrangements**

NCFTE, 2009 envisaged health, physical education, yoga, and vocational education as important aspects of the core curriculum at the primary, secondary, and senior secondary school levels for overall (physical, social, emotional, mental, and spiritual) development of learners. In this program, various practicals have been introduced including Physical Education, Art in Education, Yoga etc. which requires resource persons and experts in these areas and proper institutional facilities and arrangements, but these institutions do not have such experts and facilities and even recruiting such experts for one practical and making institutional arrangements also adds to their expenditure. None of them are following such classes or assign the duties for such practical tasks to their regular teachers who lack expertise in the field, and it will also add to their workload.

- **Difficulties in Evaluation**

As most of the fieldworks involve a variety of observations and digital aspects, it is difficult on the part of the internal and external evaluator to check the authenticity of these observations and fieldwork. Also, it is equally difficult to evaluate the digital profiles of all activities due to the workload of teachers and lack of ICT arrangements. Further, the internal mark was just a reflection of percentage of attendance in class rather than actual evaluation.

- **Different Pay Structures**

Teacher attrition rate is high due to uniformity and unremunerative pay structures for which subject specialists are not being strained for an extended period. A little additional pay in other institutions resulted in shifting of faculties from one institution to another institution. This also leads to class adjustment with other subject teachers without any special orientation and special training.

Suggestive Measures

- All teachers training institutions must be intellectually and academically interconnected with other departments of the university to make it multidisciplinary and integrated in true spirit. Student interaction program, exchange of pedagogical content knowledge among departments shall be organized periodically.
- By correlating different subjects and covering them in proper sequence, overlapping of content can be reduced. Contents need to be specific and properly defined, integrated so that each topic can be delimited which will bring uniformity. Number of papers need to be

evenly distributed to reduce the curriculum load keeping in view the relevance of the topic and subject. Special attention shall be given to development of skills on Functional Literacy and Numeracy (FLN) and value education with model techniques.

- To provide a cooperative school environment during the internship programs of the Teacher Education, school authorities need to be officially intimate regarding the expectations from schools according to the changed curriculum. The viewpoints of school authorities need to be taken into consideration, while implementing school engagement programs.
- ICT should be a part of core theory papers along with its practical aspect and it should be introduced in the first semester to remove the requirement of learners to be computer literate at entry level.
- The teachers training institute appoint Physical, Art and Yoga instructors or the pupil-teachers need to be tagged with such institution which has expertise in such areas.
- Appropriate reference materials and good quality books should be made readily available related to the new syllabus. E-library is one of the options to keep the content updated as per changing policies and guidelines of the government and findings of latest research.
- Teaching as a profession, calls for systematic evaluation of all facets of the professional training – knowledge and understanding of educational theory, practical field skills and competencies related to learning and teaching and professional attitudes and values. Hence, a comprehensive evaluation process needs to be adopted to assess all such areas which are sine qua non to the teaching profession.
- Uniformity and remunerative may structures shall be given to teacher educators through inter-institutional coordination and minimal service period shall be enforced to address high attrition rate teacher education. Self Financing Institutions may empanel a group teacher educator to meet the requirement of teacher education. Interuniversity coordination for classes by subject teachers through interactive technologies may resolve the issues related to non-availability of specialized teacher educators.
- Teacher Training Institutional shall not be restricted to teaching only. Research and extension on education shall be given utmost priority. Research related to futuristic curriculum, emerging pedagogical, community participation; alternative assessment in close

coordination with school and department of education shall be conducted periodically. Trainees shall select topics based on actual problems in school education in consultation with local school and administrators rather than repeating a topic without its importance in subsequent academic years.

- Subject Matter Specialists (SMS), especially people having practical experiences on the subject areas should be invited to take special classes on newly incorporated contents.
- Higher education institutions with teacher education programs must establish formal collaboration or a network of government and private schools, which will avoid overlapping of practical teaching with the school events. Practice teaching for trainees shall be very flexible to complete the practice teaching in various schooling environments. Teaching to small groups using micro-teaching techniques, evening classes in residential/Ashram school/OAV, classes in urban slums, seasonal schools for migrants, and special classes for low performing students could be options. This will also help the trainee to participate in school activities such as community service, students counselling to dropout students, adult, and vocational education, etc. and get practical exposures on various activities of school. Universities with teacher education programs may adopt few schools as a unit of educational experiment.
- Detailed orientation related to new subjects and topics should be provided to the teachers.
- Activities related to Research, Development and Extension need to be promoted in curriculum transactions and use of innovative methods in various curricular areas of the schools and teacher education.

Conclusion

It may be concluded that the NCFTE, 2009 is much more relevant in the present scenario of teacher education. The problem is the area of implementation particularly in integrated courses. Institutions implementing such program should investigate the framework, exchange their ideas, make positive approaches, and pave the way for a positive change in teacher education without which the future generation cannot be groomed properly. We are also hopeful for positive change.

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Academic Adjustment in Later Adolescence: Challenges and Possible Coping Strategies

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Abstract

Academic adjustment deals with the adjustment pattern of students in their academics and education. The issues pertaining to academic adjustment are of major concern among students of later adolescence. It has a range of effects on students' overall development. However, because of rapid hormonal changes in the body of students, transitioning to adulthood, students fail to cope with environmental and societal demands. The review article is focused on analyzing the existing research and studies dealing with a wide range of adjustment issues and challenges faced by students of later adolescence that hamper their overall development. In this review article, studies on providing suitable coping strategies and solutions to academic adjustment issues were also reviewed. The academic adjustment and allied challenges must be addressed by social agents as early as possible to ensure students' overall development. The results of this review article revealed that academic adjustment has a very key role in ensuring learners' academic achievement.

Keywords: Academic Adjustment, Later Adolescence, Challenges, Coping Strategies

Introduction

Throughout life, our psychological needs are continually being aroused and we engage in behaviours that reduce or satisfy these needs. All human behaviours, conscious or unconscious, voluntary, or automatic, planned, or unplanned, wise, or foolish represent our attempts to meet our needs. The constant cycle of needs arousal and satisfaction, in a broad sense, is the process of adjustment.

“Adjustment is the process by which an organism maintains a balance between its needs and circumstances that influence the satisfaction of these needs.” (Shaffer, 1961) Adjustment is spread over a wide array of human lives. Academic adjustment is one of those. It is the

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modification of how a student participates in-class activities. It refers to the development of coping skills (Nonglait & Myrthong, 2021) and learning strategies aimed at achieving satisfactory academic results.

Students across the globe face these issues of academic adjustment every year. Freshmen students are students studying in the first year of their higher education. Freshmen students are the students living in their stage of later adolescence (Quan, Zhen, & Yao, 2014). Later adolescence refers to the age group of students up to the age of 24 (Jan, et al., 2016). First-generation students refer to the students who are the first-ever children to study higher education in their families.

Approximately 1/3rd of students are first-generation students (Choy, 2001) and over the past 25 years, it was analyzed that students of the first generation are less likely to be well-adjusted academically.

Factors such as lack of coping strategy, lack of social support and surprising standards that are overwhelming, educational demands, social stressors related to communication barriers, low socio-economic standing, disorientation & culture shock contribute to academic adjustment problems & challenges (Jemal, 2012).

Factors such as academic achievement of students, personality traits, peer interaction, and loneliness lead to the determination of academic adjustment level (Laycock, 1984), (Lidy & Kahn, 2006), (Ryan, 2011)&(Quan, Zhen, & Yao, 2014). Academic achievement is of utmost significance and issues relating to bad academic adjustment of the students (Kenney, Lac, Hummer, Grimaldi, & LaBrie, 2015).

Keeping in view the immense importance of academic adjustment in the prospects of a student's life, the present review article focuses on academic adjustment and related challenges faced by students in their later adolescence. This review article also aims to understand the steps to overcome the academic adjustment issues of adolescent students from the reviews. Also, to provide possible coping strategies that can be made available to students of later adolescence as early as possible to ensure students' academic adjustment in a proper manner.

Academic Adjustment Challenges

Colleges and universities have traditionally provided the opportunity for young people to continue beyond high school and complete their preparation for adulthood. But since these students and the campus diversity are so varied, the student drop-out rates are increasing because of mal-academic adjustment (Choy, 2001). The academic adjustment issues among adolescent students can be of the following types.

(i) Lack of coping strategy.

Coping strategies are behavioural and cognitive tactics used to manage unsuitable conditions. Coping strategies and their deficiency among adolescent learners affect their academic adjustment badly. According to Nonglait & Mrythong (2021), coping strategies decide the academic adjustment pattern of students.

Ineffective coping strategies lead to poor adjustment to college, such as coping motivated drinking affecting students' health adversely (Kenney, Lac, Hummer, Grimaldi, & LaBrie, 2015). Many of the students feel incompetent and face difficulty in coping with the lifestyle in universities which causes maladjustment among learners academically (Jemal, 2012). College students instead of adopting a positive coping strategy adopt negative coping strategies that show low self-efficacy resulting in bad academic achievement (Quan, Zhen, & Yao, 2014).

(ii) Lack of social support.

Social support is the perceptual thought of being cared for by someone. It has the feeling of a sense of assistance available from other people and most importantly that one is part of a social network.

According to Lidy & Kahn (2006), social support has a direct relationship with a student's perception of academic adjustment. Less amount of social support leads to the maladjustment of a student academically.

Most students lack social support from their family, friends and surroundings including social support from teachers and professors. This lack of social support directly affects the academic adjustment of students (Kenney, Lac, Hummer, Grimaldi, & LaBrie, 2015).

(iii) Loneliness.

Loneliness is a negative emotional experience, which tends to make individuals adopt a negative view of their surroundings and lose interest in positive exploration.

Loneliness may hinder freshmen's good academic adjustment as they lose interest in mastering learning skills and take a negative attitude towards their surroundings (Quan, Zhen, & Yao, 2014).

The analysis suggested that loneliness had a direct negative effect on academic adjustment and negatively affected academic adjustment by activating a negative coping style and suppressing a positive coping style. Both forms played a mediating role in the relationship between academic adjustment and loneliness in higher education institutes.

(iv) Poor peer interaction.

Peer interaction provides a unique context in which children learn a range of critical social and emotional skills, such as empathy, cooperation, and problem-solving strategies.

In adolescence, students face a severe crisis in terms of establishing proper peer interaction. Students are unable to form a friendship in adolescence with their peers. This poor interaction among peers leads to the development of academic anxiety and hampers academic adjustment (Nonglait & Myrthong, 2021).

Peer interaction in the initial days of college life makes problems in a student's adjustment pattern. Inability to express their anxiety, and stress among peers leads to the development of issues academically (Ryan, 2011).

(v) Maladjustment to new values and norms.

Whenever there is a transition of residence among cities or states, students face a culture shock that hampers students' overall adjustment pattern. In terms of academics, students face new norms and new classroom culture that takes time to sync in.

All freshmen-level college students face some form of adjustment problems during the transition to higher-level education. The new values, norms and unexpected standards are overwhelming in this scenario. It leads to the development of incompetency among students (Bennett & Okinaka, 1990).

(vi) Authoritative parenting style.

Parenting style refers to the overall role and behavioural aspect of parents towards their kids. It also includes their attitude to their kids in their overall growth and development process.

Parenting had a greater impact on the academic adjustment of students. Authoritative parenting leaves a negative impact on the academic adjustment of learners. It was also found that most families had an authoritative style of parenting. Therefore, students who grown up with authoritative parenting are found to have low academic adjustment levels (Kenney, Lac, Hummer, Grimaldi, & LaBrie, 2015).

(vii) Low personality traits.

Personality traits are characteristics and qualities that help define an individual as a unique individual. They're often developed throughout life and may remain consistent across many situations and circumstances. The transition from school to college can be stressful. Here personality plays a key role in the academic adjustment of the learner.

Students studying in the first year of their development of academic adjustment issues. So having low personality traits can have a severe impact on the maladjustment of a learner (Lidy & Kahn, 2006).

(viii) Psychiatric morbidity.

Psychiatric morbidity is a mental illness, which is a behavioural or intellectual sample that causes considerable distress or impairment of private functioning.

According to Kumaraswamy (2013), psychiatric morbidity can have a negative impact on overall adjustment patterns emphasizing academic adjustment.

(ix) Poor communicative skills and new language.

Communicative skills help in great interaction and coordination among different agencies of society. Communicative skills and language have complex integration to interact with society and social agencies.

Communicative skills and poor language acquisition are major barriers for students having a transition to higher education in terms of academic adjustment (Bennett & Okinaka, 1990).

Language has a key role in academic adjustment. Most students faced academic adjustment challenges because of a lack of hold of the English language (Nonglait & Myrthong, 2021).

(x) High education demands.

Educational institutions, especially the top institutions of the country have a lot of expectations from their students. For example, good team management, better relations with

authorities, peers and teachers, and better coordination among curricular and co-curricular activities.

Advanced courses provide greater intellectual challenges and place greater demands on students. The surge in academic demands has a negative impact on the academic adjustment of students, e.g. “I’m finding academic work at college difficult.” (Choy, 2001)&(Kenney, Lac, Hummer, Grimaldi, & LaBrie, 2015).

Coping strategies

To overcome the academic adjustment challenges, various researchers have suggested possible coping strategies. A few of them are as follows,

(i) Orientation program at colleges.

The orientation program makes learners acquainted with the college environment and interact with peers, teachers, and the environment of stay.

Higher education institutes should organize an orientation program when students first join college. It will enable learners to be oriented to the kind of challenges they face when making the transition into higher education (Nonglait & Myrthong, 2021).

(ii) Development of proper coping strategy.

Good coping skills can be achieved by creating more chances for freshmen to communicate and establishing a new social network. It can also provide adequate psychological counselling services (Quan, Zhen, & Yao, 2014).

According to Nonglait & Myrthong (2021), students should adopt task-oriented coping and emotion-oriented strategies in coping with academic adjustment challenges.

(iii) Improvising social support.

To perceive more social support, greater emotional stability is of utmost importance. This perception of social support is responsible for students experiencing better adjustment in academics. In order to develop proper social support, peer group support networks and partnering 1st-year students with their seniors confer a proximal source of well academic adjustment (Lidy & Kahn, 2006) and (Kenney, Lac, Hummer, Grimaldi, & LaBrie, 2015).

(iv) Peer interaction to overcome loneliness.

Bauer & Rokach (2004) found out that students on order to cope with the feeling of loneliness when making the transition to college by staying connected with their friends from school.

Attention to peer dynamics must be emphasized. For the improvement of peer dynamics, teachers must play an active role (Nonglait & Myrthong, 2021) and (Ryan, 2011).

To overcome loneliness, college educators can organize group activities and peer mentoring programs in academics. These will have a positive impact on peer relationships. Good peer relationship has a direct effect to overcome loneliness.

(v) Promoting authoritarian parenting style.

As observed in the works of Kenney, Lac, Hummer, Grimaldi & LaBrie (2015), authoritarian parenting impacts academic adjustment positively. To ensure better academic adjustment authoritarian parenting style must be promoted among parents by the teachers.

(vi) Improving better personality traits.

College counsellors have a key role in academic adjustment. College counsellors can improve better personality traits by providing services in the areas of prevention of poor adjustment and college attrition, individual and group therapy, and outreach (Lidy & Kahn, 2006).

(vii) Counselling to reduce psychiatric morbidity.

Kumara swamy (2013) suggested a few preventive measures to overcome psychiatric morbidity. A few of them include,

- a) Conducting seminars and workshops to identify the psychological problems of students,
- b) Setting up counselling centers in all colleges,
- c) Creating awareness among students by organizing interactive sessions and
- d) Regular workshops on stress management, time management, assertive training, and communicative skills.

(viii) Improving language-related issues.

The language barrier and allied issues can be overcome by relying on help from students, friends, and roommates. Asking the difficult concepts to the teacher repeatedly. Starting a newcomer club to share newcomer problems and experiences. Socialization and sensitization of peer-mates towards language issues by setting up a counselling cell at college (Sridhar & Jain, 2018).

(ix) Overcoming Culture Shock.

The culture shock according to Sridhar & Jain (2018), can be overcome by establishing newcomers club at the institutional level to celebrate their regional festivals. Also socializing with locals of the region to adapt to the new culture.

(x) Overcoming Educational Demands.

Educational demands as suggested by Jemal (2012), can be overcome by improved interaction among peers, improvised teacher-student interaction etc. Teachers can act as facilitators to overcome the conceptual understanding issues of students.

Adding to that, the organization of student-friendly classrooms and multilingual instructions can act as catalysts in overcoming academic adjustment-related challenges.

Conclusion

Academic adjustment and challenges pertaining to the same area of major concern in this developing world. The academic adjustment challenges among learners of the adolescent phase, specifically studying outside their area of the homestay are of concern to stress.

Academic adjustment challenges as found out by researchers are, lack of coping strategy (Kenney, Lac, Hummer, Grimaldi, & LaBrie, 2015; Nonglait & Myrthong, 2021 & Quan, Zhen, & Yao, 2014), lack of social support (Kenney, Lac, Hummer, Grimaldi, & LaBrie, 2015 & Lidy & Kahn, 2006), loneliness (Quan, Zhen, & Yao, 2014), poor peer interaction (Ryan, 2011; Nonglait & Myrthong, 2021), mal-adjustment to new values and norms (Bennett & Okinaka, 1990), authoritative parenting style (Kenney, Lac, Hummer, Grimaldi, & LaBrie, 2015), low personality traits (Lidy & Kahn, 2006), psychiatric morbidity (Kumaraswamy, 2013), poor communicative skills and language barrier (Bennett & Okinaka, 1990; Nonglait & Myrthong, 2021) and lastly high education demands of institutes (Choy, 2011; Kenney, Lac, Hummer, Grimaldi, & LaBrie, 2015).

The coping strategies for the academic adjustment challenges include.

Orientation programs at colleges (Nonglait & Myrthong, 2021), developing proper coping strategies (Quan, Zhen & Yao, 2014; Nonglait & Myrthong, 2021), improvising social support (Lidy & Kahn, 2006; Kenney, Lac, Hummer, Grimaldi, & LaBrie, 2015), peer interaction to overcome loneliness (Bauer & Rokach, 2004; Nonglait & Myrthong, 2021; Ryan, 2011), promoting authoritarian parenting style (Kenney, Lac, Hummer, Grimaldi, & LaBrie, 2015), developing better personality traits (Lidy & Kahn, 2006), counselling to

reduce psychiatric morbidity (Kumaraswamy, 2013), improving language-related issues (Sridhar & Jain, 2018), overcoming culture shock (Sridhar & Jain, 2018) and overcoming educational demands (Jemal, 2012).

Later adolescence proceeds to adulthood. A phase where an individual starts his/her action towards satisfying personal, family, societal and national needs. This will only take place when an individual attains his/ her highest virtue in academics. Again adjusts himself/ herself to the ever-changing needs of society and surroundings. Adolescent students facing the challenges of academic adjustment must adopt a suitable coping strategy to overcome all aspects of academic adjustment and allied problems. A well-academically adjusted individual has maximum chances to conquer future hurdles in the academic span.

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Journey from Periphery to Centre: Inclusivity of Genders from Educational Perspectives

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Abstract

Inclusivity is a necessary weapon to fight against any kind of discrimination. Education is supposed to be the best tool to better any society and a society cannot be imagined without human beings. If this society remains under the dungeon of prejudices and segregation regarding genders, the order of the society will be dismantled. One of the burning issues in our society till now is gender inequality. Even after the proliferation of education, this problem threatens society by hindering its progress. Without the cooperation of all the genders a society cannot prosper. Gender issues do not only encompass the issues of women; rather these address the problems faced by transgender people and even by men in our patriarchal society. So, present paper aims to highlight the problems faced by different genders in different arena of their lives and at the same time it also suggests some measures to cope with such situation along with the inspiring thoughts of few eminent personalities, some instances who have challenged gender inequality and the recommendations of some commissions. This paper is in support of gender inclusivity in the main stream of education to strengthen this society as well as our beloved nation.

Key Words: Gender Inequality, Socialization, Gender Harmony, Women education, Transgender

Introduction

Inclusivity is an umbrella term which covers a vast space. If it is related to society it encompasses people across all the sections of the society irrespective of caste creed religion race gender class and age. But in the case of formal education there is a specific age limit. Inclusivity also denotes the persons with special need and empathetic care. Despite all the initiatives undertaken by government and other non-government organizations to build an inclusive nation,

inclusive society is till now a utopian concept to a large portion of society. India is still witnessing discrimination against backward or so-called lower caste people, women, physically or mentally challenged persons, homosexuals, transgender people, androgynous people, raped women, divorced or left women, widows, acid victims, single mothers, orphans, people without proper birth identity etc. In most educational institutions, children of single mothers or children without parental identity are prohibited to be admitted. Transgender people are still segregated from the main stream of education as well as society. Moreover, despite the flourish of campaigns, policies, and laws to promote women education, women are also marginalized in multifaceted ways. So, to better the condition of inclusive education gender issue is overly sensitive.

For a long time, women were hardly considered as human beings. After that when their separate existence as Homo Sapience was recognized, they became included as second-class citizens— the second sex. Now, after a long struggle the concept of inclusion of transgender people in various fields including academic and official fields, political field etc. has come into existence. ‘Transgender’ itself means ‘beyond gender.’ A castrated man who is known as a ‘Khoja’ in our historical past belongs to this category. The people who are born with ambiguous genitals where external genitals appear as same, i.e., what the society understands as ‘Hijra’ or ‘Kinnar’ are also transgender people. Transsexual people who have changed their sexes are also referred to as transgender. Transgender people are also those who have not changed their sexes but regard themselves as their opposite genders and characterize themselves according to the opposite genders. Hopefully now-a-days in many forms a separate option has been included for transgender people. First, this should always be remembered before passing any comments or creating any kind of segregation that gender is a social construct based on biological or sex differences. It demands a changing worldview to reshape genders not by just mere role playing. Inclusive approach can cement these gaps among various genders or in other words inclusive education can blur the manmade watershed among genders.

Viewpoints of Eminent Personalities

India has witnessed the activities of many renowned educationists who have shown their concerns regarding gender bias and education. Pandit Ishwar Chandra Vidyasagar, Raja Rammohan Roy, Swami Vivekananda, Rabindranath Tagore, Begum Rokeya, Jyotiba Phule and

Savitribai Phule advocated the importance of women education and undertaken active steps by propagating and founding many schools for girls. All of them emphasized the fact that without women education and progress no civilization can flow smoothly. According to **Vivekananda**, a bird cannot fly with one wing (Singh, 2014). Same thought was reflected from the speeches of **Begum Rokeya**, “We (women) constitute one half of the society and if we are left behind, how can the society progress? If a person’s one leg is tied how far can he go?” (Khatun, 2019). **Tagore** also advocated the thought that nobody has the right to snatch the right of women. More or less all of them stand by the inclusion of women in education. Not only in education but also in every step of life men and women both need each other’s assistance. Tagore’s *Chitrangadamay* may be cited here. Chitrangada neither wants to be goddess to Arjuna, nor does she want to be a slave to his lover. She wants to be with him in every will and woe being a companion (Lahiri, 2021). Such thought can balance our society. Eminent psychologist **Sudhir Kakar** (1989) points out, marital ‘relations seem impelled more by hostility than tenderness or love... pervaded as much by hatred and fear as by desire and longing.’ In this situation, neither of the spouse can live peacefully nor their offspring. Its adverse impact affects education also.

Objectives

The aim of the paper is to promote gender equality in our society however the paper emphasizes two objectives as follows:

- To highlight the problems faced by different genders so that every human being is aware of the facts and may try to avoid such causes which hinder gender inequality.
- To discuss some probable remedies to cope with gender discrimination so that everybody can take care of those measures to promote gender harmony.

Problems Faced by Different Genders

To discuss any issue at first it is necessary to identify specific problems. If we know the roots of the problem only then it will be possible to uproot those faults. From the introductory passages one may assume that gender related problems are faced only by women or transgender people. But it is not true. Men also face problems due to various social issues. LGBT is also related to this gender-based problem. Lesbians, gays, bisexual people also face humiliation like transgender people. How gender plays its role in different spheres of education has been discussed here.

Before going to school the basic education and first process of socialization of a child begin at home. So, family plays a vital role in a learner's life. The values of the family members have effect on child's upbringing. Children imitate their elders. So, it is necessary to mend their behavior first. When the children see their parents perform different instinct roles, in their inner world they also start to imagine themselves in the position of their parents. That is why children are compared to the mirror of the family. They learn to differentiate the etiquettes between boys and girls. Like, girls should not shout or even speak loudly but it is quite common to boys. Imitating their parents, the girl child plays with dolls and utensils. From childhood she becomes child-mother or child-wife and on the other hand the boy child regards him as the master of the house. This conception transcends its temporal boundary and proves to be fatal when these children enter their domestic sphere. Moreover, if there are both boys and girls in a family, in most of the families girls are paid less attention than boys. Biasness is there even regarding women education. Many girls are forcefully married by their family members just after passing Secondary, Higher Secondary, or hardly Graduation. Married women seldom get chance to continue higher education or to better their career. Boys are also not free from stereotypes. They are also often pressurized to choose subjects or career and life partner according to the choice of their guardians. Dress code is also imposed upon boys and girls forcefully. These discriminations separate them from their selves being otherized by familial norms and values. The persons who want to change their sex often lack the support of the families. They are regarded as mentally imbalanced. Androgynous people are also often criticized and mocked due to their gestures. And in the case of the transgender people, they often face humiliation from their kith and kin. If a person lacks familial support or faces such kind of biasness his or her education and career are hampered.

After family, the second stage of socialization starts from society. As human beings are social creature and no person can live alone in this world, everybody is to abide by social norms and beliefs. But it creates problem when it makes a rift between various people in the name of caste creed religion or gender. Women are still considered as inferior to men. There are several customs for different status of women like virgin, married, widow etc. They are assigned to showcase their marital status through symbols. In the contrary men are free from these rituals. This double standard is maintained throughout every sphere of society. Education is also such a

concern. Gender plays a very vital role in education. For a long time, education was completely phallogocentric apart from some exceptions. Later, women education became proliferated. Still there are so many countries; even there are many states in India where women education is still a luxury. Apart from women, transgender people are still marginalized in the field of education. Seldom are they given the chance of admission in educational institutes. Even if they are admitted, their position remains in the periphery of the group. Often they are taunted, humiliated, and even persecuted. Gender inequality is deeply embedded in our mind. Even in 21st century we are raising slogans '**Beti Bachoo, Beti Padhao**' and women education is still promoted highlighting the logic that if women are educated, they can properly bring up a child. Still women are 'reproductive vessels' when it is said that girl child must be protected to build up a new generation. These stereotypical roles are narrowing the path of education.

After a certain age child are admitted into schools. From then their most wanted socialization starts. School is the second home for a student. So, the educational institutions must focus on their deviations from the code of equality if any. Discrimination inside the campus may be caused due to many reasons. Behavior of the teaching and non-teaching staffs towards different genders, separate rules for separate genders may cause grievances among the marginalized genders. Such segregated systems are also not good for the mental health of the learners. These kinds of biased activities will lead them to a discriminatory society where gender harmony will always be threatened. Many institutions imposed over restrictions on female students regarding their gestures, dress code etc. in many institutions transgender candidates are refused to give admission and even if there are some candidates belonging to these categories most of the time face humiliation and separate reception.

Most of the guardians and teachers still like homosocial group and are against heterosocial group. Thus, free mixing between boys and girls cannot be possible. So, they may fail to understand each other's emotion. Instead of friendship there may grow rivalry between them. In lieu of co-operation there takes place competition between them. And in the case of transgender people both boys and girls are most often biased. Our social prejudices are responsible for this rift.

Teachers are supposed to inculcate values among the learners along with their family members. But sometimes they deviate from their designation due to their behavior when they

make any sexist comment. They sometimes fail to mind their language and to mend their attitude. If a learner hears such comments, it poses a negative impact on their character.

Curriculum is another room of concern in this context. Education system encircles curriculum. But sometimes curricula are found to be defective. Sometimes gender bias is observed regarding subjects. Many schools still offer home science for girls. In many girls' schools there is no provision for science or commerce stream. Regarding optional papers gender-based segregation is also noticed; like, mechanics, pisciculture, agriculture, wooden works, computer application etc. are there in boys' school and stitching, woolen works, horticulture, singing, dancing, cooking etc. are there in girls' schools. And if it is a co-educational school the 'womanly' subjects are excluded. Girls may perform manly task in certain cases but if a boy wants to perform the 'womanly' task he is taunted. More traumatic situation is there for transgender students because they cannot express their wish properly in schools. Neither they are degendered, nor are they included in the main stream of gender identity.

Formal education cannot be imagined without text book. From text books the learners learn the maximum things. So, if a text book projects biased notions, the learners can never come out of their stereotypical identity. UNESCO has pointed out such faults in three aspects—content, visuals, and language. The stories or the example presented in the textbooks primarily project gender binary. From the books the learners may assume that only boys become soldier, police, farmer, driver, pilot etc. and the girls can be either housewife or they may be good cook, tailor, teacher, nurse, receptionist, air hostess etc. In the nursery rhymes girls are presented as perfect mothers running household chores smoothly. They cook food and the boys go to catch fish. There are so many examples in children books where it is obvious that the women must be kept inside limiting their scope within the domestic world. Even in fairy tales men get privileged in the name of chivalry. Princes will rescue princesses. Very few stories have projected women as saviour like Kiran Mala or Parul. Step mothers are most of the time portrayed as evils and sometimes as demons. All such contents, visuals and examples permanently leave biased impression on the children's mind. Moreover, the transgender students can hardly find their stories in such books. Language is another tool of segregation here. In English language if no gender is mentioned specifically in the case of third person, the sentence uses 'he.' Thus, girls

are automatically eliminated from the scenario. And the other genders can find no traces of themselves. In this way they are forced to regard themselves as neuter gender.

Over an extended period, education or job-related forms contain two options for gender-male and female. Now in many forms a separate option is given i.e., 'other' or transgender. But there are so many forms till now which contain no such options. Then these people face difficulty in marking an option because those options cannot project their accurate identity. Moreover, in many forms only women are to mention their husbands' names but very few forms ask men to mention the names of their wives. Now-a-days in many forms there is provision for imputing the name of mothers. But this is not mandatory for every form. Many forms till now are satisfied with father's name as the guardian's name.

Gender prejudices are found in co-curricular and extra-curricular activities also. In many schools separate fields are assigned for separate genders, like singing, dancing etc. for girls and flute playing or table for boys. In work education girls' schools offer stitching or woollen work and boys' schools offer soap making, wooden work, electrical work, model making etc. the boys who dance well are often regarded as womanly in nature.

The same abovementioned stereotype is found in the time of selection of vocational courses. Most of the women are not encouraged for it and even if they are boosted, they are asked to choose tailoring or food processing whereas the boys are championed to opt mechanical work, hardware, polytechnic etc.

Steps to be taken to eradicate gender discrimination

Where there are problems, there must be solutions too. However, some measures can be traced to eradicate gender discrimination.

Mass awareness is desperately needed. Family, society, educational and working places all must be aware of gender harmony and the evils of gender discrimination. Family is the first unit in the life of an individual. So, change should first begin at home. Parents should raise their children surpassing their gender identities and role playing according to that. To ensure this, mass awareness must be formed in many ways. Media should play an active role in breaking these stereotypes. Bollywood movie *Ki & Kaissuch* is an example where the gender role playing is completely dismantled projecting the husband managing household and wife managing corporate world. Daily Soap 'Diya Aur Baati Hum' also has broken the notion of gender-based

professions portraying the husband perfect in cooking and the wife as a successful IPS Officer with the support of her ninth-grade pass husband. Recently an advertisement of surf excel is telecasted where it is shown that a mother teaches her young son how to wash clothes. Another advertisement of Red Label tea also promotes transgender people selling tea and the bond bloomed between an aged lady and the transgender teashop owner. Some advertisements from the side of government also projected transgender people as ambassador of social welfare like using seat belts. So, mass awareness must be increased and prejudices towards transgender people should also be changed. Only then an inclusive society will be built up.

Role of a teacher is especially important in this context. After all, ‘the destiny of India is being shaped in her classrooms’ (Kothari Commission, 1964). A teacher must be free from gender bias. S/he must give importance to each gender. S/he must be incredibly careful in selecting words so that no such comment is uttered which encourages gender disparity. S/he should also teach the students to come out of biased notions. If it is a co-educational institution, the teacher should encourage to seat boys and girls together rather than segregating them. When s/he is assigned some tasks it must not follow the stereotypical gender roles.

Curriculum also can play a key role in this context. While framing curriculum the concerned authority must remember that there should not be any element of discrimination. Both boys and girls must have the same options for choosing subjects in higher secondary education. It should also be ensured that no gender-identified subjects will be offered. If there are subjects like home science, stitching, craft, music etc., these should not be imposed on the girls alone. Both sexes must be encouraged to opt the subjects according to their wish, not being guided by the preconceived notion about ‘manly’ task and ‘womanly’ task.

The writers of the text books also have responsibilities in this field. They must keep some key facts in their mind. Language should be used carefully so that there is no scope for gender discrimination e.g., the boys should not express any feminine emotion. Such kinds of sentences increase the gap between male and female. The examples should be matched with all genders e.g., there should not be maximum examples from the ‘activities’ of a boy. Such examples should be included so that girls and other genders can also relate those things with their daily experiences. Even the writer must be alert when s/he is using numerical data, e.g., ‘Suppose ten mangoes are distributed among seven boys and three girls.’ (Chakrabarty, P.K., 2016). Such

examples should not be mentioned where the girls feel inferiority complex. The examples that can discriminate the genders should not be included, like, the association of the boys with cricket and the association of the boys with cricket and the association of girls with dolls. The content and examples must maintain gender neutrality and gender inclusivity.

In forms there must be provision for transgender people. Women are not imposed to mark husbands as their guardians. If information is necessary all the genders should be treated equally.

Parents should not be indifferent to their girl children. They should raise their children with equal care and prepare them for future blurring the concept of gender roles so that they may in future help their partners. If there is any transgender child in family, parents must take exceptional care for the child. Parents should not discourage the activities of androgynous children. If they want to change their sexes, parents should support them.

Co-education can be another tool to promote gender harmony. Due to its heterogeneous setting heterosocial bond among different genders can grow easily. This healthy bonding can reduce the gender related violence like sexual violence, marital abuse, women assault etc. as in an appropriate co-educational ambience the boys, girls and transgender students can understand each other's nature, emotion, thought etc. and this is good for their future life where they are supposed to accept the companionship with their opposite sexes.

Recommendations of various Commissions/ Committees

In the post-independence period government took positive initiative to proliferate women education through various commissions, committees, policies etc. like Mudaliar Commission, NPE 1986, Commission on Empowerment of women 1977, Durgabai Deshmukh Committee (1958-59), Bhaktabatsalam Committee, Kothari Commission, CAGE 2005 and so on.

Radhakrishnan Commission (1948) recommended that both the male and female teachers must be paid same salary and to be provided with same kind of facilities.

Mudaliar Commission (1952-53) proposed co-education in every phase of education. **National Policy on Education (1986)** emphasized that girls should also opt vocational or technical courses and initiatives must be taken for this purpose. But all these abovementioned commissions could not totally break the stereotypes and so they to a certain extent did not go against some separate subjects for women. But from 2000 onwards India witnessed a paradigm shift in gender equality. The most important report was of **CAGE committee on Girls'**

Education and Common School System to Reduce Gender Disparity, 2005. They championed co-education. Along with that they first significantly pointed out the sources of gender discrimination. According to the committee, the notion of women education must be changed. The aim of women education must not be limited within the logic of successful upbringing of the children by a mother, or to reduce the infant mortality rate etc. Such logic deepens the root of gender discrimination rather than uprooting it. They also proposed that girls should get an equal education like a boy. They must be taught pure mathematics and sciences rather than restricting their education within elementary math, home science etc. Their revolutionary recommendation was if a girl is taught about maternity, a boy should also be taught paternity. In section 7(ii) of **DAKAR Framework for Action (2000)** it has been stated that “Eliminating gender disparity in primary and secondary education by 2005, and achieving gender equality in education by 2015, with a focus on ensuring girls’ full and equal access to and achievement in basic education of good quality”(Chakrabarty, 2016). **The Report of UNICEF 2009** highlighted the fact, “Even when they do manage to enter school, girls’ self-esteem and self-confidence are often restricted by gender stereotyped curricula, teachers’ low expectation from girl pupils and school environment characterized by aggression, ridicule, harassment or corporal punishment”(Chakrabarty, 2016).

Legal steps have been taken for the transgender people also. Now, homosexuals have achieved legal right. In 1994 transgender and transgendered people first got the right to vote. **The Transgender Persons (Protection of Rights) Bill, 2016** has been passed for their development and protection ensuring their constitutional rights.

Inspiring Examples

There are so many persons who face identity crisis regarding their gender. Even if a woman dares to perform any ‘manly’ task, she is often criticized. Still from the ancient period there had been such men and women who challenged this gender binary. Today we also find many examples, who are breaking gender stereotypes. However, we may cite a few persons in this regard. **Shikhandini** in *Mahabharata* changed her sex and went to the battlefield of Kurukshetra being Shikhandi. **Razia Sultan** overcoming the opposition of patriarchal society ruled over India. There are so many examples in history like **Queen Elizabeth, Rani Durgawati, Preetilata Waddedar** etc. even in our recent past we have witnessed **MaryComb**’s

struggle to become a boxer challenging the socio-familial hindrances. **Manabi Bandopadhyay** is a leading instance for transgender people. She is the first transgender to have achieved the post of principal. Born as Somnath Banerjee, she had to face a lot of ridicule from not only society, but also her father who was extremely conservative. *Hindustan Times* has recorded her speech: “For me, it’s a long battle against ignorance. There was a time when I and even my father was threatened with (dire) consequences as I am a transgender. I spent my childhood in Nadia and it’s a comeback to my home with pride and dignity after a long battle (May 28, 2015).” **Laxmi Narayan Tripathi** is the first third gender person to represent Asia Pacific in the UN 2008. At the assembly she spoke about the plight of sexual minorities.

Conclusion

Gender sensitivity is required in every sphere of society. Education enlightens us. Education helps us come out of darkness of prejudices. Education is highly affected due to gender related problems. At the same time gender bias can be eradicated through the proliferation of proper education. So, both maintain reciprocal relationship. Inclusivity is also needed to understand the current socio-economic scenario. Each gender must have fellow feeling, love, care, empathy, and co-operation for other genders. Only then a successful society, a fruitful prosperous nation will emerge to raise its head high.

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**A Study on Prosociality and Attitude towards Teachers of Higher Secondary School
Students Studying in Schools Managed by Different Religious Communities**

Mahejabin* & Md Akhtar Raza**

Abstract

The main purpose of present study was to know the prosociality and attitude towards teachers of higher secondary school students studying in schools managed by different religious communities. The two tools used in the present study first, prosocial behavior scale which was constructed and standardized by the researcher and second tool Student's Attitude towards Teacher Scale' tool was used in the present study developed by Pandey and Shukla (2010). In the present study, Higher Secondary School Students Studying in Schools Managed by Different Religious Communities were included. After the data analysis the researcher found that there is significant correlation between Prosociality and Attitude towards Teacher of higher secondary school students, studying in school managed by different religious communities. It was found that Hindu-Muslim, Hindu-Christian and Muslim-Christian managed schools students have not significant difference in Prosociality, but Hindu-Sikh, Muslim-Sikh and Sikh-Christian managed school students have significant difference in Prosociality. It was also found that Hindu-Sikh, Hindu-Christian and Muslim-Sikh managed schools students have not significant difference in Attitude towards Teachers but Hindu-Muslim, Muslim-Christian and Sikh-Christian managed schools students have significant difference in Attitude towards Teachers.

Key Words: *Prosociality, Attitude towards Teacher & Different Religious Communities*

Introduction

In the present time, whenever we simply look around and open a newspaper or listen to the news on the radio or television, we find human violence and cruelty everywhere. The values of

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helping, caring, and sharing are decreasing from our society so fast. Prosociality becomes hope to bring happiness in our lives.

Prosociality term refers for prosocial behavior. This is a new concept of Social Psychology. McDougal introduced this terminology in 1908 to tender emotions generated by paternal instinct and first Wisp (1972) used this concept in scientific context as an antonym of antisocial behavior (Cited in Penner et. al , 2004). For a long time, prosocial development remained a neglected area in psychological research. In fact, until 1970s, most of the related research remained focused on moral development only because little is known about the effect of school experiences on children's prosocial behavior.

Prosocial behavior is a positive pattern or action of behavior. It is voluntary done by an individual for others welfare but in which those helping behavior is not include, which is done by due to professional or institutional obligation. It is a concern about the right feeling for others. It is prompted by empathy, moral value, and a sense of personal responsibility. It encourages other benefit rather than personal gain. Prosocial behaviors include a broad range of activities, such as helping, sharing, donating, cooperating, and volunteering. Social Psychologist defined it in different way, some definitions are - Eisenberg and Mussen (1989) defined, "Prosocial behavior as a voluntary action that is intended to help or benefit another Individual or group of individuals.". Brown & Cook (1986) considered, "Prosocial behavior as helping, sharing, or cooperating others. Thus, prosocial behavior includes a wide variety of acts performed by one or more individuals and may also benefit those performing the acts." (Cited as Shing, 2008)

According to Baron & Byrne (1987), "Prosocial behavior refers to acts that have no obvious benefits for the individual engaging in them and even involve risk and some degree of sacrifice. Such acts...are based on ethical standards of conduct."

Thus, these definitions refer to the term Prosocial behavior as positive action done by an individual for others benefits. It is prompted by empathy, moral value, and a sense of personal responsibility rather than a desire for personal gain. These behaviors include a broad range of activities, such as: helping, sharing, donating, cooperating, and volunteering.

Rationale of the Study

Prosocial development, for a long time, remained a neglected area in psychological research. In fact, until the 1970s, most of the related research remained focused on moral

development only because little is known about the effect of school experiences on children's prosocial behavior. After 1970s, so much researches prove it as important factor in Education. Finding suggests that structuring classes to provide children with opportunities to help others may promote prosocial behavior. Bizman, Yinon, Mivtzari and Shavit (1978) found that Israeli kinder gardeners enrolled in classes that contained younger peers were more altruistic than those enrolled in classes that were homogeneous in age. Hertz-Lazarowitz et al.,(1989) observed that Israeli students in active classrooms, in which cooperation and individualized learning were emphasized, helped peers more than students in traditional classrooms.

Clarke-Stewart(1981) suggested that attendance at Community Day Care has a temporarily accelerating effect on social development and found that prosocial behavior was higher for children with non-parental care. Schenk and Grusec (1987) found that home care children were more likely than day-care children to behave prosocially in situations involving an adult stranger, whereas the two group were similar on helping unknown children.

Yarrow et al.,(1973) find out that warm, supportive interaction with teachers was associated with preschool children's modeling of teachers' prosocial actions. Howes, Matheson, and Hamilton (1994) found that children classified as securely attached to their current and first preschool teacher were rated as more considerate and empathic with unfamiliar peers than were children classified as having an insecure relationship with their teachers.

Solomon and Colleagues (Solomon, Battistich, Watson, Delucchi, Schaps, & Battistich, 1988) developed a program in which teacher were trained to maintain positive personal relationships with their students by using a child-centered approach to classroom management that emphasized inductive discipline and student participation in rule setting. They found significant progress in implementing the program, student showed positive gain in personal, social and ethical value, attitudes and motives, and a reduction of substance abuse and other problem behavior.

Hertz-Lazarowitz, & Shran (1984); Hertz-Lazarowitz, Sharan, & Steinberg (1980) found that the use of cooperative educational techniques in classroom activities has promoted acceptance of others, as well as cooperation and prosocial behavior. Feshbach,(1978), Wise & Cramer,(1988), Bandura et al.(2001, 2003) finds out positive relation between scores on

achievement tests and children's empathy or sympathy and between academic self-efficacy and prosocial behavior.

In India very few studies have been done on the prosocial behavior, 'Prosocial Development in Children-Helping, Sharing and Comforting' was studied by Neerja Chadha (2007), similarly 'Recipient need and anticipation of reciprocity in prosocial exchange' was exposed by L. Krishna(1988), and 'Assessment of prosocial behavior among primary school children' was researched by G. Puspa, G. Vedagiri, & N.D. Ramaswamy(1994).

After reviewing the related literature researcher has perceived that each religion talks about and practice prosociality, hence researcher feels that prosocial behavior plays very important role in education to improve academic achievement, positive attitude towards others, and social harmony towards different religious communities and culture. However, in India researcher found that the existing knowledge gap in this area. Hence, it is the demand of time to conduct this study.

Objectives of the Study:

1. To study the correlation between Prosociality and Attitude towards Teachers of higher secondary school students.
2. To study the Prosociality of Higher Secondary School Students Studying in Schools Managed by Different Religious Communities.
3. To study the Attitude towards Teachers of Higher Secondary School Students Studying in Schools Managed by Different Religious Communities

Null Hypothesis

H₀₁ There is no significant correlation between Prosociality and Attitude towards Teachers of higher secondary school students

H₀₂ There is no significant difference among Prosociality of higher secondary school students studying in schools managed by different religious communities.

H₀₃ There is no significant difference between Attitude towards Teachers of higher secondary school students studying in school managed by different religious communities.

Methodology

The higher secondary schools managed by different religious communities of Varanasi city constituted the population of the study. In the present study, 'stratified random sampling

techniques' was used to draw the sample from the population. The 1019 students were taken from higher secondary schools managed by different religious communities.

Tool used:

- 1. Prosocial behavior Scale:** For assessing the prosociality of higher secondary school students the prosocial behavior scale (PBS) was used. It is constructed and standardized by the researcher, which consists of 36 items.
- 2. Student Attitude towards their Teacher Scale:** This tool was used to assess the attitude of the students towards their teacher. This scale was constructed and standardized by P.C. Shukla and Geeta Pandey (2010). It consists of 90 multiple-choice in which 45 is positive and 45 is negative.

Statistical Treatments:

Collected data were organized and analyzed by using descriptive statistic, parametric tests F-test and t-test and Product moment correlation to analyzed data with the help of SPSS 21.0 version.

Data Analysis and Interpretation:

Inferential Analysis:

Objective- 1: To study the correlation between Prosociality and Attitude towards Teachers of higher secondary school students.

Null Hypothesis:

H₀₁ There is no significant correlation between Prosociality and Attitude towards Teachers of higher secondary school students.

Table 1: Correlation between Prosocial Behavior and Student's Attitude towards Teacher

Variables	N	df	r	Result
Prosocial Behavior and Student's Attitude towards Teacher	1019	1017	.678**	S

** . Correlation is significant at the 0.05 level (2-tailed).

As can be seen from table no. 01 the obtained r-value was found to be .678, which is greater than the tabulated value .062 at .05 level of significance with df 1017 Hence, r-value is significant and null hypothesis is rejected. Thus, it can be concluded that there is significant correlation between Prosocial Behavior and Attitude towards teacher of higher secondary school students, studying in school managed by different religious communities.

Therefore, it may be interpreted that there is significant and moderate correlation found between the Prosocial Behavior and Attitude towards teacher of higher secondary school students, studying in school managed by different religious communities.

Findings of the present study supported previous studies conducted by Solomon, Watson, Delucchi, Schaps, & Battistich, (1988), Battistich, Watson, Solomon, Schaps, & Solomon, (1991), Solomon, Battistich, Watson, Schaps, & Lewis, (2000).

Objective: 2- To study the significance difference among Prosociality of Higher Secondary School Students, Studying in School Managed by Different Religious Communities.

Null Hypothesis:

H₀₂ There is no significant difference among Prosociality of higher secondary school students studying in schools managed by different religious communities.

Table 2: F-ratio of Prosociality of Higher Secondary School Students

Variable	Sum of Square		df	Mean Square	F	P
Hindu Muslim Sikh Christian	Between Groups	1488.801	3	496.267	3.084	≥ 0.05
	Within Groups	163319.927	1015	160.906		
	Total	164808.728	1018			

As can be seen from table no. 02 that the obtained 'F' value was found to 3.084 which is greater than the tabulated value 2.613 at .05 level of significance with between group df 3 and within group df 1015. Hence, F-value is significant and null hypothesis is rejected. Thus, it can be concluded that there is significant difference among Prosociality of higher secondary school students, studying in school managed by different religious communities.

After getting significant difference in F-value it is necessary to conduct post hoc analysis to identify the groups which vary significantly.

Table 3: t-ratio of Prosociality of Higher Secondary School Students, Managed by Different Religious Communities

Variable	N	Mean	SD	df	t-ratio	p
Hindu	337	131.18	11.738	586	.506	NS
Muslim	251	131.72	13.385			
Hindu	337	131.18	11.738	485	3.022	≤.05
Sikh	150	134.83	12.530			
Hindu	337	131.18	11.738	616	.411	NS
Christian	281	131.60	13.208			
Muslim	251	131.72	13.385	330.038	2.344	≤.05
Sikh	150	134.83	12.530			
Muslim	251	131.72	13.385	521.661	.103	NS
Christian	281	131.60	13.208			
Sikh	150	134.83	12.530	318.548	2.500	≤.05
Christian	281	131.60	13.208			

Table 3 reveals between Hindu and Muslim community at the post hoc analysis stage t-ratio was found to .506, which is less than the t-tabulated value 1.96 at .05 level of significance with df 586. Hence, t-value is not significant and null hypothesis is accepted. Thus, it can be concluded that there is no significant difference between Prosociality of higher secondary school students, studying in school managed by Hindu and Muslim community.

Table 3 reveals between Hindu and Sikh community at the post analysis stage t-ratio was found to 3.022, which is greater than the t-tabulated value 1.96 at .05 level of significant with df 485. Hence, t-value is significant and null hypothesis is rejected. Thus, it is concluded that there is significant difference between Prosociality of higher secondary school students, studying in school managed by Hindu and Sikh community.

Table no. 03 reveals between Hindu and Christian community at the post analysis stage t-ratio was found to .411, which is less than the t-tabulated value 1.96 at .05 level of significance with df 616. Hence, t-value is not significant and null hypothesis is accepted. Thus, it is concluded that there is not significant difference between Prosociality of higher secondary school students, studying in school managed by Hindu and Christian community.

Table no. 03 reveals between Muslim and Sikh community at the post analysis stage t-value was found to 2.344, which is greater than the t-tabulated value 1.96402 at .05 level of significance with df 399. Hence, t-value is significant and null hypothesis is rejected. Thus, it is concluded that there is significant difference between Prosociality of higher secondary school students, studying in school managed by Muslim and Sikh community.

Table no. 03 reveals Muslim and Christian community at the post analysis stage t-value was found to .103, which is less than the t-tabulated value 1.96 at .05 level of significance with df 530. Hence, t-value is not significant and null hypothesis is accepted. Thus, it is concluded that there is no significant difference between Prosociality of higher secondary school students, studying in school managed by Muslim and Christian community.

Table no. 03 also reveals Sikh and Christian community at the post analysis stage t-value was found to 2.500, which is greater than the t-tabulated value 1.96 at .05 level of significance with df 429. Hence, t-value is significant and null hypothesis is rejected. Thus, it is concluded that there is significant difference between Prosociality of higher secondary school students, studying in school managed by Sikh and Christian community.

Therefore, it may be interpreted that Hindu-Muslim, Hindu-Christian and Muslim-Christian managed schools students have not significant difference in Prosociality, but Hindu-Sikh, Muslim-Sikh and Sikh-Christian managed school students have significant difference in Prosociality.

Objective: 3- To find out and compare the Attitude towards Teacher of higher secondary school students, studying in schools managed by different religious communities.

Null Hypothesis:

H₀₃ There is no significant difference between Attitude towards Teachers of higher secondary school students studying in school managed by different religious communities.

Table-04 F-ratio of Attitude towards Teacher of Higher Secondary School Students, Studying in Schools Managed by Different Religious Communities

Variable	Sum of Square		df	Mean Square	F	P
Hindu Muslim Sikh Christian	Between Groups	48208.898	3	16069.633	5.978	≥ 0.05
	Within Groups	2728291.891	1015	2687.972		
	Total	164808.728	1018			

As can be seen from table no. 04 the obtained F-ratio was found to 5.978, which is greater than the tabulated value 2.613 at .05 level of significance with between group df 3 and within group df 1015. Hence, F-value is significant and null hypothesis is rejected. Thus, it can be concluded that there is significant difference in Attitude towards teacher of higher secondary school students, studying in school managed by different religious communities.

After getting significant difference in F-ratio it is necessary to conduct post hoc analysis to identify the groups which vary significantly.

Table-05 t-ratio of Attitude towards Teacher of Higher Secondary School Students, Managed by Different Religious Communities

Variable	N	Mean (M)	SD	df	t-ratio	p
Hindu	337	323.31	53.233	586	3.378	≤.05
Muslim	251	337.80	50.036			
Hindu	337	323.31	53.233	485	1.528	NS
Sikh	150	330.95	49.829			

Hindu	337	323.31	53.233	616	.670	NS
Christian	281	320.44	52.790			
Muslim	251	337.80	50.036	399	1.330	NS
Sikh	150	330.95	49.829			
Muslim	251	337.80	50.036	530	3.891	≤.05
Christian	281	320.44	52.790			
Sikh	150	330.95	49.829	429	2.041	≤.05
Christian	281	320.44	52.790			

Table no. 05 reveals Hindu and Muslim community at the post analysis stage t-ratio was found to 3.378, which is greater than the t-tabulated value 1.96 at .05 level of significance with df 586. Hence, t-value is significant and null hypothesis is rejected. Thus, it can be concluded that there is significant difference in Attitude towards Teacher of higher secondary school students, studying in school managed by Hindu and Muslim community.

Table no. 05 reveals Hindu and Sikh community at the post analysis stage t-ratio was found to 1.528, which is less than the t-tabulated value 1.96 at .05 level of significant with df 485. Hence, t-value is not significant and null hypothesis is accepted. Thus, it is concluded that there is not significant difference in Attitude towards Teacher of higher secondary school students, studying in school managed by Hindu and Sikh community.

Table no. 05 reveals Hindu and Christian community at the post analysis stage t-ratio was found to .670, which is less than the t-tabulated value 1.96 at .05 level of significant with df 616. Hence, t-value is not significant and null hypothesis is accepted. Thus, it is concluded that there is not significant difference in Attitude towards Teacher of higher secondary school students, studying in school managed by Hindu and Christian community.

Table no. 05 reveals Muslim and Sikh community at the post analysis stage t-ratio was found to 1.330, which is less than the t-tabulated value 1.96 at .05 level of significance with df 399. Hence, t-value is not significant and null hypothesis is accepted. Thus, it is concluded that there is not significant difference in Attitude towards Teacher of higher secondary school students, studying in school managed by Muslim and Sikh community.

Table no. 05 reveals Muslim and Christian community at the post analysis stage t-ratio was found to 3.891, which is greater than the t-tabulated value 1.96 at .05 level of significance with df 530. Hence, t-value is significant and null hypothesis is rejected. Thus, it is concluded that there is significant difference in Attitude towards Teacher of higher secondary school students, studying in school managed by Muslim and Christian community.

Table no. 05 also reveals Sikh and Christian community at the post analysis stage t-ratio was found to 2.041, which is greater than the t-tabulated value 1.96 at .05 level of significance with df 429. Hence, t-value is significant and null hypothesis is rejected. Thus, it is concluded that there is significant difference in Attitude towards Teacher of higher secondary school students, studying in school managed by Sikh and Christian community.

Therefore, it may be interpreted that Hindu-Sikh, Hindu-Christian and Muslim-Sikh managed schools students have not significant difference in Attitude towards Teachers but Hindu-Muslim, Muslim-Christian and Sikh-Christian managed schools students have significant difference in Attitude towards Teachers.

Major Finding:

1. There is significant correlation between Prosociality and Attitude towards Teacher of higher secondary school students, studying in school managed by different religious communities.
2. There is significant difference in the level of Prosociality of higher secondary school students studying in schools managed by different religious communities
3. After post hoc analysis, it was found that Hindu-Muslim, Hindu-Christian and Muslim-Christian managed schools students have not significant difference in Prosociality, but Hindu-Sikh, Muslim-Sikh and Sikh-Christian managed school students have significant difference in Prosociality.
4. There is significant difference in Attitude towards Teacher of higher secondary school students studying in schools managed by different religious communities.
5. After post hoc analysis, it was found that Hindu-Sikh, Hindu-Christian and Muslim-Sikh managed schools students have not significant difference in Attitude towards Teachers but Hindu-Muslim, Muslim-Christian and Sikh-Christian managed schools students have significant difference in Attitude towards Teachers.

Conclusion

Based on above findings of the study following conclusion have been drawn by the researcher: Finding of this study indicates positive correlation between Prosocial Behaviour and Attitude towards Teacher of higher secondary school students, studying in school managed by different religious communities.

Schools Management effects on students' level of Prosociality and different religious managed schools possess high level of prosociality. There is no significant difference between Prosociality of Hindu-Muslim, Hindu-Christian and Muslim-Christian managed school students but there is significant difference between Prosociality of Hindu-Sikh, Muslim-Sikh, and Sikh-Christian managed schools' students.

Schools Management effects on students Attitude towards Teacher. There is no significant difference between Attitude towards Teachers of Hindu-Muslim, Muslim-Christian and Sikh-Christian managed schools' students but there is significant difference between Attitude towards Teacher of Hindu-Muslim, Muslim-Christian and Sikh-Christian managed schools' students.

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Influence of Teachers on Learning of Hindi in Mizoram

Swati Dwivedi* & Lalhmasai Chuaungo**

Abstract

One of the characteristics of a successful language teacher is their level of proficiency in the language they are teaching. Proficiency in a language is an essential requirement for effective language teaching. It is also obvious that teachers have a crucial role as the primary model, source, and facilitator of learning and skill-acquisition in the target language. Multilingual exposure not only improves language competencies but also it is catalytic to higher achievements across the curriculum. Hindi is taught as a third language in the state of Mizoram, India. It is the general scenario across the country that the teaching and learning of third language is implemented lackadaisically. Lack of enthusiasm among the Hindi teachers and students has been found during a wide-ranging research work on status of Hindi language in the schools of Mizoram. With reference to the teachers' proficiency in the Hindi language, the study was following a participatory interactive-observation design. This qualitative study reveals that wherever teachers lack Hindi language skills, students also performed below to the expected learning indicators.

Key Words: Hindi Language, Language Proficiency, Language Learning and Teachers' Influence.

Introduction

The power of language and its impact of learning in general has been recognized across the globe. In the last century itself the famous sociolinguist M.A.K. Halliday (1993) proposed a language-based theory of learning which emphasized that learning a language means learning the foundation of learning. There is relation in learning process in generals and learning of language. Recognizing such ideas NCTE in 2015 had also proposed a core course in the curriculum of teacher education (B.Ed.) with the title "language across the curriculum (LAC)." LAC is not only a course of study for prospective teachers in India but also it had been an

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educational movement in European continent during last four decades of 20th century(Gray et al., 2018). Similar, parallel and related in its aim another movement had also been flourished in American continent with the name writing across the curriculum (WAC)(Bazerman et al., 2005). LAC has also been recognized as the approach to second language teaching under wider landscape of content and language integrated learning (CLIL)(Grenfell, 2004). Which further relate with communicative language teaching (CLT) in the theories of second language teaching (pedagogical content knowledge) and constructivist approaches to teaching and learning in general pedagogical theories(Richards & Rodgers, 2016). All such evidences prove the centrality of language in learning and proposes that if a school aims to improve the learning of students across the curriculum, improvement in the language learning is prerequisite. Presently it is a known fact that more languages, more knowledge(NCERT, 2005). Multilingualism is a resource in the classroom. Despite change in political ideology the present government of India gives place to historical three language formula (TLF) as the key feature of National Education Policy 2020(Govt. of India, 2020). NEP 2020 not only gives place to TLF, but it puts more emphasis on flexible inclusion of languages, importance to local languages and learning of more than three languages during the school years. “Promoting multilingualism and the power of language in teaching and learning” is one of the twenty-two points guiding principles of NEP 2020. The power of language in teaching and learning can be harnessed only in the case if teachers recognize and exercise that power, particularly the language teachers. Role of language teachers and language proficiency of the teachers of other subjects are the center point of discussion in the rejuvenation of the educational system.

According to Elder (1994) there are ‘four aspects of language and language-related ability’ that determine a language teacher’s ability to teach a language effectively:

1. The ability to use the target language as both the medium and target of instruction
2. The ability to modify target language input to render it comprehensible to learners
3. The ability to produce well-formed input for learners
4. The ability to draw learners’ attention to features of the formal language.

Language proficiency is the basis of teachers' ability (Mitchell, 1988). "A teacher affects eternity" is the most used and misused quote of Henry Adams (1907). The quote represents the whole philosophy of teacher's impact on students' learning. This is true in both ways in educating and miseducating, and this can clearly be observed particularly in the case of second language learning. Horwitz (1996) points out that non-native teachers experience feelings of uneasiness and inadequacy in the target language, and that this anxiety has negative effects on language teaching. Wrong concepts taught by the teachers may be corrected by the students themselves during the time of higher learning, but wrong pronunciation learned at the school time may occur in natural talk of the expert of the same language, even though being the expert or teacher of that language person may be knowing the correct pronunciation which occurs during conscious use only. Coming out of the philosophical purview of the impact of the teachers on students' learning, behaviorism considers the teacher herself a great stimulus for students' learning (Joshi, 1976). Therefore, teachers' gestures, postures, movement, and voice modulation are analyzed as affecting factors. The classical conditioning proposes that the simple smiles of the teacher may make the classroom environment suitable for learning (Eggen & Kauchak, 2000). If some of us discard behaviorism (B), those may believe in constructivism (C), even without taking B Vs. C in contrast. Moving to the other side of the aspect, social constructivist theory of learning which emphasizes the role of modelling, feedback, and children's imitation of adult speech (Bandura, 1986). Modelling plays a great role in learning and is greater in the special case of language acquisition (first as well as second). First language acquisition largely depends on modelling. In the case of second language teaching and learning, hiring a native speaker is the oldest and the newest trend across the globe. In European countries a native speaker can be a teacher of language with less qualification who may assist a subject teacher to teach in that language (Grenfell, 2002). In the history of the pedagogy of language teaching, modelling has been the part of structural approach as well as communicative approaches (Richards & Rodgers, 2016). The researchers concluded that teachers are one of the most important factors in L2 learners' motivation (Dörnyei, 1994). Since modelling is one of the major factors in the teaching and learning of second language, the authors of the paper tried to analyze the role of teachers in learning of Hindi in Mizoram.

Methodology: The research paper is methodologically rooted into interpretive paradigm of research. The result and findings are based on a participatory observation and interaction with the stakeholders. Observations, field diary and informal discussion/conversations(interviews) were the major tools of the research. The findings are triangulated with the main research work by the authors on Hindi as a Subject at Elementary Level of Education in Mizoram: An Analytical Study which was grounded on mixed method approach. Hence this paper is reporting the context of major work on status of Hindi language in Mizoram.

Result and Discussion

The question, ‘what is a language teacher’? became more important if it is put up with reference to the teacher of second or third language (Gardner, 2020). A teacher can make a big difference in a student’s learning. The degree of a student's success or failure varies greatly among individuals and influenced by the teachers. The role of a language teacher goes beyond only imparting knowledge; it also includes helping pupils understand the task at hand, stay motivated, and develop discipline. Further the teachers roles and characteristics are discussed with regards to the Hindi teachers in Mizoram.

Teachers’ Self-efficacy and Students’ Interest in Hindi

Self-efficacy, as a socio-affective concept, was introduced by Bandura (1977). Language teachers' beliefs in their abilities to perform language teaching-related tasks that enhance learning, or their self-efficacy beliefs, are increasingly acknowledged to have a critical role in determining how their knowledge is translated into action (Wyatt & Dikilitaş, 2019). Teachers’ self-efficacy beliefs can be defined as teachers’ beliefs in their abilities to support learning in various task-, domain- and context-specific cognitive, metacognitive, affective, and social ways (Wyatt, 2008, 2016).

During the time of introduction and informal talk, some of the Hindi teachers refused to talk in Hindi. Teachers claimed that they were not comfortable in Hindi language. In those cases, teachers were totally depending on the textbook and just completing the prescribed exercises were all about Hindi language teaching. The students of that school were not equipped with the basic skills of reading, writing, and speaking in Hindi. But it was found that wherever the teacher’s self-efficacy was high, students were performing well. When the teachers were

comfortable in Hindi the students also showed enthusiasm and eagerness to learn Hindi, to talk in Hindi and to speak in Hindi.

Multilingual Exposure of the Teacher

Multilingualism is the ability to communicate effectively in more than one language (Okal, 2014). Multilingualism can also be defined as the existence of multiple languages within a society. It can be considered as valuable resource for society (Webb & Kembo-Sure 2000 cited in Okal, 2014). The ability to communicate in more than one language (multilingualism) opens a lot of options. It facilitates faster and easier communication and it was found that the practice of multilingualism fosters creativity (Fürst, & Grin, 2021) and mental flexibility.

It has been found that the teachers who got their education out of Mizoram, either in North Eastern states or other states (not necessarily in Hindi states) were more comfortable in using Hindi language during informal conversation with the researcher. Their students had also demonstrated relatively well their skills in Hindi language on achievement test as well as in informal conversation with the researcher.

Language Capacity of Teachers of other Subjects

The learning environment of the student is equally important. According to the literature, learning environment is categorized into three broad aspects; physical environment (such as materials, building, or classroom), psychological environment (such as attitude and value), and academic environment (such as learning strategy or academic performance), and these affect the overall academic success in acquiring second language (Kiatkheeree, 2018). A learning environment has significant effect on the level of language proficiency of the learners. For example, students and teachers' interaction in a classroom, as part of learning environment, is regarded as important factor in promoting students' achievement. It became more important when the other teachers of the school were using the second/third language. It created more positive environment for language learners. More teachers who spoke Hindi in a school, more was the achievement of the students. Students performed better on test of Hindi language in those middle schools where other subject teachers (Non-language teachers) were possessing good proficiency in Hindi language. This finding support the idea of language across the curriculum (Vollmer, 2006).

Multilingual Environment of the Classroom

Multilingual classrooms refer to contexts which show linguistic and cultural pluralism. Learners in multilingual classrooms are influenced by cross-cultural experiences. Mitchell (2012) depicts multilingual learners as students who are influenced by cross-cultural experience and whose daily lived reality necessitates the negotiation of two or more languages. Therefore, multilingual classrooms or contexts are characterized by the coexistence of several languages. Multilingualism is an asset to be nurtured in the classrooms. The more students were eager to read and write in those schools, where students were admitted from diverse linguistic backgrounds. It was necessary that students from Hindi speaking family are catalytic to peer learning but also students from Assamese, Bengali or Nepali background were also able to create good environment for Hindi language learning. Though this is a factor of peer influence, teachers' role is also vital in maintaining multilingual environment in the classroom. Otherwise, students from minority language could not influence the language learning of the students of majority language. They might be feeling shy of using their languages in school-premises also before their batchmates. As per the result from the pedagogical practices scale the Hindi teachers were found unaware of the benefit of multilingualism. The scenario may also be interpreted with the fact and result of the test on the notice board of the schools which revealed that the students Hindi or related language background were scoring less in Hindi language. The fact also raised a need of teacher training in culturally responsive pedagogy (Kimanen et al., 2019; Monyai, 2021; Nasri et al., 2019).

Teachers' Attitude toward Hindi Language

Teachers' attitude plays a significant role in the success or failure of teaching and learning of second or third language. Teacher's attitude towards Hindi language can motivate or demotivate the learners. Demotivation is "specific external forces that reduce or diminish the motivational basis of a behavioral intention or an ongoing action" (Dörnyei, 2001). Demotivation is considered as negative influences which cancel existing motivation. It claimed as 'teacher owned' (Dörnyei & Ushioda, 2001). It means the lack of motivation was attributed to the actions or responsibilities of the teacher. Teachers' competence and teaching styles are considered as a demotivating factor (Meshkat & Hassani 2012). Unsuccess in second or third language acquisition is believed to be strongly correlated with demotivation. There is also a third term

used in language acquisition “Amotivation”. According to Dörnyei (2001), amotivation happens when people think the outcomes are unrealistic and unreasonable. The idea that there is no sense in doing anything results in amotivation. In other words, the learner understands that the target is unachievable and beyond his or her capacity.

Teachers’ behavior in the form of hypercriticism towards the Hindi language plays a crucial role in Hindi language learning. It was found in informal talks with the Hindi teachers that many Hindi teachers claimed Hindi is very tough language for the students in front of them. The scenario was revealed through the Achievement test of reading and writing which was to be conducted in class VIII (eight) as the part to find out the status of Hindi language in Mizoram. The reading test was following a vocabulary test based on the textbook of class VII (seven). The first test of vocabulary was a group test but reading test was an individual test. Therefore, researcher conducted the reading test on selected students. Students were selected with the help of Hindi teachers of schools. It was observed that at many schools, teachers declared that they(students) could not read “none is able to read”. At some other places where teachers declared a particular student as inefficient to read, exceptionally a few of such students were able to perform on reading test at moderate fluency. Such negative attitude and self-efficacy may cause negative environment to learn a target language. This research further makes a way to test the Hindi language teachers in Mizoram empirically on some variables like attitude and self-efficacy.

Teachers’ Motivation

It is not only students’ motivation but also teachers’ motivation that affects students’ academic achievement. There is a positive association between a teacher's motivation and the motivation of their students, suggesting that teacher motivation is transferable (Dörnyei, 2001). Teacher’s motivation is very important for effective teaching- learning of language, especially when the language is not the first language. The amount of eagerness and passion possessed by the teacher influences language learning. The better working condition is the motivating factor for the teachers. It influenced the learners positively. It can be in terms of incentives that teachers receive.

In Mizoram, the exclusive concept of “substitute teacher” was found. The substitute teachers were those teachers who were teaching at the place of the recruited teachers when a deputed teacher is officially or unofficially on long-leave. A significant number of substitute teachers were teaching Hindi at schools. Teachers with less salary and temporary job status were not motivated to teach and thus adversely affect the learning of the Hindi language (Khanal et al., 2021; Malik, 2013).

The second concern was about the class size. Teachers taught large number of students were less motivated due to work load. Although there were two Hindi teachers in most of the schools, the allocation of teachers was not appropriate. It was found irregularity in allocation of Hindi teachers in most of the schools. Schools with large number of students had only one Hindi teacher (regular/substitute). While schools with lesser number of students had two Hindi teachers. It was found that work load of teachers impacts negatively on teachers’ motivation and ultimately on students’ motivation on learning Hindi language.

Conclusion

Language learning is a battle of the heart as much as of the mind, so this is the responsibility of the teacher to encourage and motivate the learners. If the learner feels that the target language is very difficult and hard, it affects adversely. Teachers contacted during this study openly accepted that Hindi language is very tough for their students, and they can’t read, write, and speak in Hindi. The unfavorable external factors involve a variety of elements, including the classroom atmosphere, teaching situations, methods, and teacher's behavior negative words or hypercriticism of the language used by teacher influenced teaching learning process and demotivate the language learners. It was also reported by some educationist in Mizoram to the researcher that government of Mizoram did not set quality parameters during recruitment of Hindi teachers. Training, recruitment, and deployment of Hindi teachers seems to be more political and economic intervention than linguistic and educational one. Triangulating the results with the unpublished research work by the authors It can be concluded that the status of Hindi language in Mizoram is not satisfactory. This is the scenario in the state where Hindi is the language in school since its statehood and the state is possessing exclusive Hindi teacher training college affiliated to Central Hindi Institute, Agra.

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Equalization of Educational Opportunities: The Need for Reassessing and Re-Evaluating Issues for Sustainable Development

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Abstract

Change is unavoidable in society, also, it raises a myriad of problems. From 'traditional' to 'modern,' from 'rural' to 'urban,' and from 'agrarian' to 'industrial,' cultures have gone through various phases of development. However, India is still classified as a developing country despite the advent of globalization, industrialization, and modernization. What could be the fundamental root causes of the country's inability to develop? One of the major setbacks is the inequality and lack of educational opportunities across societies. There have been massive shifts in every sphere, including cultural, economical, political, social, psychological, and religious issues. The chapter extensively explores the underlying concerns in educational disparities that affect the social life pattern despite such tremendous positive transformation and progress.

Keywords: *Equal educational opportunities, Sarva Shiksha Abhiyan, Kasturba Gandhi Balika Vidyalyaya, DPEP, National Education Policy, Education Commission.*

Introduction

India got its independence on the 15th of August 1947, following a series of freedom struggle against the British colonial rule. Since then, the nation has undergone colossal changes in every domain, be it cultural, economic, political, social, psychological, or religious aspects. The constitution of India holds it as a Sovereign, Socialist, Secular, Democratic and Republic. Despite such huge positive transformation and development, there remain some underlying problems that disturb the social life pattern. The Education Commission of 1964-66 (also known as the Kothari Commission), states that one of education's important social goals is to maximize opportunities, allowing underprivileged backward classes and individuals to use education as a trigger for improving their situation. Every society that values social justice and is eager to

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improve the common man 's lot and cultivate all the talent available must assist in ensuring progressive equality of educational opportunity for all segments of the community. This is the only guarantee that an equitable and human society will be built in which the exploitation of the weak is significantly reduced.

Problem Statement and Literature Review

Disequilibrium in educational opportunities

The Sargeant Report (1944) recommended that equal educational opportunity may be provided at the primary stage through free schooling along with other facilities that include free supply of books, midday meals, medical aid, and provision for scholarships. The Indian Constitution also guarantees equal educational opportunities to all people of the country.

In the Report of the Education Commission (1964–1966), it was discovered that free schools run by the state and local governments, where most people send their primary-aged children, are typically of inferior quality. Unquestionably superior private schools exist, but only the middle and upper classes can afford to attend them due to their high tuition costs. Quality education is only available to a small minority, who are personally chosen not based on aptitude but based on their ability to pay fees, as opposed to being accessible to at least the capable children from all social strata. As a result, it is exceedingly difficult to identify and develop the entire national pool of competencies; the stance is undemocratic and goes against the egalitarian ideal.

According to the UNICEF Annual Report (2018), in the primary grades, approximately 11% of children do not attend any classes. Children from the poorest homes are five times more likely than those from the richest to drop out of elementary school, so poverty continues to be a significant barrier. Gender equality in primary education has only been achieved in 49% of countries, and educational opportunities for females are notably constrained. In war zones, almost 27 million students are not in school.

According to Deep Chand and SailuKarre in the journal ‘Contemporary Voice of Dalit’ (2019), only 67.7% of scheduled caste students in rural India can attend school, compared to 81.3 % of non-SC students. This means that more than 20.8 % of scheduled caste students in rural India never got the chance to attend school. Even though SC students' educational attainment has increased over the past ten years in India, these children are not receiving satisfactory schooling (Sedwal and Kamath, 2008).

In the Report of Education Commission (1964-66), it was found that free schools maintained by the state and local administration to which the masses send their children at primary level, are often of inferior quality. Some private schools are unquestionably superior; yet because many of them charge extortionate fees, they are only accessible to the middle and upper classes. Rather than being accessible to at least the competent children from all strata of society, quality education is exclusively open to a small minority, who are personally selected not based on aptitude, but based on their ability to pay fees. Hence, the capacity to identify and develop complete national pool of competencies is severely impeded; the position is undemocratic and contradicts egalitarian ideal.

The UNICEF Annual Report (2018) revealed that approximately 11% of primary school children do not attend school at all. Poverty remains a formidable obstacle, with children from poorest homes five times more likely than those from the richest to drop out of elementary school. Education chances for girls are particularly limited and gender equality in primary education has only been realized in 49% of nations. About 27 million are out of school in warfare zones.

In the Journal Contemporary Voice of Dalit (2019), Deep Chand and SailuKarre mentioned that more than 20.8% of the scheduled caste students in rural India never got to attend schools; only 67.7% of the scheduled caste students can receive education in comparison to 81.3% of non-SC students. Although the quantum of educational increase of the SC students has been better from the last decade in India, there is no satisfactory schooling being furnished to these students (Sedwal and Kamath, 2008).

Methodology

The study was focused primarily on qualitative research methodologies owing of the nature of data analysis. The research was meticulously carried out, and both historical and current data were verified to reach conclusions that allowed for the successful production of some statistically significant findings. The authors used a conceptual approach to thoroughly investigate the mechanism of disequilibrium in educational opportunities. A comprehensive strategy was used to provide a conspectus on the research topic. The paper followed a pattern of gathering and examining underlying issues while incorporating general knowledge and

awareness based on meticulous analysis of material from authentic written sources, including books, journals, and documentation.

Findings or Results

Problems in Equalization of Educational Opportunities

Lack of Schools and Colleges: India is the second most populous country in the world. It is still a developing country. There is a large population of young children, particularly in rural areas who have never attended school. Educational institutions are mostly set up in the urban and metropolitanized cities due to the available facilities and advancement. Some places have very few primary and secondary schools and that too, are located far from the residence of children. This creates an imbalance in educational opportunities.

Difference in Economic Status: About half of the population in the country suffers from poverty. Moreover, the number of government schools is very less in comparison to the private schools. As a result, it becomes exceedingly difficult for poor parents to pay the school and college fees. So, they do not send their children to pursue education. Although education has been made free in government schools for children from 6-14 years of age, they remain over-packed, which deprives them of attaining education.

Differences in the Standards of Educational Institutions: Government schools usually follow the fixed standard of curriculum set by the State Board or CBSE. However, there are a few which receive extra benefits. Similarly, in case of private schools, the quality of education differs. On one hand, there are those private schools which provide a high-quality education with advanced technologies, teaching aids, apparatus, and curriculum. Whilst there are those schools which cannot even provide the basic quality education due to lack of capital. Thus, it forms a barrier between the students of developed educational institutions and the students from underdeveloped schools, which makes the competition unfair with the unequal distribution of educational opportunities.

Differences in Home Environment: Since the Independence of India, there has been a huge improvement regarding educational awareness among families. On one side, there are children from educated and rich families, whose parents can provide them with support, encouragement, guidance, and expensive tuitions. On the other side, there are children with poor and illiterate parents. They may be the first generation in the family to have the privilege of obtaining

education. Hence, they do not get the needed support, guidance, and encouragement at home. This leads to inequality in educational opportunities and experiences.

Gender Disparity: Even though India has become a modernized nation, it still is conservative in some respects. There continues to be wide discrimination with regards to the education of boys and girls at all stages. In some sections of society, parents hesitate to send their daughters to school, believing that it is unworthy to invest the amount on the education of their girl child as they are expected to get married off soon and be housewives. Moreover, it is considered immoral to be in close contact with the boys. Much importance is given to educating the boys, while education of the girl child is denied or ignored. Although, recent statistics show that the rate of enrolment of the girl child is gradually increasing, it is yet much lesser in comparison to that of boys.

Difference between Rural and Urban: Educational awareness in rural societies is very less. The members are accustomed to agricultural activities and other primary occupation, being neglectful to education. The few children who can attend school are provided with the least basic education. Rural schools are not well equipped and lack infrastructure facilities. Urban societies, on the contrary, are more focused on education. Educational institutions are well equipped with better facilities and quality education. Graduates from urban educational institutions are more likely to have opportunities in higher education and jobs. Thus, the educational opportunities are unevenly dispersed.

Differences between Upper and Lower, Backward and Advanced Classes: There is a wide disparity between the upper and lower class in all the spheres of society, including education. The upper class or advanced section of society tries to exploit or have control over the weaker section. The caste system is still prevalent in many parts of India, where the lower class is systematically and intentionally deprived of education. Social issues such as Casteism, Racism, Communalism and Socio-economic discrimination hinders the educational opportunities of children belonging to the weaker section of the society.

Lack of Motivation towards Education: This is one of the major obstacles in the equality of educational opportunities. A lot of students may be working hard and sincere in learning, but they are often a time get de-motivated due to boredom and laziness. As a result, they lose

interest and feel that formal education is insignificant. Some quit school halfway as dropouts and hence, miss out the benefit of equal opportunities in education.

Discussion

The government has put several efforts in equalizing the opportunities of education through various schemes, policies, committees, and commissions. Let us study some of the crucial steps taken in bridging the wide disparity in educational opportunities. The Education Commission (Kothari Commission) of 1964-66 has laid out certain measures to be adopted for ensuring equal educational opportunities for all.

- Efforts should be made for improving the standard of education in general schools.
- Common School System should be established.
- Regional imbalances in the provision of educational facilities should be shrunk to the maximum.
- Special schools like Public schools should give admission to students based on merit.
- Efforts should be made for providing good educational facilities among children of rural, backward classes and tribal areas.
- Special encouragement and emphasis should be laid on the education of girls.
- Free supply of books and writing materials at the primary stage.
- Provision of book banks, textbooks and libraries should be made in all the secondary and higher educational institutions.
- Tuition fees should be removed at the primary and lower secondary stage. There should also be such provisions for the needy and deserving children.
- Provision of transport facilities should be made to reduce the cost on hostels and scholarships.
- Free studentship should be provided to prevent segregation of social classes.
- Provision of scholarships should be made at all stages and sectors.
- Educational facilities should be expanded for the especially abled children. Efforts should be made for developing integrated programs to enable them to study in regular schools.
- Reduction in the differences in educational development of different districts and states.

The National Policy on Education (NPE), 1986, also emphasizes the following objectives for the equalization of educational opportunities-

- It aims in removing the disparities and provide equal educational opportunities to those who have been denied of education so far. It puts more emphasis on girls and women education, the minorities, especially abled children, and the Scheduled Castes and Scheduled Tribes.
- The policy has recommended provision of incentives like free books, uniforms, transport facilities and attendance allowance.
- No tuition fees up to the primary level in government schools of all states.
- Reservations for SCs and STs, the backward sections as well as coaching and remedial teaching classes should be provided.
- Provision of post-metric scholarships.
- Area Intensive programs aim at providing basic infrastructure and facilities in areas inhabited by educationally backward minorities, lacking adequate and quality elementary and secondary schools.
- Education of especially abled children should be integrated in regular or general schools.

The Government of India has also adopted several schemes and measures for bringing about a balance in providing quality education to all irrespective of their origin.

Sarva Shiksha Abhiyan (SSA)- The Sarva Shiksha Abhiyan, which was launched in 2000, aims to achieve educational equality by prioritising primary education. It has made provisions for free education for children aged 6 to 14. SSA's mission-related objectives include the following:

- a. To enrol every child in school by 2003. 'Back-to-School' Camp, Alternative Schools, and Education Guarantee Center are a few ways.
- b. To centralize primary education to deliver a satisfactory, high-quality, and feasible primary education. With an emphasis on education for life, the goal is to provide all children with elementary education of satisfactory quality.
- c. To eliminate all gender and social category gaps in primary and elementary learning.

The Kasturba Gandhi BalikaVidyalaya (KGBV)- The Indian government introduced the Kasturba Gandhi BalikaVidyalaya in August 2004. The primary goal of the KGBV is to provide girls from minorities, SCs, STs, and OBCs with high-quality education and opportunities. Its objectives are as follows:

- a. Establishing residential schools where at least 50 girls, primarily from minority and SC communities, can enroll in the elementary level of the school.

- b. To provide these schools the infrastructure they necessitate.
- c. To prepare and procure the tools and materials required for teaching and learning in schools.
- d. To set up suitable systems for evaluation and monitoring, as well as for the necessary academic support.
- e. To encourage and assist the families in preparing the girls for residential schools.
- f. In upper primary, a focus will be placed on girls, particularly adolescent girls who cannot attend regular schools.
- g. Given the program's targeted nature, girls from minority or SC, ST, OBC communities would receive priority enrolment in such residential schools (75%), followed by girls from below the poverty line families (25%).

District Primary Education Program (DPEP)- The Government of India launched the District Primary Education Program (DPEP) in 1994. To accomplish the following goals:

- a. To ensure that all children have access to primary education, it covers about 18 states and 272 districts. It aims to increase the number of kids enrolled in the program who are between the ages of 6 and 14 from every section of society.
- b. To reduce school dropout rates and slow down stagnation. All students' primary dropout rates are to be decreased to less than 10%, according to this plan. A Minimum Level of Learning (MLL) program should be attended by kids. The program seeks to increase students' average achievement rates by at least 25% in basic literacy and numerical skills and at least 40% in all other competencies.
- c. To lower dropout rates and learning achievement gaps between worker and worker groups in society to less than 5%.
- d. To increase the institutions and organisations at the national, state, and district levels that are responsible for planning, monitoring, and evaluating primary education. It primarily aims to increase community involvement, decentralisation, and teacher and societal capacity building.

Right to Education Act, 2009- The Indian Parliament enacted the "Right to Education Act," also known as "The Right of Children to Free and Compulsory Education Act," on August 4, 2009. However, on April 1, 2010, it became effective. The terms "Free" and "Compulsory" education refer to the duty of the relevant government to: (a) guarantee free elementary

education to all children aged six to fourteen; and (b) ensure that all children aged six to fourteen are required to attend and complete their elementary school.

National Education Policy (NEP) 2020- The policy aims to “ensure inclusive and equitable quality education and promote lifelong learning opportunities for all by 2030.” It seeks to include the inclusion of diversity- Early Childhood Care and Education: The Foundation of Learning, Foundational Literacy and Numeracy: An Urgent & Prerequisite to Learning, Curtailing Dropout Rates and Ensuring Universal Access to Education at All Levels, Curriculum and Pedagogy in Schools: Learning should be Holistic, Integrated and Enjoyable, and Engaging, and Special Educators.

Limitations and Future Research

With the rapidly changing society and constantly emerging challenges, the study has great scope for further research in future. This research points towards some of the following aspects that need to be addressed:

- Mandatory schooling and universalization of primary education has been recommended by every commission, act, and policy even earlier in the 1900s, but how far will it be successful in achieving its goal remains a question.
- Disparity in educational opportunity is a product of several other social problems, and without the addressal and intervention of these, establishing balance is not possible.
- Are all social strata guaranteed access to quality education?
- Are the commissions and policies being updated in conjunction with a thorough ground check and considering the nation’s economic, social, political, and ideological structure?
- The State and Centre is not solely responsible for tackling these issues; is there cooperation of the families and society?

Conclusion

A plethora of distinct societal problems have evolved over the course of history. What is now regarded as a social concern may not have been so just a few years ago. Poverty, illiteracy, population growth, gender discrimination, domestic violence, alcoholism, drug abuse, terrorism, juvenile delinquency, and other major social issues still exist in our societies. As a result, social conventions, norms, and values mirror the problems of disorganization that have plagued society throughout its history. Nonetheless, there is no such thing as a problem without a solution. The

overall welfare of society as well as the growth of everyone depends on equilibrium in educational opportunity. Despite the government establishing numerous programs and policies, one of the greatest setbacks of Indian society has been its failure to achieve the aim of 100% literacy and “Education for All”. For the formation of an equitable society, equal educational possibilities are crucial. There is a discrepancy between the amount of competent and productive labour that society requires and what is available. The growth of a close relationship between the two is conceivable only through equal distribution of educational avenues.

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**Factors Motivating Student's Behavior on Online Education: from the perspective of
business schools**

Rukshar* M. Altaf Khan**

Abstract

With the advancement of society, science and technology, online learning is now a part of everyday life, and it is in high demand. The aim of this research is to identify factors that motivate students towards online education and their impact on students' motivation. Google Forms was used to collect data from an online questionnaire. A sample of 85 students from the business school (BBA and MBA) was collected. In this study, the quota sampling technique was used. SmartPLS-4 software was used to analyze the data. Correlation and multiple regression analyses were used to validate the proposed hypothesis. The findings indicate that time management, e-empowerment, and self-efficacy all have a significant direct effect on student motivation.

Keywords: *Online education, Business schools, time management, e-empowerment, self-efficacy, and student's motivation.*

Introduction

Online learning has already made a name for itself in the corporate world in the innovative age where we currently live in. Currently, every organization is concentrating on putting various strategies into place to hire a very high-quality, productive, and motivated crew. A business school, often known as a school of business, is an institution that offers bachelor's or graduate-level program in business administration or management. Those students have been attending business schools for subjects like entrepreneurship, administration, strategy, and finance. They have also taken classes in the subject of accounting, leadership, public relations, and marketing. For simply, we can say that business schools instruct students on how to run a business. Such type of institutions has produced a number of business managers. Some CEOs of significant companies, however, never attended business school or even don't have degrees. According to the Global Education Monitoring Report (Paper, 2022), the education 2030

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framework for action, which is the road map for achieving the fourth SDG, emphasizes the critical role of non-state actors in education.

Sir Richard Charles Nicolas Branson started the “Virgin Group” later also known as virgin megastores. Currently, the “Virgin megastores owns four hundred above separate organization in various field. Branson quit academy at the age of 16 years. As a results, he is lacking not only degree but also A-levels, which are equivalent to Senior high school in United Kingdom. Intermediate students can register in BBA and MBA is available to graduate students. Further Graduates can also pursue doctoral degrees. FT Lexicon reports that B-School, “A college or part of a college where students can study economic and financial subjects and learn about managing a business.”

A “Business School” can also be referred to as a management school, or simply as a B-School or biz school. Because the B-school curriculum is demanding, students must work extremely hard to keep up. But at the end, students who want to become a businessman have a difficult, lucrative career. So, for the majority, networking with classmates, professors, and executives who can be useful in the future is more important than getting A+ marks. Prospective students should carefully compare the costs of attendance to the potential advantages of earning a B-School degree, considering the prestige and specialization areas of each B-School.

Online learning is a new trend in education that primarily follows the 70/20/10 guideline. Most businesses now assess employees' knowledge and abilities through specialized change and group creation using online learning. This guideline states that the model for learning and growth should be based on 70% work experience, 20% of this was learned through collaboration with others, and just 10% was learned through formal education. A team of researchers from the Greensboro Center for Creative Leadership devised this rule. Furthermore, research revealed that just 10% and 20% of knowledge obtained by productive people was developed by employees through traditional learning and educational events. Oral learning, social learning, mentoring, and other activities all contributed to the workers. The practice turned out to be the most effective strategy in the end. The challenges of offline learning are reduced by online education.

Asian Management Institutes India's top business school has been named as being in Bangalore. The top institutions were listed in the 2022 Financial Times (FT) master's in management (MIM) rating, which was released on Monday. The Institute's standing in the world

has improved this year. IIM Bangalore moved up from 47th place in the world rankings in 2021 to position 31 in 2022. In this ranking, IIMBs improves the school's recognition and standing in India. One of the best business schools in India with a distant learning MBA programd is IMT Ghaziabad. The institute was established over ten years ago and is accredited by the DEB University Grant Commission. The institute has established a solid reputation for itself in a noticeably short period of time.

Literature Review

(Hou &Ahn, 2022) showed that excellent course design and active support behaviours after school have a significantly positive influence on customer satisfaction. The training content of the product owner plays a role in decision-making, especially in the initial phase. In the usage phase, the product side can build a community based on the analysis results, then conduct individual coaching and regular evaluations, which can effectively improve customer satisfaction and post-class expectation confirmation. (John et al., 2022) examined the social media marketing (SMM) persuasive process, especially in higher education. A conceptual model for the development of SMM communication was created and tested by theoretical information acceptance model. The self-conducted survey of a sample of international student travellers in New Zealand examines the effects of Argument Quality (AQ), Source Credibility (SC), Audience Involvement (AI), and Audience Engagement (AE) on their attitude formation and decision-making, with a focus on the mediating effects of audience involvement and engagement in SMM communication.(Sarfraz et al., 2022) examined the association between student attitudes and academic performance as measured by the “Unified Theory of Acceptance and Use of Technology” components (performance expectancy, effort expectancy, and facilitating condition). He looked at how technology trust influenced both students performance UTAUT constructs. (Erguvan, 2021)suggested that that all faculty members are aware of contract fraud to the extent that in the online learning environment, at least a quarter of their students utilise essay mills, and that various cultural practices, such wanting things to be simple, being lazy, and feeling pressure to get a college degree, encourage contract cheating. (Cao et al., 2021)explored how shared responsibility affected the relationship between marketing and other coping abilities in students of B-School and happiness of them through the quick transition to e-learning and expectations for educational accomplishment. Regardless of staff readiness, student readiness or

experience, the COVID-19 epidemic has drastically changed in education sector. (Adam & Gunarto, 2021)the findings indicated that Social-Media and Word of Mouth had a favourable and substantial influence on choice toward attend a private institution in Palembang City. The powerful tool is social media for promoting online education in higher education. When students are satisfied with the lectures, they will spread the word about the material and suggest the courses to others. (Broeren et al., 2021)evaluated the potential impact of a classroom intervention on students' ability to use memory exercises in a higher education classroom. The purpose of this study examined whether these effects may be applied in a classroom context employing key marketing communication ideas.(Gopal et al., 2021) the findings showed that the study's four independent variable- teachers' quality, course design, rapid feedback, and student expectations all had significance impact on students' satisfaction, and student's satisfaction had significance effect on students' performance. (Salem, 2020) the goal of the study is to analyzed social media marketing in higher education and identify the methods and strategies utilized by universities to promote their brands and determine the effects of social media marketing impacts on boosting higher education institutions' competitive advantage. (Muneeb et al., 2021) examined the effect of internal marketing orientations (IMOs) on higher education institutions' competitiveness in the UAE. College competitiveness and faculty performance are mysteriously correlated with the IMO characteristics of Information Generation (IG), Information Diffusion (ID) and Information Responsiveness (IR) are inexplicably linked to college attractiveness and performance of faculty member. IR. Compared between IG, ID and IR was originated to be a more substantial prognosticator of college competitiveness. (Novikova et al., 2020) investigated the role of social and communication tools, specifically media and online social platforms such as Instagram, Twitter, LinkedIn, or YouTube, or other communication search platforms such as Enterprise Europe Network, in technology transfer between Ukrainian universities and international research universities. It was proposed to develop a modern strategy for commercializing the outcomes of research institutions' innovative activities by leveraging the resources of the media sectors.. (Mazurek et al., 2019)explored that social media is viewed as 'youth' media which is articulated in the target market and the messages' content. Social Media is not meant to advance science and research. Investigators and the management are supposed as interested parties who are unconscious of the opportunities of SM for enhancing the reputation

and image of schools. (Wei et al., 2019) provided the direction to further improve the O2O framework. Regarding the instructors, some focus more on the formal aspects while ignoring platform knowledge and the improvement of students' talents. Some kids sign up, participate, and gain free access to the platform. (Pevzner et al., 2018) showed the possibilities of internet marketing as a modern diversity management tool at universities and other educational institutions. In today's information society, Internet marketing is becoming a technology that makes it possible to attract the attention of customers using educational services to academic programs of higher education institutions to promote these programs, and to provide an opportunity to attract potential students from various social groups to the university. (Su et al., 2018) proposed a framework from the perspectives of social marketing, community platform, & CRM, with the goal of increasing interactions and strengthening fans loyalty, as well as gaining more fans for participating activities. (Dora & Saudi, 2020) examined the role of digital marketing designed for MSMEs in order for them to be sustainable and sustainable. According to the study's findings, MSMEs need to use innovative and creative digital marketing strategies in order to thrive in the New Normal era. (Andrade et al., 2019) aimed to investigate the influence of the existing B-Schools' method for online learning, specially, how deans handle faculty difficulties. The findings indicate a demand exceeds supply.

The objective of this study is to identify factors motivating students towards online education and their impact on students' motivation.

Conceptual Framework Model

Previous studies have been indicated that excellent course design and active support behaviours after school have a significantly positive influence on customer satisfaction (Hou & Ahn, 2022). Thus, we can argue in the context of factors motivating student's behaviour that the student's motivation will be affected by the time management, e-empowerment, and self-efficacy.

As a result, the conceptual framework model below is set up.

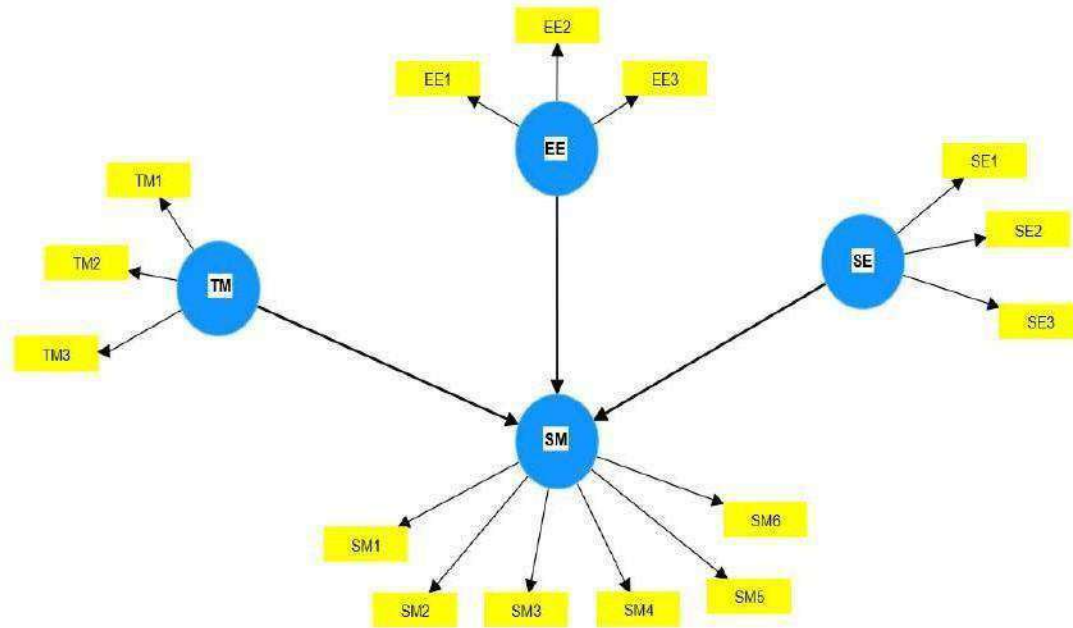


Figure 1: Conceptual Model

Research Hypotheses Development

Time Management

The most crucial component that affects an individual's overall performance and accomplishments is time management. The goal of this study is to ascertain how students' academic success and time management abilities relate to one another (Nasrullah & Khan, 2017).

H1: Time management has a significant impact on the student's motivation.

E-Empowerment

Present study is based on students' e-empowerment enhance by motivational factors of online education. (Weber, 2020) suggested that the Learner Empowerment Scale is a viable and trustworthy method for determining a student's level of interest.

H2: E-empowerment has a significant impact on the student's motivation.

Self-Efficacy

(Mojavezi, 2015)analysed how teacher self-efficacy (TSE) affected students' motivation and performance.

H3: Self-Efficacy has a significant impact on the student's motivation.

Student's Motivation

Student behaviour in academic contexts is attempted to be explained by modern ideas of academic motivation. Though each theory has its own constructions and own explanations, these ideas are connected in several ways. This theoretical study of motivation described a number of theories of motivation, and it used the influence of emotions as a unifying element to tie them all together. (Seifert, 2004).

Research methodology

This study's instrument was created by adapting “five-point Likert scales” ranging from "strongly disagree" to "strongly agree" from previous research. Questionnaire was created and distributed to B-school students in Delhi, NCR via online Google forms. The questionnaire is composed of two parts: Part A is related to demographic profile of the respondents. Part B is based on hypotheses proposed in this study. A filtrate questions were asked to determine whether B-school students are motivated by online education. Only if the respondents answered yes did they proceed to the next section of the questionnaire. Only those respondents who were highly motivated by online education were included in this study, which used a quota sampling technique to obtain responses. Only 85 of the 139 responses received were motivated to pursue online education, so only 85 responses were used for analyzing data.

Analysis of Data

“Demographic Profile” of Respondents

Table 1 shows the demographic profile of respondents. There were 61.7% female and 38.3% male among those who responded. 38.3% of respondents were under the age of 20 years, 61.7% were between the age of 20-30 years, 0% were over the age of 30 years. In terms of educational background, 78.7% were undergraduates, 2.2% were graduates, 19.1% were postgraduate students interested in online learning.

Table 1: Respondents’ Demographic profile

Profile	Characteristics	Frequency	Percentage
Gender	Male	52	61.7
	Female	33	38.3
Age	Up-to 20 years	33	38.3
	21 to 30 years	52	61.7
	30 to above	00	00.0

Education Qualification	Undergraduate	67	78.7
	Graduate	02	2.2
	Post graduate	16	19.1

Process of Data Analysis

SmartPLS 4 was used in the study to analyze the data. SmartPLS-4 was chosen for analysing the data because it is better for evaluating small sample size. A two-stage evaluation procedure for measurement and structural models was introduced.

Measurement Evaluation Model

The measurement evaluation model was examined using “internal consistency,” “convergent validity” and “discriminant validity.” Table 2 shows the “mean”, “standard deviation”, and “loadings” of all constructs in the study.

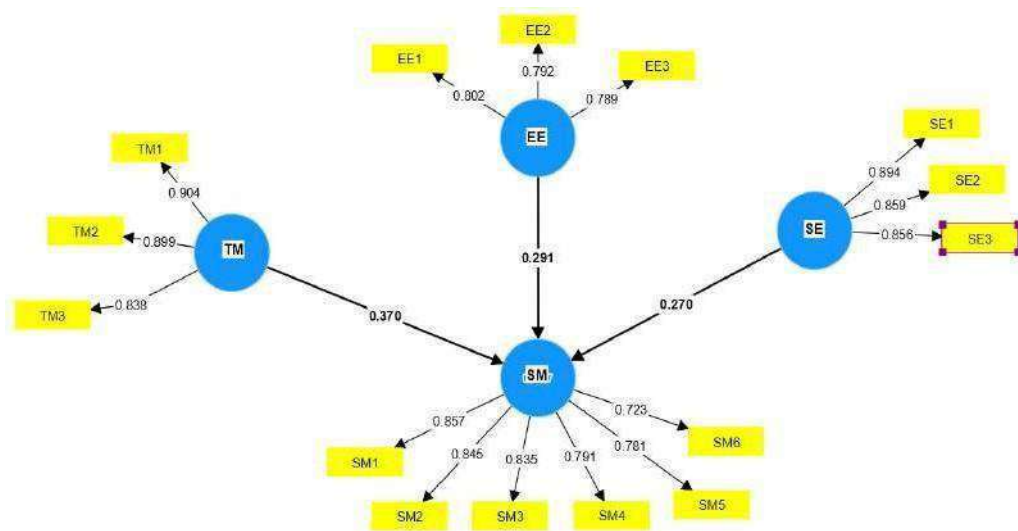


Figure 2: Measurement Evaluation Model

Table 2: Sample Mean, Standard Deviation and Constructs' Loading

Constructs	Item	Sample Mean	Standard Deviation	Constructs' Loading
Time management	TM1	0.904	0.022	0.904
	TM2	0.897	0.026	0.899
	TM3	0.836	0.038	0.838
E-empowerment	EE1	0.802	0.046	0.802

	EE2	0.790	0.052	0.792
	EE3	0.784	0.065	0.789
Self- efficacy	SE1	0.892	0.028	0.894
	SE2	0.861	0.033	0.859
	SE3	0.849	0.053	0.856
Students' motivation	SM1	0.857	0.034	0.857
	SM2	0.843	0.037	0.845
	SM3	0.836	0.029	0.835
	SM4	0.789	0.051	0.791
	SM5	0.781	0.046	0.781
	SM6	0.718	0.082	0.723

Table 3 shows that all four constructs exceeded the necessary requirement, with loading value exceeding 0.708, Composite Reliability exceeding 0.7 and Average Variance Extracted above 0.5, (Sarstedt et al., 2020). Cronbach's Alpha value, which measures internal consistency was also greater than 0.7, (Farrell & Rudd, 2009). This showed the constructs' convergent validity.

Table 3: Constructs' Cronbach's Alpha, Composite Reliability and Average Variance Extracted (AVE)

Construct	Cronbach's Alpha	Composite reliability	Average Variance Extracted (AVE)
Time management	0.855	0.912	0.776
E-empowerment	0.708	0.837	0.631
Self-efficacy	0.840	0.903	0.757
Students' motivation	0.893	0.918	0.651

Discriminant validity shows “the extent to which the measure is reasonably distinguishable from related constructs within the nomological network.” According to Table 4's Fornell-Larcker Criterion, the constructs TM-0.881, EE-0.794, SE-0.870 and SM-0.807 had square roots of AVEs, that were higher than the values of correlation between each construct and other constructs. Thus, the Fornell-Larcker criterion was used to determine the discriminant validity.

Table 4: Discriminant Validity “Fornell-Larcker Criterion”

	TM	EE	SE	SM
Time Management	0.881			
E-Empowerment	0.670	0.794		
Self-Efficacy	0.611	0.674	0.870	
Students’ Motivation	0.730	0.721	0.692	0.807

Table 5: Cross-loading Criterion and Discriminant Validity

	Time management	E-empowerment	Self-Efficacy	Students’ motivation
TM1	0.904	0.548	0.554	0.665
TM2	0.899	0.579	0.551	0.657
TM3	0.838	0.651	0.509	0.605
EE1	0.519	0.802	0.443	0.606
EE2	0.617	0.792	0.673	0.583
EE3	0.453	0.789	0.490	0.522
SE1	0.589	0.636	0.894	0.653
SE2	0.520	0.573	0.859	0.638
SE3	0.474	0.541	0.856	0.491
SM1	0.643	0.644	0.569	0.857
SM2	0.552	0.594	0.561	0.845
SM3	0.649	0.670	0.665	0.835
SM4	0.579	0.550	0.525	0.791
SM5	0.633	0.592	0.588	0.781
SM6	0.431	0.372	0.388	0.723

The cross-loading criterion is shown in Table 5, where all constructs loaded higher than when they were cross-loaded with other constructions across the columns. Thus, the cross-loading criterion was used to determine discriminant validity.

According to Henseler, Ringle and Sarstedt (2015), “The HTMT value is less than 0.90, discriminant validity has been established between two reflective constructs”. Table 6 shows that the discriminant validity of the measurement model was established.

Table 6: Discriminant Validity “Heterotrait-Monotrait (HTMT) Criterion”

	TM	EE	SE	SM
Time Management				
E-Empowerment	0.861			
Self-Efficacy	0.715	0.868		
Students’ Motivation	0.824	0.886	0.775	

Structural Evaluation Model

Before evaluating the structural model, multicollinearity must be investigated to ensure the validity of those results. The Values of Variance Inflation Factor (VIF) must be under 5. In this study VIF value **1.362 to 2.862** range. To establish whether the hypotheses are relevant, bootstrapping approach was used to evaluate the structural model (5,000 resamples).

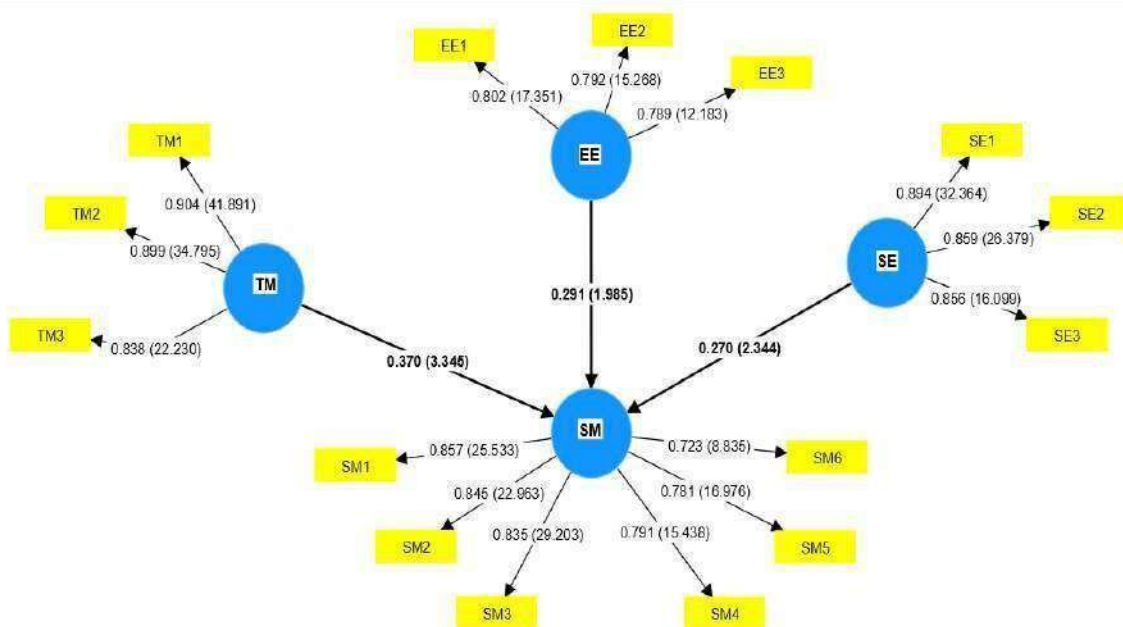


Figure 3: Structural Evaluation Model

Table-7 shows that the H1, H2 and H3 (Hypotheses) were accepted, representing that Time management ($\beta = 0.353$, $t=3.45$, $p<0.05$), E-empowerment ($\beta = 0.322$, $t=1.985$, $p<0.05$) and Self-efficacy ($\beta = 0.261$, $t=2.344$, $p<0.05$) significant and positive impact on students' motivation.

Table 7: Results of Hypotheses

Hypotheses	Path	B	SD	T	P-value	Results
H ₁	TM->SM	0.353	0.111	3.345**	0.001	Accepted
H ₂	EE->SM	0.322	0.147	1.985**	0.047	Accepted
H ₃	SE->SM	0.261	0.115	2.344**	0.019	Accepted

Note- β = path coefficient, SD= standard error, T= t-statistics and results of hypotheses.

Table 8: R², R² Adjusted and F²

	R ²	R ² Adjusted	F ²
SM	0.667	0.654	
TM			0.207
EE			0.112
SE			0.109

Table 8 represents that the R² is the coefficient of determination, and the values between 0 and 1 are considered significance. Results need to be more than 0.19 to indicate a weak coefficient. Results above 0.33 are considered good and above 0.67 are considered excellent. The outcomes of this study are good because the R² value is 0.667, or 66.7% and the R² adjusted value is 0.654. As a result, we can conclude that the relationship between variables is significant.

A statistical concept called F² (effect size) uses a quantitative scale to quantify the strength of the link between two variables. According to Cohen (2013), When F-square value is 0.35 effect size is high, F² 0.15 is medium and F² 0.02 effect size is minor. Effect size of Time management=0.207, E-empowerment =0.112 and Self-efficacy =0.109, which is less than 0.35 so that effect size is large.

Results and Discussion

Motivational factors of online education are especially important for higher education. This research attempted to identify the effects of time management, e-empowerment, and self-efficacy on student motivation for online education in business schools. Online learning

environments primarily consist of an e-learning system focused on adaptability, deliverability, freedom, and independence of learning. Students feel comfortable when they reach the learning-teaching environment anytime, anywhere. If we analyze the results of this research, Time management showed significant effect on students' motivation (H1 accepted), Similar to the findings of other studies, university students' academic success is closely tied to their time management skills (Nasrullah & Khan, 2017). E-empowerment had significant effect on students' motivation (H2 accepted) similar to earlier findings that the relationship of learner empowerment and interest has significantly/ positively affect motivation (Weber, 2020). Self-efficacy was significant in affecting students motivation (H3 accepted), which agrees with the findings of earlier research by (Mojavezi, 2015).

Theoretical Implications

Theoretical considerations for this study include enhancing comprehension of students' motivation for online education in B-School of India. Few studies have examined the motivating factors of online education that affect student motivation. The results of this study add to previous research by offering a through framework for describing the role of motivational factors in engaging business school students in online education. The results showed that time management, e-empowerment and self-efficacy had significant direct effects on students' motivation. This study confirms the significance of being educational, interesting, and qualitative from the student's point of view.

Managerial Implications

Our findings have several managerial implications. Identifying the specific characteristics that impact student motivation level for online education, the Indian government can be more confident that their investment in e-learning will pay off in terms of increasing enrolment ratio and literacy levels in India. Students should consider choosing an appropriate e-learning platform that can engage students in online education. This study provides some key recommendations for online service providers to deliver quality course content through the best e-learning platform, while also being entertaining, flexible, and motivating for students.

Limitations and Further Research

This study has various restrictions that must be considered. Sample size is insufficient to generalize to India's large population. As a result, the future may require a large sample size.

This study focuses on students from b-schools in Delhi NCR, and future research may focus on students from other states' rural and urban areas from various collages rather than b-schools. Furthermore, quota sampling was used to select respondents for the study. As a result, in future studies, random sampling techniques could be used to collect data, and the results could be improved.

Conclusion

Online learning has already made a name for itself in the corporate world in the innovative age we are currently living in. Every organization is focused on implementing different strategies to hire an extremely high quality, productive and motivated workforce. The results showed that time management, e-empowerment and self-efficacy have a significant direct impact on student motivation. Our findings demonstrate a favorable correlation between TM, EE, and SE and students' motivation. Furthermore, e-learning has advanced significantly in recent years compared to previous years, and educational institutions are also taking part in this technological advancement, which is always connected to their students via the internet. Online education is becoming increasingly important and is becoming required for all colleges, including business schools.

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Challenges, Opportunities and Experiences on Open Book Examination: From Students' Perspective

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Abstract

Present paper aims to find out the students' perception on open book examination (OBE). The open book examination was conducted during the Covid-19 pandemic in many higher learning institutes in the country. For many students prior to pandemic the term "open book examination" was not perceived. The results revealed that 62.18% students were familiar with the format of OBE prior to sitting the exam. Most of the respondents (53.85%) report of experiencing problems during online open book examinations, the major problems mention was network issues/poor internet connectivity, difficulty in finding relevant material to answer questions, takes time to answer the questions, unfamiliar and lack of learning material/reference. 59.94% of students opines that open book examinations performance was better than closed-book examinations (CBE). The students perceive their experiences on OBE as 'good' by (50.96%), 'very good' (16.03%), 'fair' (15.06%), 'excellent' (11.86%) and 'poor' (6.09%). However, 55.13% of respondents have chosen closed book exam over open book exam due to familiarity with the traditional closed book examinations.

Keywords: open book examination, closed book examination, higher education, Covid-19

Introduction

Examinations are an essential means in learning. It is an assessment aimed at measuring students' abilities, skills, aptitude, or knowledge in one or more topics of studies. Today, across the globe, various methods and means are being used for assessing learners especially in higher educational institutes. But, in India, educational institutes are well acquainted with the traditional closed-book examinations. However, the Covid-19 pandemic has forced the institutes to resort to new methods to evaluate the students across the nation. During the pandemic, the closure of schools,

colleges and universities not only interrupted the teaching of students, but the closure also coincided with a key assessment period and many examinations were postponed or cancelled across the nation. Despite all these challenges, the Indian higher education institutes have reacted positively and managed to ensure the continuity of teaching-learning, research, and services to the society with some blended modes. In higher education, during the pandemic, many colleges and universities were replacing traditional exams with online assessment tools. The approach adopted to conduct online examination varies as per the convenience and expertise among the educators and the compatibility of the learners (Pokhrel. S & Chhetri. R 2021). In April 2020, the University Grant Commission (UGC) issued a guideline for higher education to conduct examination focusing on Open Book Examination (OBE) stating that “Open Book Examinations” gives a fair and equal opportunity for students appearing in online/offline/blended mode. During the OBE, students will be allowed to refer to books, study materials and notes to answer the questions. In compliance with the UGC guidelines many Indian education institutes have adopted the open book examinations. As OBE help students to remove the fear of examination as it minimizes cramming (Kalidar et al., 2018) and it takes away the pressure of retaining information among the students, allowing them to focus on understanding the concepts better (Vallikat, 2021).

According to Saif(2021), an “Open Book Examination” is an assessment which is designed in a way that allows students to refer to class notes, textbooks, or other approved material while answering questions. It can also mean that students are provided with the exam questions prior to writing the exam. OBE represents an assessment tool that promotes more active learning. Feller (1994) believed that the closed book examinations test what students can memorize while open book examinations measure high-level skills such as conceptualizing, problem solving and reasoning, corresponds almost completely to a total, real-world situation.

Many other potential research opportunities in the study of open book examinations do exist. And educators agree that open book examinations reduce anxiety and students appear to consistently prefer open-book exam despite the acknowledge understanding that they generally require the exhibition of higher-order thinking skills (Brightwell, Dainiel, & Stewart, 2004). Eilertsen & Valermo (2000) viewed open book test to encourage thinking at higher cognitive levels and promote study and teaching methods.

Considering the OBE efficacy and compatibility to students, many universities and colleges in the state, Assam have adopted open book examination to evaluate the students during the Covid-19 pandemic. However, when the declaration of conducting examinations through open book was publicized many misperceptions were created among the students and teachers in the state. This sudden adoption of new assessment method seemed to be incredibly challenging for the students because they are mostly familiar with closed book examinations. Thereafter, students construct or find model answers to these questions and memorise them (Swart, J. & Sutherland, T. 2014) for the traditional closed book examinations. Therefore, this open book examination was completely a new concept where the students have never sat before. Despite many challenges, the students have positively retorted to open book examinations.

Review of Literature

Several studies on open book examination have been done to know its impact on students of higher learning institutes. Eurbuonyanun, et al. (2021), compared 4th year medical students of online surgery clerkship assessment scores to the traditional written examination. The percentage of correct scores using open-book examination was compared to the results of the traditional closed-book examination. Their result reveals that the traditional groups medical students who took the online open book examination had a significantly higher mean score in both multiple choice questions and essay examinations, but a significantly lower mean score in short answer examinations. The medical students preferred a traditional closed-book examination over an online open book examination (62.2%), however, students believed (66.7%) that OBE approach was the best solution to prevent contamination during the Covid-19 pandemic.

Saif (2021), investigate the impact of OBE on the learning habits of undergraduate students. Findings revealed that a high percentage of the participants preferred open-book examinations despite being more familiar with closed-book examinations. They felt that preparations for open book examinations consume less time and require less memorization and left more rooms for logical thinking.

Loi & Yuan, (n.d) in their study found that majority respondents prefer open book to closed book examination. The reason 90.6% stated they have less to memorise in open-book examination and 43.0% who prefer open book examination agreed that it was less time consuming for preparation. On the contrary, students preferring closed-book examination state the reason that examinations

were usually easier, more direct, clearer, and closer to questions in tutorials and their text books. Furthermore, they could save time needed to locate the information they need to answer the questions during the examination.

Das(2017), examine the effectiveness of OBE in terms of achievement of Standard VIII students in language subjects along with the examination anxiety. Result of the study showed the open book examinations are not productive or fruitful in enhancing the achievement of student in any language subjects and report that it did not help to reduce the examination anxiety of students.

Swart& Sutherland(2014), Student perspectives of open book versus closed book examination – a case study in satellite communication. The purpose of this article was to highlight the advantages and disadvantages of open book examination while classifying its application in higher education. It was implemented in a telecommunications module at two vastly different universities and student feedback was sought at the end of the course. Student feedback highlighted that majority of these students would prefer open book examination as compared to closed book examination.

A comparative study on students' achievement, Ioannidon(1997), found that there is no significant difference in total examination score between the students who took closed-book examination. Open book examination is not a better measure of different abilities than the closed-book examination when the examination is designed to address critical thinking and higher order skills. The findings recommended that other factors like the teaching-learning process applied throughout the course affect students' achievement. From this perspective, it was mention that open book and closed-book examination have their advantages and disadvantages.

Methodology

The population comprised all the students of higher learning institutes of the state, Assam, who have sat for online open book examinations during Covid-19 pandemic. A sample of 312 students have participated in the survey, where 92.2% of the respondents were of the age group from 21-25 years of age and 59.62% (186) were female and 40.38% (126) were male respondents. The respondents were perusing MA/M.Sc/M.Com and other diploma courses. The findings are based on primary and secondary sources. Primary data are collected through questionnaire and secondary data from various journals, research papers, articles, and other academic publications.

Objectives

The main objective of the paper aims to find out from the experience of students on an open book examination, their performance, challenges, and opportunities towards open book examinations. The paper also aims to seek the student's preference over open book examinations versus closed-book examinations.

Result and Discussion

Course of Study sat for Open Book Examination

In the state, this was the first time an open book examination was held in universities and colleges. Although it was challenging to be evaluated by new assessment tool most of the students had a new experience. The courses of study sat for OBE was B. A/B. Sc/B. Com end semester examinations (95.51%) while 4.49 percentage of students sat for other diploma courses. The respondents mention that this exam was the first open book examination they have sat for. And 38.46 percent of respondents reveal of not perceived of the term "open book examination" prior to Covid-19 pandemic however, majority (61.54%) of the respondents report of been aware of open book examinations.

Familiar with the Format of Open Book Examination

Any new evaluation tools to be effectively used for assessment, it must be part of the learning experiences (formative assessment). What is permitted and prohibited during the exam should be clearly explained (Doghonadze, 2018). The Indian students are well-acquainted with traditional closed-book examinations while open book examination was a new assessment for many students that require high analytical skills. Teachers too lack effective teaching techniques for open book exams. Even some students were unfamiliar with the type of questions they are going to face in OBE. Therefore, it is important to know whether the students were familiar with the format of open book examinations before sitting for online OBE. Regarding this, 62.18 percent of students responded of been familiar while 37.82 percent students were not familiar with the format of open book examination prior to sitting in the exam.

Problem/s experience during Open Book Examination

Most of the respondents (53.85%) reveals of experiencing problems during online open book examinations, while 40.71% report not facing any problem and 5.44% mention 'not sure'.

The widespread problem(s) experienced by students during online open book examinations are provided in the table below:

Table No. 1, Showing the problem(s) faced by the Respondents

Sl. no.	Types of Problem	Frequency (N=168)	Percentage (%)
1	Takes time to answer questions	27	16.07
2	Difficulty in finding relevant material to answer questions	51	30.36
3	Network issues/poor internet connectivity	134	79.76
4	Unfamiliarity	23	13.69
5	Lack of learning material/reference	23	13.69

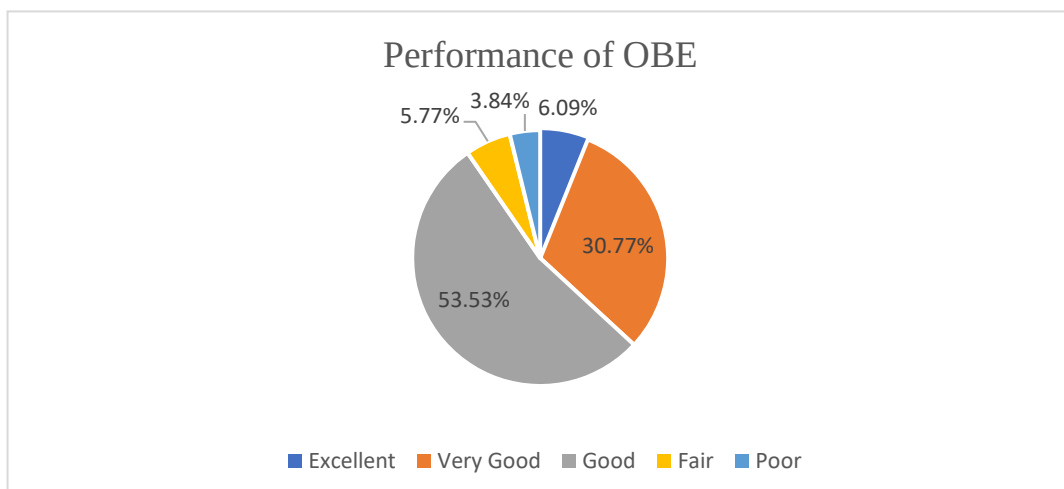
The above table depicts the types of problems experienced by the respondents during the online open book examination. The highest number of students had network issues /poor internet connectivity. Other problems revealed by the students are difficulty in finding relevant material to answer questions (30.36%), takes time to answer the questions (16.07%) and others opine of unfamiliarity and lack of learning material/reference (13.69%).

Sufficient study material/reference books for Exam

The finding revealed that 51.28 percent of students didn't have sufficient material/reference books for the online OBE. In the study of Ashri & Sahoo (2021), they reported of two broad challenges to online OBE that are lack of digital infrastructure and non-availability of requisite study material. This problem is quite common in several studies of OBE.

Performance/percentage obtained in Open Book Examination

The responses of students in respect to their performance/percentage obtained in open book examination are provided in the figure below:



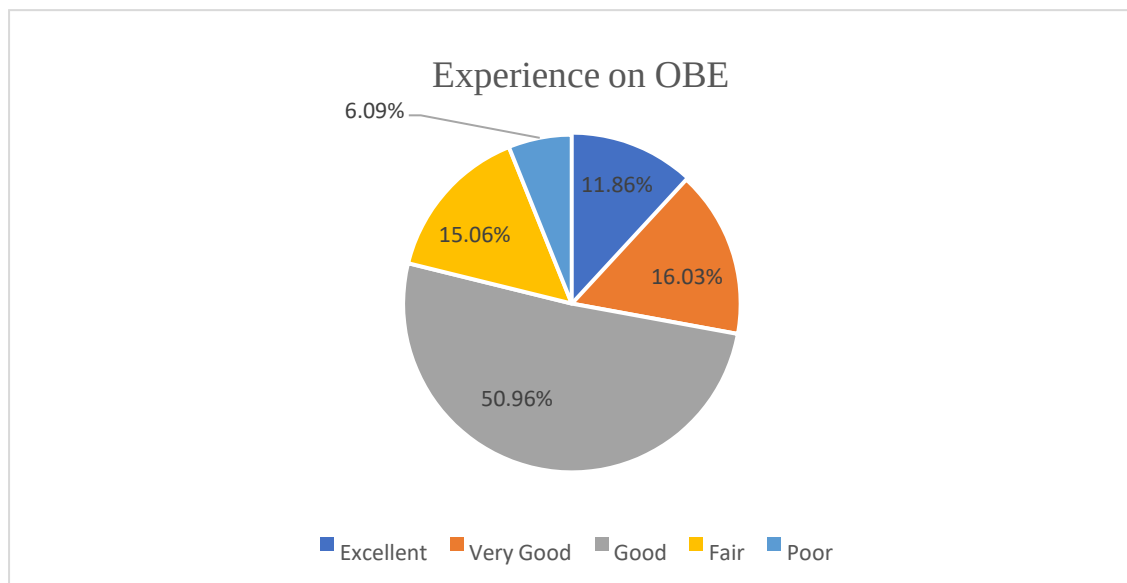
From the above figure, it may be assured that majority of the students' performance in online open book examinations was good (53.53%), follow up by particularly good (30.77%), excellent (6.09%), fair (5.77%) and poor (3.84%).

Based on the performance/percentage obtained by students in OBE, most of the respondents (59.94%) mention that OBE performance was better than closed-book examinations whereas 12.18 percent of respondents opined that open book examinations performance was not better than closed-book examinations. On the contrary, 27.88 percent of respondents report of similar percentage obtain in both type of examinations.

In response to performance, 65.06 percent of students expressed of been satisfied and the reason(s) provided for satisfaction was "secure expected marks" by majority of students (53.20%) and "secure more marks than expected" by 46.80 percent of students. On the other hand, 34.94 percent of students remark of dissatisfaction with their performance stating the reason as "secure less marks than expected" in open book examination.

Overall Students' experience on Open Book Examination

The perception of students regarding their experience in open book examinations are shown in the figure below:



From the above figure it is observed that majority of the students have perceived their experiences on open book examination as 'good' (50.96%), which is followed by 'very good'

(16.03%), 'fair' (15.06%), 'excellent' (11.86%) and 'poor' (6.09%). This result shows that most of the students had a pleasant experience in open book examinations and it can be generalized that though this was the first OBE the students have positively retorted to the open book examination held during the Covid-19 pandemic.

Students' preferences over Open Book & Closed Book Examinations

Regarding the preference to choose over open book examination versus closed book examination, most (55.13%) of them have choose closed book examination over open book examination. However, the reason(s) provided for choosing closed-book examinations were familiar with closed-book examinations (77.91%) and less stressful (33.14%). On the contrary, 44.87 percent of students who prefer open book examinations stated the reason(s) as less time consuming for preparation (72.91%), no memorization (70.00%) and less stressful (62.86%).

From the result, it may be interpreted that majority of the students preferred closed-book examinations due to familiarity. However, this finding is also consistent with the result of Eurbuonyanun, (2020) and inconsistent with the findings of Saif, A. (2021), Swart, & Sutherland, (2014); Loi, L.S., & Yuan, W. (n.d).

Advantages and Disadvantages of Open Book Examination

The respondents were asked to specify the advantage and disadvantage of OBE based on their experiences. In this question, only 187 (59.34%) students have given their responses. The following advantages specified by students were less stressful, good for scoring marks, no memorization required etc. The disadvantages were: time consuming for answering questions, needs time in locating the answer, difficulties in finding relevant materials to answer questions, not convenient for online mode especially in remote areas. In the study of Mahmoudzadeh et. al (2015) most of the students stated that OBE was more suitable for high cognitive learning and solving real life situations, with less stress in preparing for and during the exam.

Conclusion

This paper discussed the challenges, opportunities, experiences on OBE from students' perspective. Though this was the first-time open book examination students sat for, the result showed that students have obtained a good percentage, and many were satisfied with their performance. Majority of the students have better result in open book examinations compared to their previous closed book examinations. The students are assured of having a pleasant experience

on open-book examinations. However, many of the students preferred closedbook examinations over open book examinations because they are familiar and convenient with the former type. From various studies on open book examinations showed that, however, the biggest challenge of all is perhaps lack of familiarity to open book examinations of the students. The key to successful implementation and increased students' learning rests with how well the examination testing protocol and process are described to the students (Green, Ferrante & Heppard., 2016). The open book examinations are quite challenging. At times it can be more difficult than closed book examination as student is not used to open book exams which require a different study approach/strategy. It may therefore, necessary oftaking a formative examination with the new assessment methods to help the students be familiarized. Facilitating the shift from closed book to open book examinations there is a need to use of a more interactive mode of teaching in the first hand. As in the traditional style of teaching, students are passive learners who either listen to lectures or take notes. In open book examinations, the focus shifts from the reproduction of information to the processing information and the questions assess a students' abilities rather than his knowledge of course content (Saif 2021). Traditional closed book examinations have been carried on for decades and are still preferred by many educational institutes. For open book examinations, some teachers fail to develop questions that require high analytical skills and lack effective teaching techniques. Even students are unfamiliar and are not able to prepare themselves in a proper way which creates a sense of confounding of what to learn. Therefore, introducing an open book examination could become a choice of many institutions if the matter addressed above can be resolved.

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A Review of ICT Initiatives, Targeting the Inclusive Development in India

Sujata Pal* R.N. Sharma**

Abstract

The last two decades have witnessed a revolution due to the onset of technology in India. The focus of Information and Communication Technology (ICT) is now moving from simple automated information to making lives simpler and connected. Advancement of technology has widened the horizon of the term inclusion. It is not just restricted to disability but also incorporates heterogeneity, diversity, and individual differences. It is a multi-dimensional concept and can be understood from a variety of fronts, which includes educational inclusion, economic inclusion, political inclusion, socio cultural inclusion. Inclusive society is a bond built through cooperation and collaboration between the advantaged and disadvantaged sections of society regardless of their strength and weakness. The new digital aspiration is the driving force behind the transformation of society. The digital transformation is not just enhancing productivity and efficiency on economy but also promotes connectedness with citizens. The penetration of digitalization in every aspect of life shouldn't disguise the fact that the digital divide is still very wide.

Keywords: inclusive society, ICT initiatives, major challenges

Introduction

Ever since the time of independence India believes and enacts on the principle of Antyodaya which means development of the people at the bottom of the pyramid. A sustainable society requires both economic growth and inclusivity i.e., creation of opportunities for all. It is a long-term perspective. The introduction of the Five-year plans from the very inception had a significant bearing towards the development of the country in various fields but poverty, unemployment, lack of proper infrastructural facilities, epidemic had crippled the society. Such a diasporic condition could be seen in almost all the third world countries. The wretched condition grabbed the attention at the international level.

At this point of time *the Salamanca Statement (1994)* for the first time introduced a new concept to the world i.e., inclusion. Initially the scope of inclusion was restricted to the field of education only. It was observed that education is the only remedy to all maladies. Consequently, the adoption of the United Nation convention on the *Rights of Persons with Disabilities (2006)* intended to protect the rights and dignity of persons with disabilities. There was a shift in approach in society from medical model of disability to social model of disability which ensures not just equality but also equity and empowerment. The main aim of society is not just to focus on the issues faced by persons with disabilities but to involve the community at large alongside. In the following years SDG have broadened the horizon of inclusion and have integrated with the advancement of technology. Inclusion as a goal reflects a globalized network world where freedom of movement and full participation of all is taken for granted and enshrined as a right. Inclusive Development Index (IDI) has been designed as an alternative to GDP that reflects more closely the criteria by which people evaluate their countries economic progress. Inclusive Development Index is a parameter that refers to both pace and the pattern of the economic growth of a country. It ranks countries economic performance based on three pillars: growth and development, inclusion and integrational equity and sustainability. It thus becomes evident that economic development and inclusivity through access to resources goes hand in hand. Fig 1 explains the elements of Inclusive growth that marks a positive environment which embraces diversity, greater engagement of people's participation and making people feel valued.

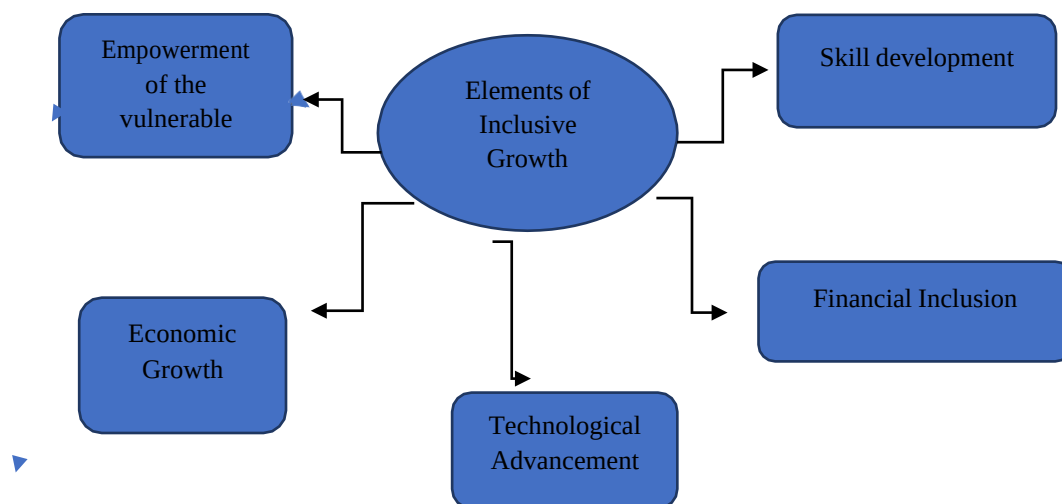


Fig 1: Elements of Inclusive Growth

Keeping the broader perspective in mind the Indian Government took the initiative to launch the National e-Governance Plan in 2006 with 31 Mission Mode Projects (MMP) was one of the few structured initiatives focused on making the government citizen centric. Since then, the programs have transitioned from being focused on not only service delivery but also creation of a knowledge-based society and inclusive and sustainable ecosystem. India being one of the signatories of SDGs and her commitment towards sustainable goals is clearly reflected in the national development agenda as the motto of NITI Aayog states “Sab ka sat Sabkha Vikash” (collective efforts for inclusive growth).

At the same time, digitalization a revolutionary initiative helped to transform our country into an empowered economy and foster inclusive growth. Digital technologies broke the traditional barriers of communication, interaction, and access to information for persons with disabilities. Accessible to ICT can uplift the level of persons with disabilities across different domains including education, employment, e-governance for documentation, financial inclusion etc. The advent of internet has heralded a new age not only of information sharing in general but also a proliferation of web-based services that caters to the need of both disabled and non-disabled communities. Through internet users can remotely participate in a wide range of activities such as lifelong education, professional growth and development, economics, government services and consumer activities. Websites and mobile devices have experienced a gigantic boom in popularity and prevalence over the past two decades and are expected to impact more to the socio-economic participation of society.

ICT initiatives in inclusive educational setting

Inclusive education is not just a process of dumping the disabled students in regular schools rather a bondage built through cooperation and collaboration between the disabled and nondisabled students in a normal classroom regardless of their strength and weakness. Education for students with disabilities has long been a serious cause of concern in India. Over the years policy makers have very diligently incorporated technology in the new educational approach i.e., inclusive. ICT is an umbrella term that includes communication device encompassing radio, tv, computer networks. The primary aim of ICT in the field of education is to cater to the needs of diverse learners.

Benefits:

- to accomplish one's task at its own pace
- motivates and develops confidence for ICT among the CWSN learners
- versatile education set up like distance education, satellite classrooms promote a better and a faster learning
- help learners in multiple means of engagement and representation
- simplifies the task of measuring learning performance
- low-cost building technology infrastructure

The innovation of the various assistive technologies like **Speaking dictionary, Hemingway app, Jolly phonics, Ginger** have improvised the learning process among the disabled learners, yet these technologies are under process to come to the reality surface.

The *ICT Policy Mission 2009* is a devise that supports and sustains ICT enabled activities and processes to improve access, quality, and efficiency in the school system. It ensures quality and excellence of global standards by providing up-to-date ICT facilities like dedicated internet, enough computers, LCD projectors, virtual classrooms etc. for effective and efficient teaching learning process. The policy seeks to achieve twin goals by bringing the power of ICT in reality:

- harnessing the capability of human resources of the whole of India
- emerge as a Global Hub and destination for IT services by 2020

The real impact of digitalization on Indian education system has been significant. It is a broad technical sphere that helps any student to attain and gain information from any corner of the country. India is progressing well towards digital education and acts as a supplement and doesn't completely overpower physical mode of education. The four main quadrants of Digital Education System in India are given in figure 2.

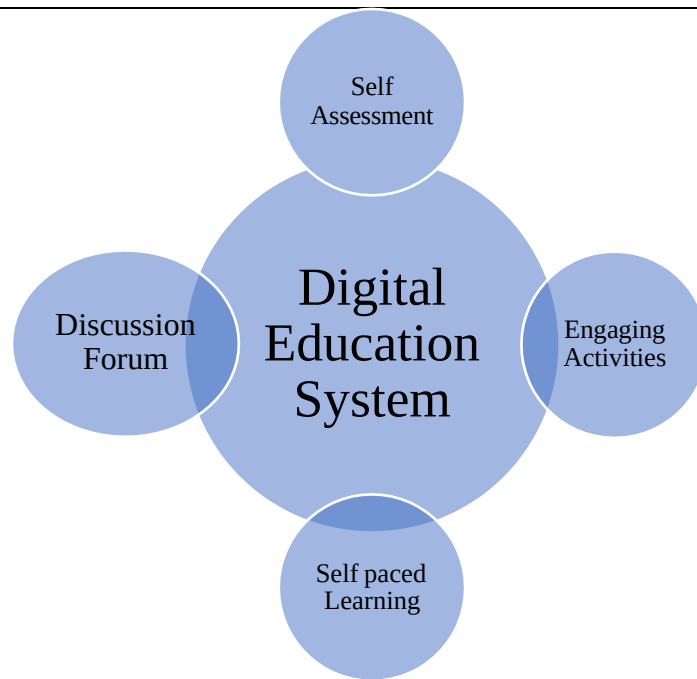


Figure 2: Four main quadrants of Digital Education System in India

The various digital initiatives taken by Indian Government to make education an inclusive one is as follows:

- **e-PG Path Shala-** it is an initiative of the MHRD under its National Mission on Education through ICT and maintains high quality content, curriculum based and interactive e-content in 70 subjects across all disciplines.
- **PM e-Vidya-** it aims to deliver high quality education to all students in the country. It helps students who don't have access to internet. It even provides special type of e-contents for visual and hearing-impaired students.
- **DIKSHA portal** – it was created as a digital medium for all the country's teacher. It was initially started to train teachers through online mode. Apart from it reading materials are available for both teachers and students. It ensures quality and continuity of education.
- **National Digital library (NDL)** provides access to a large number of e- resources which facilitates learners in a long way. It provides free access to many books and is designed to hold content in at least 10 most widely used Indian languages. Many

universities are even collaborating with other libraries in and outside India and promotes inter library facility for both normal and visually impaired students. courses.

- **SWAYAM (study webs of Active Learning for Young Aspiring Minds)**- it is India's own MOOCs platform offering free online courses on all most all disciplines. Integrating this online mode with conventional education, the learning outcomes of students are expected to improve in coming days.
- **SWAYAM PRABHA**- it is an initiative to provide 32 high quality educational channels through DTH across the length and breadth of the country on 24x7 basis. It enables us to deliver e-education in a most cost effective and inclusive manner.
- **FOSSEE (Free/Libre and Open-Source Software in Education)** – it aims to reduce dependency on proprietary software in educational institutions. It encourages students and teachers to use them in education and in research.
- **IRINS (Indian Research Information Network System)**- it is a free software service developed by INFLIBNET that provides an opportunity to create scholarly network.

All these are some initiatives which has been undertaken by the Government of India from time to time and is based on three principles- access, equity, and equality. Its main aim is to enhance holistic advancement through quality education and to take the best teaching learning resources to all, including the most disadvantaged.

ICT initiatives towards inclusive professional empowerment:

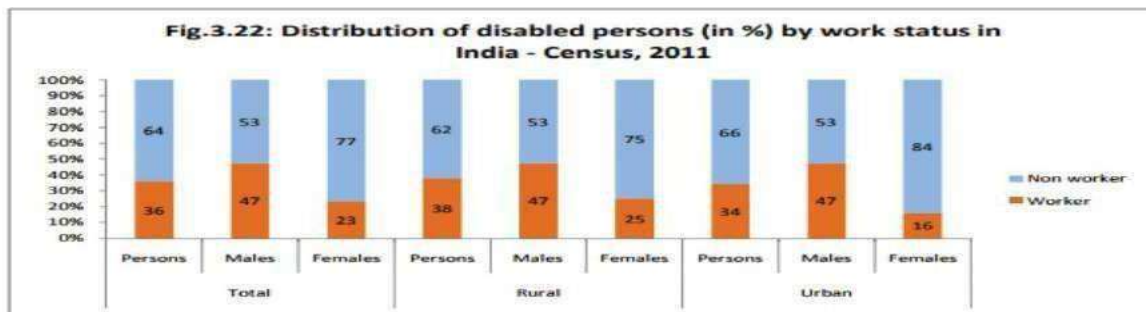
ICT is continuously developing and transforming human life into a more informative, knowledgeable, and productive one. The Indian government is playing a vital role in restoring the ICT enabled solutions adapted to the needs of persons with disabilities, thereby increasing the availability and accessibility of assistive technologies in social, economic, professional, and other domains. It aims to offer a real opportunity for social inclusiveness. The table 1 below gives us a clear picture about the literacy rate among the disabled in India.

Table 1: The literacy rate among the disabled in India

Educational Level	Persons	Males	Females
Illiterate	1,21,96,641	56,40,240	65,56,401
Literate	1,46,18,353	9,34,835	52,70,000
(i) Literate but below Primary	28,40,345	17,06,441	11,33,904
(ii) Primary but below middle	35,54,858	21,95,933	13,58,925
(iii) Middle but below Matric/ Secondary	24,48,070	16,16,539	8,31,531
(iv) Matric / Secondary but below Graduate	34,48,650	23,30,080	11,18,570
(v) Graduate and above	12,46,857	8,39,702	4,07,155
Total	2,68,14,994	1,49,88,593	1,18,26,401

* Source: Office of the Registrar General & Census Commissioner, India

Figure 3: Working status of the disabled as per census 2011



Source: Census of India, 2011

As per Census 2011 given in Fig (3.2.2) about 36% of the persons with disabilities are working (Male-47% and Female23%). Among the workers with disabilities 31% constitute agricultural labourers, 50% of PWDs are engaged in service sector whereas 4% of children with disabilities below the age of 14 are engaged in various manual labour.

Skill Development

As India moves progressively towards becoming a global knowledge-based economy, it must meet the rising aspirations of its youth. It can partially be achieved through focus on advancement of skills that are relevant to the emerging economic environment. The challenge pertains not only to a huge quantitative expansion in skill training for youth, but also to the much

more important task of raising their quality. With a goal to create opportunities, space, and scope for the development of the talents of the Indian youth and to enhance their technical expertise ICT focuses on Youth Skill Development as one of its pillars.

Smart India Hackathon is a nation-wide initiative that offers students and especially engineering young minds a platform to think creatively and outside the box and inculcate a culture of product innovation and mindset of problem solving. This would help the private sector to hire the best mindset across the nation. It is a platform to harness creativity and crowd sourcing intelligence for improving governance and quality life. It is a bright future commitment to greater diversity promoting inclusive society.

Finding a suitable job opportunity for a graduating student with a disability was once a big challenge but now we even have private companies like IBM and Wipro who are offering equal opportunities to all for living up to their potential. Such an approach helps towards a wider sensitization and inclusivity. **e-Kalpa** is an initiative by the ministry of human resources as part of National Mission in Education through Information and Communication Technology (NMEICT). It offers digital learning environment for design. These courses have been designed in a manner that offers independence and also as an aid to self-help group.

Scheme for Higher Education Youth in Apprenticeship and Skills (SHREYAS) the objective behind this scheme is to provide industry apprenticeship opportunities to youth and will enable them to get gainful employment and contribute to country's progress. Its functional aim is to establish a close link between education and industry/service sectors on a sustainable basis. The main purpose of this scheme is to establish an 'earn while you learn' system into higher education.

Faculty Development:

In 2019 MHRD launched an **Annual Refresher Program in Teaching (ARPIT)**, an initiative for online professional development of higher educational faculty using the MOOCS platform SWAYAM. The main objective behind this effort is to take the best teaching learning resources to the most disadvantaged sections of society. It has revolutionized professional development of

faculty thereby furnishing to large target audiences by accelerating ICT and online technology platform SWAYAM.

Leadership for Academicians Program (LEAP) was launched in 2019 which aims to build higher managerial capabilities of existing higher education leaders and administrators and to draw fresh talent into the management of higher education systems. It was launched to design and deliver a structured scheme of leadership development for higher educational institutions.

National Initiative for School Heads and Teachers Holistic Advancement (NISHTHA) is a capacity building program for improving quality of school education through Integrated Teacher Training. It aims to build competencies among all teachers and school principals at the elementary stage. The functionaries are trained in an integrated manner on learning outcomes, school-based assessment, new initiatives in education addressing diverse needs of children through multiple pedagogies.

Intel India has been working with various Central and State education bodies offering programs in higher education teacher training, informal education, and science promotion. The Intel Teach Program is a professional development program that helps teachers across India enhance 21st century learning through the effective use of ICT.

Entrepreneurship Development: With the gradual development of new innovative sectors India should aim to develop an ecosystem that encourages innovation across the industrial spectrum. The heavy loss incurred in the economy during the pandemic phase India must rebalance its economy through domestic consumption, import demand and higher value-based activity and hence entrepreneurship is vital for the future of the developing countries. ICT plays a vital role in promoting sustainable enterprises and inclusive based society.

Entrepreneurship and Skill Development Programs (ESDP) is one such program that aims to motivate youth from different sections of society and considers entrepreneurship as a career option. It enhances awareness and knowledge of participants about various entrepreneurial aspects, skill domains and policies of the government.

Scheme of Fund for Regeneration of Traditional Industries (SFURTI) its main aim is to provide sustained employment for traditional industry artisans and rural entrepreneurs. It will not

only provide skill development training and capacity building but also focuses on brand building, e-Commerce initiatives and innovation, and new media marketing.

ICT Initiatives Towards Inclusive Empowerment in Tertiary Sector:

Digital competency and mobile learning are conceptualized as the most significant factor for digital social inclusion and realizations of citizens for nations towards a modern knowledge-based society. Internet, mobiles, Tv sets are continuously contributing towards social inclusion. These technologies have provided an opportunity to persons with disabilities to grow and empower themselves to become a part of the mainstream section of society.

Internet as an assistive technology for people with disabilities builds their capacity to communicate with each other at a distance. The various web-based application like screen reader, speech recognition, voice to text message helps them to gather and understand public information and news, participate in leisure interests with others and write to authorities and friends; thus, empowering them to get access to socio cultural activities. Mobile technology enables the accessibility of everyone including persons with disabilities to gain services at the time of need thus unleashing huge potentials for independent living. Technologies ought to be designed to be inclusive so that they can facilitate greater inclusion and universal accessibility. Apps like **Wheel Map** and **Be My Eyes** help the users to connect with volunteer who can aid them with everyday task. Apart from it there are sophisticated apps like **Google Fit**, **Apple Health**, **Quardio** ensures independence to disabled people and gives the option to take charge of their difficult situation. Digital TV have expanded its features so that it becomes viable to disabled people also. The unification of sign language interpretation or voice over audio channels sets an example of digital inclusion.

e- Banking and e-Post office system have also stretched their arms in adopting inclusivity. Inclusive digital banking means getting access and use of financial services like websites regardless of differences or disability. It offers greater flexibility, choice, and personalization. Banks have designed their websites and apps following inclusive principles. Modernization of post offices and banks is enhanced through central server-based system using computerization.

e- GOVERNANCE for e-INCLUSION e-Governance is the application of ICT to government functions and procedures with the purpose of increasing efficiency, transparency, and citizen participation. Globally e-Governance system has evolved towards wider acceptance. Almost all the countries have embraced e- governance as part of their long-term policy. e- Governance and e- inclusion are complementary services that would help to overcome all the social political barriers and foster social development. Some of the major initiatives related to public service delivery are as follows:

- Common Service center
- UMANG (unified mobile application for new age governance)
- e-District Mission Mode Project
- UPI (Unified Payment Interface)
- Co-win
- My Govt
- e-Seva

e-Governance is a gateway to social inclusion and transparency. In the face of widespread inequality and widening social gap the government aims to create a variety of condition for the growth of citizens through good governance, a pre requisite for human well beings and development. It aims to uplift all sections of society by fixing accountability, transparency and making the process more inclusive.

Major Challenges

In recent times the use of ICT has been deeply interwoven into the fabric of society and has become a crucial element for personal development and institutional empowerment for inclusive social progress and economic growth, yet there are major barriers and challenges that hinders the progress of the society from becoming all-inclusive which are as follows:

- the cost of assistive technologies is still a significant barrier that prevents persons with disabilities from fully accessing health care services, educational benefit and living life independently.
- Limited accessibility and use of ICT remains a significant barrier which is observable in all areas of development

- Lack of effective implementation of policies that benefits the disabled
- The challenges to promote the employment of persons with disabilities are concerned attitudinal barriers are still highly prevalent in the work place.
- Many e- governance projects are based on PPP mode, hence it's a challenge to integrate the professional approach of the private sector with the social concern of the government.
- The non-availability of digital services in local languages and lack of proper internet services is still a major concern in rural areas. There should be adequate awareness building program so that they can be a part of it and reap the benefit of internet.

Conclusion

The purpose of building inclusive societies is to foster harmony among all persons by enabling persons with disabilities to be integrated in mainstream environment. To maximize the potential benefit of ICT it requires a proper understanding of the full range of challenges and barriers faced by persons with disabilities in each local context. We should work to ensure that societies are open and inclusive to all. Inclusion requires change in strategies, approaches and policies so that needs and priorities of all are recognized from the initial stages of every decision-making process. We should aim at advancing inclusion models that help to improve access to the core rights and participation in societies. Accessibility to ICT is the final step of inclusion ensuring that the services are available at the door step and thereby mitigating all kinds of hurdles.

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**Use of ICT and Edmodo as Learning tool: A Correlational study among Teacher
Trainees**

Sangeeta Suman*

Abstract

Information and communication technology (ICT) plays a key role in Education at present era. Everyone recognizes use of ICT at their own purpose. ICT gives opportunity to learn the required global 21st century skills. ICT helps teacher to teach attractive and effective content. Edmodo is a learning tool which is like Google-classroom and Facebook. Edmodo is the platform where learning takes place with less effort. This study aims to investigate the correlation between the ICT user and user of Edmodo. The sample for the study is 76 pupil teachers. These pupil teachers are user of Edmodo. The researcher administered two self-constructed tools of Use of ICT and Edmodo as a learning tool. The study focuses on correlation between them.

Keywords: *Communication, Edmodo, Perceptions of Pupil-teacher, Virtual classroom.*

Introduction

Learning is the core of Education. Without learning, Education is worthless. Learning is essential element of education. Learning facilitates through technology. Researches indicate that integration of technology in learning makes learning more effective. During COVID-19 situation many students start learning through using various technologies in their teaching learning. Integration of technology in teaching learning found suitable for learners (Ratheeswari,2018).Pedagogical innovations with ICT can improve academic achievement of students at exceptionally low cost (Kremer and Holla,2008). Technological integration of education with help of ICT can mitigate weakness of teaching and helps students to learn at their own interest.

Edmodo is a learning tool used for the purpose of teaching. Edmodo is used as facilitator between teacher-student. Edmodo has a specific feature to connect parents as well. This study is based on the pupil teacher using Edmodo as their virtual classroom. Primitively, researcher

conducted a survey, based on use of ICT. After teaching via Edmodo conduct another survey based on Edmodo as learning tool. The purpose of the study is to highlight that use of ICT and Perception of Edmodo as a tool of learning is how much correlated to each other? Does any relationship exist between the use of them?

Research Questions

1. What is the level of pupil teachers using ICT in their teaching-learning process?
2. What are the perceptions of pupil teachers towards Edmodo as learning tool?
3. What is the relationship between use of ICT and perceptions of Edmodo as learning tool?

Objectives

1. To study the pupil teachers level of ICT awareness.
2. To study the pupil teachers' level of perception towards Edmodo as learning tool.
3. To find out the relationship between ICT Awareness and perceptions of Edmodo as a learning tool.

Methodology

The study is adopting descriptive survey and data analysis through quantitative approach. For this study 76 pupil teachers are selected randomly. The set criteria of the selection are pupil teacher must have to use Edmodo as teaching tool. Two survey tools were designed by researcher, first tool to measure use of ICT in daily life. There were approximately 40 items which covered the dimensions Hardware, Software and online resources uses. The Cronbach alpha value for this tool is .930 which is excellent. Second tool is designed to measure Perception of Edmodo as a learning tool, in which total 15 items were included. The Cronbach alpha value for this tool is .869 which is good and reliable for conducting research.

Sample description

In this study 62 males (81.6%) and 14 females (18.4%) pupil teacher were participated. Out of which 39 Rural (51.3%) and 37 Urban (48.7%) are the residence they belong from. They were also from different pedagogies i.e., 48 Science/Maths/Computer (63.2%) and 28 Social Science/Commerce (36.8%). The pupil teachers were using Edmodo after taking some awareness about tool by the Researcher.

Data analysis, Findings and Discussions

Objective 1

- To study the pupil teacher level of ICT awareness is defined in Table 1 is as below:

Table 1: Variables for use of ICT tool

	Variables	N	Mean	SD
Gender	Male	62	129	27.11
	Female	14	118	16.13
Residence	Rural	39	126	27.60
	Urban	37	129	24.00
Pedagogy	Science/Maths/Computer	48	129	26.32
	Social Science/Commerce	28	125	25.00

The mean value and standard deviation can be found in Table 1 the mean for male pupil teacher is 129 and standard deviation is 27.11 and for female pupil teacher the mean value is 118 and 16.13 standard deviation. The variation found in mean and standard deviation on basis of residence is 126 mean and 27.60 standard deviation for rural and 129 mean and 24.00 standard deviation for urban. The Pedagogy wise variation shows 129 mean and 26.32 standard deviation for Science/Maths/Computer and 125 mean and 25.00 standard deviation for Social Science/Commerce.

Table 2: The Scale for calculation

Level of ICT use	The score Range	Frequency
Extremely high	200-176	4
High	175- 142	18
Moderate	141-108	35
Low	107- 74	18
Very low	73- 40	1

The scale used for measuring the level of ICT use can be viewed in Table 2. The score of 200-176 represents very high score and frequency of this score is 4; 175-142 represents high score and frequency of this score is 18 where as 141-108 represents moderate score and maximum frequency i.e., 35 approx half, found in this range. While low score range is 107-74 and

frequency of score is also 18. A score range which is very low score range is 73-40 and minimum frequency found in this range i.e., 1.

Table 3: The Level of ICT use

Variables	N=76
Level of ICT use(Mean)	127.67
Minimum	73
Maximum	194
Standard Deviation	25.86

Table 3 shows the levels in use of ICT among pupil teacher and found it through survey is 127.67 where minimum value is 73 and maximum value is 194. Their standard deviation value is 25.86. Therefore, it can be considered that level of ICT use is moderate among them. Pupil teacher have basic and moderate knowledge of ICT tools. They were not too techno savvy.

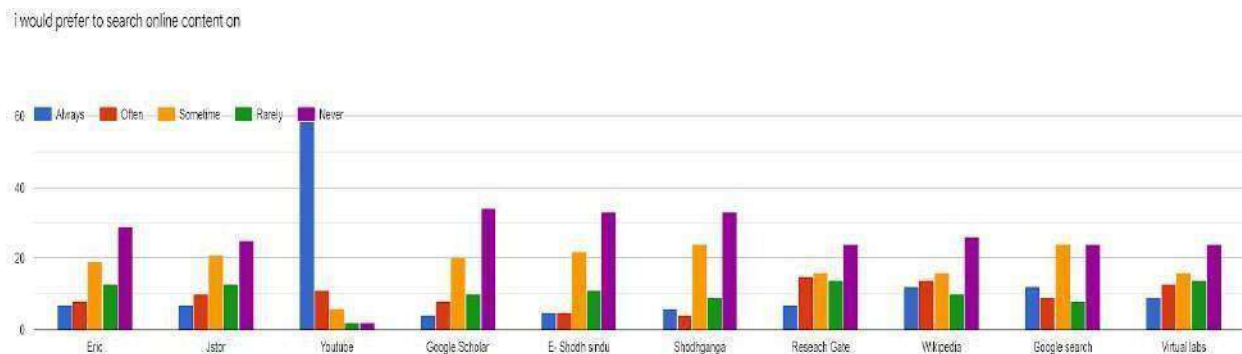


Figure 1: Use of Online search resources by Pupil teacher

In this item, as shown in Figure 1, total 76 pupil teachers were giving their response with how they search their online content? via Eric, Jstor, Youtube, Google Scholar, E-shodhsindhu, Shodhganga, Research Gate, Wikipedia, Google search, Virtual Labs like online platforms. For every search, rating is based on likert scale (Always, Often, Sometimes, Rarely, Never). Rating on Always maximum found in Youtube(59), Google search(12) and Wikipedia(12).

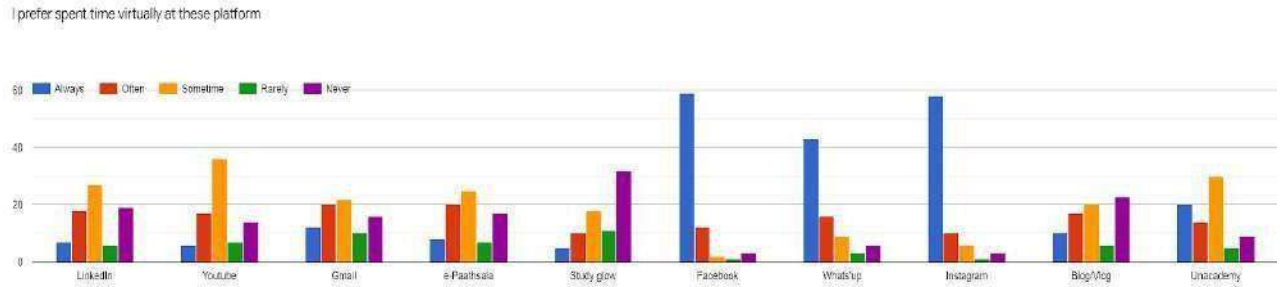


Figure 2: Use of Online platforms by pupil teacher

In this item, as shown in Figure 2 total 76 pupil teacher were express their views in they like to prefer spent their time virtually on LinkedIn, Youtube, Gmail, E-paathshala, Study glow, Facebook, What's up, Instagram, Blog/Vlog, Unacademy. For each search, rating is on likert scale (Always, Often, Sometimes, Rarely, Never). Rating maximum on Always in Using Facebook(59) and Instagram(58) and then on What's up(43). It shows that they love to spend their time on social media platform.

Objective 2

To study the pupil teachers level of perception towards Edmodo as learning tool discuss as:

Table 4: Variables in Perceptions of Edmodo as a learning tool

	Variables	N	Mean	SD
Gender	Male	62	59.94	6.31
	Female	14	58.43	5.19
Residence	Rural	39	59.97	7.25
	Urban	37	59.32	4.72
Pedagogy	Science/Maths/Computer	48	59.55	6.19
	Social Science/Commerce	28	59.85	6.09

The value of mean and standard deviation(SD) of pupil teacher perception of using Edmodo as a tool of learning is shown in Table 4 for male the mean value is 59.94 and standard deviation is 6.31. and for female mean is 58.43 and standard deviation is 5.19. The mean and standard deviation for Rural mean value is 59.97 and SD 7.25 and for Urban mean value is 59.32 and SD is 4.72. Calculation of mean and SD for pedagogy wise mean for Science/Maths/Computer is

59.55 and SD value is 6.19 and for Social Science/Commerce mean value is 59.85 and SD is 6.09.

Table 5: The Scale to measure Level of Perception of Edmodo as a tool for learning

Level of Perceptions	The score Range	Frequency
High	55- 75	61
Moderate	54- 35	15
Low	34 -15	0

This score range is used to measure level of Perception of Edmodo as a learning tool in Table 5. The score range from 55- 75 represents high level of perception and found maximum frequency 61 and score range from 54-35 represents moderate perception and frequency is 15. While from score range 34-15 there is no frequency found.

Table 6: Level of Perceptions of Edmodo as a learning tool

Variables	N=76
Level of Perceptions of Edmodo(Mean)	59.65
Minimum	45
Maximum	71
Standard Deviation	6.12

Table 6 shows that the level of Perception of Edmodo as learning tool is high, i.e., 59.65 from the above study data reveals that the minimum value is 45 and maximum value is 71. The standard deviation is 6.12. The conclusion of the study is there is high level of perception of Edmodo as a learning tool in pupil teacher.

Objective 3

To find out relationship between Hardware one of the dimensions of ICT use and perceptions of Edmodo as a learning tool discuss as:

In this the Correlational value is .671 which means moderate positive correlation exists between Hardware, one of the dimensions of use of ICT and Perception of Edmodo as a learning tool.

Table 7: Correlation between Hardware and Perception of Edmodo as a learning tool

		Hardware	Level of Perception for Edmodo use
Hardware	Pearson Correlation	1	.671**
	Sig. (2- tailed)		.000
	N	76	76
Level of Perception for Edmodo use	Pearson Correlation	.671**	1
	Sig. (2- tailed)	.000	
	N	76	76

To find out relationship between Software one of the dimensions of ICT use and perceptions of Edmodo as a learning tool discuss as:

Table 8: Correlation between Software and Perception of Edmodo as a Learning tool

		Software	Level of Perception for Edmodo use
Software	Pearson Correlation	1	.632**
	Sig. (2- tailed)		.000
	N	76	76
Level of Perception for Edmodo use	Pearson Correlation	.632**	1
	Sig. (2- tailed)	.000	
	N	76	76

In this the Correlational value is .632 means moderate positive correlation exist between Software, one of the dimensions of use of ICT and Perception of Edmodo as a learning tool at .000 significant level.

To find out relationship between Online resources one of the dimensions of ICT use and perceptions of Edmodo as a learning tool discuss as:

Table 9: Correlation between Use of Online resources and Perception of Edmodo as a learning tool

		Use of Online resources	Level of Perception for Edmodo use
Use of Online resources	Pearson Correlation	1	.831**
	Sig. (2- tailed)		.016
	N	76	76
Level of Perception for Edmodo use	Pearson Correlation	.831**	1
	Sig. (2- tailed)	.016	
	N	76	76

In this the Correlational value is .632 means moderate positive correlation exist between Online resources, one of the dimensions of use of ICT and Perception of Edmodo as a learning tool at .000 significant level.

To find out the relationship between ICT Use and perceptions of Edmodo as a learning tool discuss as:

Table 10: Correlation between use of ICT and Perception of Edmodo as a learning tool

		Use of ICT	Level of Perception for Edmodo use
Use of ICT	Pearson Correlation	1	.890**
	Sig. (2- tailed)		.000
	N	76	76
Level of Perception for Edmodo use	Pearson Correlation	.890**	1
	Sig. (2- tailed)	.000	
	N	76	76

There is high positive correlation between the two variables i.e., Use of ICT and Perception of Edmodo as a tool of learning and the significance at the .000 level. The value of correlation is .890 which is considered highly positive correlation.

Conclusion

ICT in education plays a vital role in society to get sustainable development. Teaching-learning becomes more effective and interesting with the help of ICT. In this study researcher has found the moderate level of ICT user can also use and like Edmodo as learning tool. Although, there is a correlation between ICT user and Perception of Edmodo as a learning tool, it represents the person who is more techno-friendly with ICT tools can use and like Edmodo more than others. Therefore, it can be says that pupil teacher having moderate level of ICT knowledge and use can have high level of perception towards Edmodo as a tool of learning.

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**Effectiveness of Vedic Mathematics-based Instruction on the Achievement in
Mathematics based on Gender.**

Yogeshwari. S* H. Indu**

Abstract

There exists a problem of learning Mathematics with not only the students but all the folks learning mathematics. Intricate and complex formulas and the skeptical series of methods to solve a problem might be the main reason for this problem of learning mathematics. To eradicate this fear, plenty of easy methods are being brought up by experts. Vedic Mathematics which is a combination of verses and sutras which was persuaded by ancient Indians and was believed to be a competent system of mathematics. This Vedic mathematics is being encased by the experts for the current generation to ease learning and minimize the fear of learning mathematics. This paper is focused on the effectiveness of Vedic mathematics on achievement in mathematics based on gender. The study involves 160 third-grade students, involving two groups, one group taught through the conventional method (control group) and the other taught through Vedic mathematics (experimental group). A 'Pre-test and post-test nonequivalent control group design' were used in this study. The Vedic method applied in the experiment group employed for girls and boys separately indicates the group taught through Vedic mathematics obtained significantly higher mean achievement.

Keywords: Vedic mathematics, Achievement in mathematics, III grade students.

Introduction

Education is the platform to bring out the potential inherent in man and makes him much more productive in his actions. Mathematics plays a vital part in the field of education and acts as the basis for logical reasoning, abstract thinking, and the development of reasoning. Mathematics is the foundation for all other subjects, so learning mathematics is more important. Learning mathematics mainly depends on the methods followed in teaching and learning as well. "The

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school curriculum aims to enable learners to acquire knowledge, develop understanding and inculcation of skills, positive attitude, values, and habits conducive to the all-round development of their personality” (National curriculum framework for school education,2005).The present school curriculum is framed such that it acquires all the values given above but it can also focus on the various methods that can be used to teach and joyfully learn mathematics. Most of the students consider mathematics as a difficult subject as it takes more time, needs more patience and it takes more effort to solve problems. For these reasons, most of the students started to ignore learning mathematics. Vedic mathematics is an ancient treasure that makes learning mathematics easier and quicker. This study analyzed the effectiveness of Vedic mathematics on achievement in mathematics among boys and girls third-grade students.

Vedic Mathematics

Vedic mathematics was reframed by Indian monk Jagadguru Swami Sri Bharati Krishna Tirthaji Maharaj.Vedic mathematics involves sixteen sutras and thirteen sub-sutras through which all mathematical calculations can be performed cheerfully. Mathematical problems which involve many formulas and numerous steps can be solved in a few steps by Vedic mathematics. In addition to basic arithmetic operations, higher mathematics which includes algebra, trigonometry, calculus, integration, differentiation, and many others can also be solved using those sixteen Vedic mathematics sutras cheerfully and blissfully. Since this study involves only third-grade students only basic arithmetic operations such as addition, subtraction, and multiplication are taught to the children. **Smitha (2014)** conducted research focusing on the ancient Indian Treasure of Vedic Algorithm, the ‘Anurupyena’ method. The study consisted of 240 secondary students selected based on the cluster sampling technique. The results found that the Vedic sutra, ‘Anurupyena’ used for calculating cubic values is more efficient in increasing the speed of calculation among the students, and the data shows that the applications of ‘Anurupyena’sutra is more useful than the traditional method of teaching mathematics in improving computation speed. **Rajesh (2015)** attempted to analyze the efficiency of teaching mathematics through the Vedic method in the achievement of secondary students. The study consisted of 200 students as the sample, taking 50 boys and 50 girls from class 9, and the same number of samples including both genders from class 10 was taken for the study. The schools were identified purposively, and the groups of the experiments were selected using the method

of randomization. The researcher selected a stratified random sampling technique. This study revealed that boys of standards 9th and 10th acquired high mathematics achievement through the Vedic mathematics-assisted teaching method similarly girls have acquired high achievement. **Dhivyadeepa (2014)** investigated the effect of the Vedic method of teaching mathematics in performing subtraction. This study involved 24 students in grade four. The parallel group experimental method was used in the study. Research self-made achievement test was given for pre and post-tests. The study revealed that the achievement means scores between the pre-test and post-test of the controlled group have no significant difference. It was also found that the difference in the mean score achievement between the pre and post-test of the experimental group was significant, and it also showed a difference in achievement scores between post-tests of the experimental and controlled group significantly. **Vyas (2020)** has tried to find the effect of teaching mathematics through Vedic methods taking into consideration a few variables for ninth-grade school students. Post-test equivalent group experimental design was adopted in the research. A sample consisting of 160 students was selected for the study and 40 students in each group were chosen. It was found that the Vedic method of teaching mathematics was more efficient than the conventional method in the student's achievement in mathematics subjects and in the achievement of girls in mathematics subjects. It was also found that Vedic mathematics was more effective for lower-achievement students in mathematics. The researcher also found that the Vedic mathematics method was more effective in the achievement of boys than girls. **Amulya (2021)** tried to find the effectiveness of the Vedic method on multiplication for sixth graders. The researcher used the Vedic method of multiplication as an independent variable to know the effect on the achievement of students in an experimental setting of 58 students of class vi who were selected randomly by using the blindfold method from four upper primary schools. The schools were selected using the purposive sampling method. 28 students were taken and taught through the conventional method (control group) and 30 students were chosen and taught through the Vedic method (experimental group). A randomized pre-test and post-test equivalent group design were chosen in the research. Two self-made achievement tests on multiplication have been used one each for the pre and post-test. The results showed that the Vedic method of multiplication is effective while considering the achievement of students over the conventional method in the post-test. It was also revealed that the boys and girls in the experimental group had

achieved high in the post-test and the difference between the two means had no significant difference.

Statement of the Problem

The researcher tried to find whether there is more effectiveness in teaching mathematics through the Vedic method based on the sub-sample gender. The researcher also tries to know the differential effect of gender on achievement in mathematics for both the conventional method of teaching and Vedic Mathematics-based instruction. Hence, the researcher framed the title of the problem as ‘Effectiveness of Vedic mathematics-based instruction on the achievement in mathematics based on gender.’

Need of the Study

Vedic mathematics redefines the negative perspective of most learners towards learning mathematics. Vedic mathematics consists of sutras that help in solving problems in an easier and quicker method. Through Vedic methods, the problems are solved in very few steps, and this automatically increases the confidence level of the students, and they feel fear-free in learning mathematics which will help in making mathematics learning, an interesting and joyful one.

Objectives of the research

1. To test the effect of Vedic mathematics-based instruction on achievement in mathematics of the students taught through the conventional methods and those who were taught through Vedic methods for sample employed for girls and boys separately.
2. To compare the mean achievement post-test score in mathematics of third-grade students based on their gender for both the students taught through the conventional method and those who were taught through Vedic methods for sample employed for girls and boys separately.

The Hypothesis of the Study

1. Vedic mathematics-based instruction does not affect the achievement in mathematics of the students taught through the conventional methods and those who were taught through Vedic methods for sample employed for girls and boys separately.
2. There exists no significant difference in the mean post-test score of achievement in mathematics of third-grade students based on their gender for both the control and experimental group separately.

Sample Selected

The samples selected for the study were 160 students from the third grade of which 80 were taken as the experimental group and 80 as the controlled group. The sample chosen for the study is given in Table 1.

Table 1: Sample Distribution of the Study

Type of school	Experimental group	Controlled group	Total
Boys	39	25	64
Girls	41	55	96
Total sample	80	80	160

Tools

In this study, the researcher framed and validated a Vedic mathematics-based instruction package. An achievement test in mathematics was also constructed and standardized. The researcher also established the reliability and content validity of the prepared tool. The achievement test consists of 40 items initially. Based on the expert's opinion, the items were reduced to 32 items. The pilot study was done with a sample of 100 samples of third-standard. Based on item analysis and item discrimination few items were deleted and the remaining 20 items were considered for the study. Vedic mathematics-based instruction module was constructed based on the Tamil Nadu state board syllabus prescribed for the third grade. The module unambiguously explains the steps followed in the Vedic method to calculate the basic arithmetic operation, addition, subtraction, and multiplication.

Analysis of the Data

1. **Hypothesis 1:** Vedic mathematics-based instruction has no effect on the achievement in mathematics of the students taught through the conventional methods and those taught through Vedic methods for sample employed for girls and boys separately.

The investigator, to know whether the Vedic Math method was effective for female and male students separately employed an independent sample t-test. The result of the statistical analysis is presented in Table 2.

Table 2: The Significance test of Difference in the Mean Post-test Scores of Achievement Test of the Control and Experimental Group for the subsample Gender

Sub Sample	Variable	Group	<i>n</i>	<i>M</i>	<i>SD</i>		<i>p</i>
Girls	Achievement	Control	5	11.89	2.63	7.014*	$p < 0.0001$
		Experimental	4	16.16	2.25		
Boys	Achievement	Control	2	11.16	2.21	6.965*	$p < 0.0001$
		Experimental	3	14.89	2.22		

Note. ** denotes the value is significant at the .01 level

n= no. of students, M=Mean, SD = Standard Deviation, r = coefficient of correlation, t= t value, p = probability

As per Table 2, the *t* value obtained for the significance test of the difference in the post-test score of achievement in mathematics of the students taught through the conventional methods and those were taught through Vedic methods for the subsample girl is 7.014, which is said to be more than the table value 1.96. Therefore, the difference is significant.

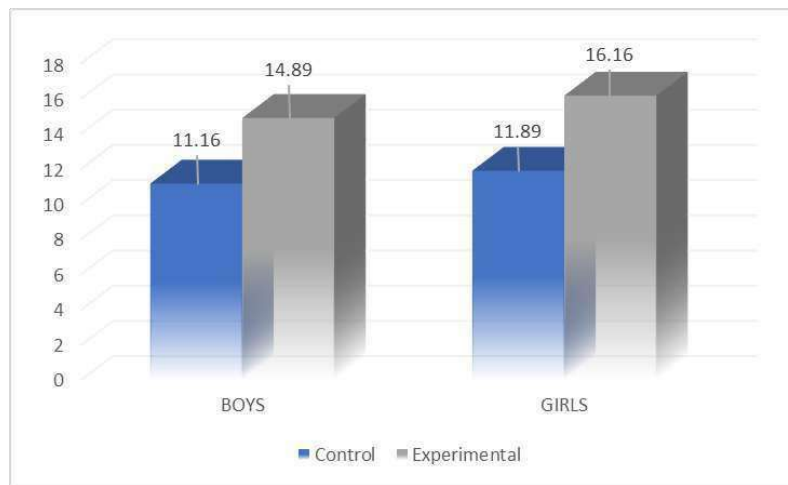
The significance test of the difference in the post-test achievement score of the students taught through the conventional methods and the students taught through the Vedic methods for male students gives a *t* value of 6.965 which is greater than 1.96 and hence significant.

Discussion

The significance test of the difference in the mean post-test value of achievement between the students taught through the conventional method (control group) and the students taught through the Vedic method (experimental group) employed for girls and boys separately shows that the students taught through the Vedic method obtained significantly greater mean achievement scores in the post-test. This denotes that the Vedic method applied in the III grade experiment

group has a considerable effect on mathematics achievement. Hence Hypothesis 1 is rejected.

Figure 1: Graphical Representation of the Difference in the Mean Score of Achievement In Mathematics of Control And Experimental Group For The Sub-Sample Classified Based on Gender



Hypothesis 2: There exists no significant difference in the mean post-test score of achievement in mathematics of third-grade students based on their gender for the control and experimental group separately.

The investigator tried to find whether the difference in the mean post-test score of achievement in mathematics differs with gender. The analysis result is presented in Table 3.

Table 3: The Test for Significant Difference in the Mean Post-test Scores of Achievements in Mathematics Test for the subsample Gender

Group	Variable	Sub Sample	<i>n</i>	<i>M</i>	<i>SD</i>		<i>p</i>
Control	Achievement	Boys	31	20.35	5.25	2.589*	0.011
		Girls	49	23.08	4.13		
Experiment	Achievement	Boys	55	24.05	4.90	5.173*	$p < 0.0001$

		Girls	25	17.60	5.73		
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Note. ** denotes the value is significant at the .01 level

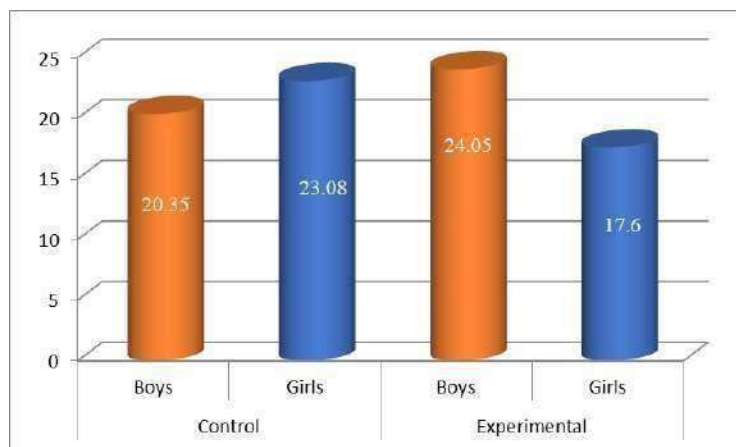
n= no. of students, M=Mean, SD = Standard Deviation, r = coefficient of correlation, t= t value, p = probability

Table 3 infers that the mean score obtained for boys in the control group is 20.35 (SD = 5.25) and for girls is 23.08 (4.13). The t value is obtained as 2.589 with a *p-value* of 0.011.

Table 3 also denotes the boys taught through Vedic mathematics have a mean post-test score of 24.05 with an SD of 4.90 and the boys obtained 17.60 (SD= 5.73). Meanwhile, the girls secured a mean score of 17.60 with an SD of 5.73. The value is found to be 5.173 at a .01 level of significance, with a p-value less than 0.0001. The obtained t value is higher than the critical ratio at a .01 confidence level. Therefore, the hypothesis is rejected.

The difference in the mean score in math achievement is represented with the help of a graph in Figure 2.

Figure 2: Difference in The Mean Score of Achievement In Mathematics Of Control And Experimental Group Based On Gender



Discussion of the Results

The t value obtained for the difference in the mean post-test score of achievement in mathematics of the girl students of the students taught through the conventional method (Control group) and the students taught through Vedic methods (Experimental group) is found to be (7.014) greater than the critical ratio at a .01 level. Hence the mean score of the girls from the

students taught through the Vedic method(experimental group) is greater than those of the students taught through the conventional method (control group).

The mean score of boys from the students taught through Vedic methods (Experimental group) is found to be significantly greater than the boys from the group taught through conventional methods (Control group). Since, the value of the test of significance, (6.965) is more than the table value at a .01 level of confidence.

The t value of the difference in the mean post-test score of the students taught through the conventional method (control group) for the sub-sample gender is obtained as 2.589 which is more than the table value at a .01 confidence level. The mean score of girls is found to be significantly greater than that of boys. At the same time, the boys from the group taught through Vedic mathematics (experimental group) are observed to be greater than the girls, with a t value of 5.173. The t-value denotes that the difference is significant at a 0.01 level of significance. Vedic mathematics is more effective in boys, denotes the analysis.

Findings of the Study

- The mean post-test scores of girls in the group taught through Vedic methods (experimental group) are significantly greater than the group taught through the traditional method (control group).
- The mean post-test scores of boys from the group taught through Vedic methods (experimental group) are significantly greater than the mean post-test score of the group taught through the conventional method (control group).
- The Vedic method of teaching mathematics is found to be significantly effective in contrast to the conventional method of mathematics teaching
- The post-test scores of girls from the group taught through the conventional method (control group) holds to be significantly greater than those of boys.
- The boys from the group taught through Vedic methods (Experimental group) have significantly greater mean post-test scores compared to the girls from the group taught through Vedic methods (Experimental group).
- The Vedic method of teaching Mathematics is more effective in boys when compared to the girls of the group taught through Vedic methods (Experimental group).

Conclusion

Vedic mathematics helps to solve problems in a much easier and quicker way. It allows the learners to understand the problem deeply and facilitate solving problems in less time. Even the most complex problems can be solved efficiently through Vedic mathematics. It increases the concentration of the students and improves their retention capacity. The inclusion of Vedic mathematics in the school curriculum can facilitate the learners to study mathematics in a fear-free atmosphere. This will increase the achievement level of the students and help them stay confident.

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A Creative Life-skills Training for Rural Girls

A. Jayasree* N. Murugeswari**

Abstract

“The soul of India lives in its villages,” the statement of Gandhiji is relevant even after seventy-five years of independence of India. Gender plays a crucial role in the life of all girls universally. Even then, opportunities are plenty in this digital era for urban girls. But life-skill support is necessary for those in rural areas to catalyse their empowerment in all fields. Thus presenting a minute model for converting the attitude of rural girls to positive premises with creative life-skills training is the prime purpose of this investigation. Education has given a lot to Indian women theoretically. But failure to have an empathetic and happy life with self-awareness leads them to various addictions. The glass ceiling also persists in peer groups with negative attitudes to all. Learning new behaviour suggested by Bandura gains importance here. These point to the objective of regulating habits from budding ages using colourful pictures to attract them to life-skill ideas. Therefore, ‘selected adolescent rural girls were given training using a module with digital paintings on life skills by the researcher’ in central Kerala. Data collected with questionnaires before and after the training and analysis proved its effectiveness. Suggestions are given on the scope for further expansion and for including such life skill modules in the syllabus in all classes by policymakers of the nation.

Keywords: Rural girls, Life-skills, Paintings, Module, Syllabus.

Introduction

Nature and pictures can appeal to all. In this computer era different styles are available in popular data sources for training girls in rural areas. As colourful images can go deeper to the memory of viewers, a module 'using digital paintings drawn by the researcher on life skills' was used to train selected rural girls. This model is an endowment by this senior citizen for society

perplexed by calamities. As per Gandhian principles on villages, rural girls were chosen for the study. The extent of the research area was limited to selected adolescent girls **of the institution in central Kerala. Life-skill is a favourite field of study to develop positive attitudes in life, but limited studies can be accessed in surveys in the above arena.** Thus there is a scope for further expansion of exploration in the area. Here “Balance for better” the idea of International Women’s Day 2019 is viewed from an innovative angle to make a positive gender balance through mental balance.

Review of Related Literature

‘Women’s studies’ is a developing field with ample scope for innovative learning in digital media or life skills to empower women in such areas. Thus, to have an overall idea about previous works in those fields, popular data sources of magazines, newspapers, Internet documents, conference proceedings, books, government documents, dissertations, and theses accessible were examined. As the same topic on digital painting **and life-skills training for rural girls was not available, each part of it was reviewed separately. For example,** Sanjaya Mishra (2019) pointed out the importance of the use of Information and Communication Technology in diverse ways to learn or teach the needy. In their book, Scott S. J and Barrie Davenport (2018) explained how to stop anxiety, worrying, and negative thinking through deep breathing and meditation. Madhu Krishnani (2016) noted that, there was a significant relationship between creativity and adjustment. But school adjustment was high in urban students, who had easy access to modern technology. Rod Judkins R (2015) had several black and white illustrations in the book on creative thinking to discuss his topics. New National Education Policy, 2020 also regards skills, including life skills an indispensable part of instruction. CBSE has some of these skills in manuals for helping teachers. UGC has developed a curriculum for life skills. All these are not elaborate syllabi on a par with the main subjects of study. A single module by a researcher with digital paintings for teaching the life skills in class six was also not seen during this review.

Methodology

Education has to uplift the mental health as well as intellectual activities of a child. Compared to urban girls, educational and economic conditions of their rural counterparts need

much improvement. The diverse complexities of life skills, digital media, and solutions to issues of rural girls are not to discuss here. Only a module is presented for the adolescent girls using the experience by the researcher as a preliminary step to help the consented participants to understand selected generic life skills. A life skill is a skill useful to manage well in daily life and here it is the selected life skills for the study. Rural is the area connected with the selected countryside and painting is the art of using paint as seen in the Oxford dictionary. Paintings explain vision, ideas, thoughts, or hopes explicitly on canvas. Any digitalized content comes under the category of the digital medium and, here it is digital paintings used in the study.

UNICEF has described ten generic life skills like self-awareness, effective communication, empathy, interpersonal relationships, ability to manage emotions, etc. UNICEF based on a 2000 KAPS survey shows that more rural teenagers (33%) than their urban counterparts (20%) face stress from parents who impose wishes upon them. Analysis in a study by Dinakaran, J. and Dr. M. Gabriel (2014) at Kemp's Town at Trichirappalli also validated that adolescent girls are very low in their life skills. As Kerala is a highly literate state and the 'queen of Arabian sea' is in central Kerala, sampling is done from the institution here. The main objective of the study was to find out the life skill awareness among adolescent rural girls before and after the training with the module using digital paintings on selected life skills. The main hypothesis focused on the point of variation amidst chosen rural girls in 'their life skills narrated using digital paintings after the training'. Mainly questionnaires were used as data collection tools to elucidate answers and for comparing training activity by pre-testing and post-testing. After the training, the data compared and analyzed using a five-point scale-closed questionnaire, T-test, etc. Training life-skill needs to replicate the natural processes (observation, social interaction, modelling) by which children learn behavior. Thus, the theoretical framework used in an interdisciplinary mode in this article is the social learning theory for learning new behavior (Bandura, 1997).

Results and Discussion

Data Analysis and reporting of results proved the training effective. It is evident in the rejection of the null hypothesis stating that there is no difference between the two observations and, accepting the alternate hypothesis approving the difference. The T-test value of 9.735 was greater than the table value of 1.68 after training all girls with all selected life skills. The mean

difference was also higher between 4.919 and 2.012. Thus, there was a significant difference in awareness of chosen life skills after the training with digital paintings. The T-test values of each of the ten selected life skills also were higher than the test values after the training. For example, the T-test value of self-awareness 8.278, was greater than the table value of 1.68. There was difference also in the mean here between 4.98 and 1.93. In the case of interpersonal relationship skills, the test value of 9.66 was greater than the table value of 1.68 after the training. This comparison could prove the effectiveness of the life skill training by the module in those rural girls. In the families of participants mothers were more educated, which is advantageous for a girl child for future development. It shows the change in families after giving more value to education for a long time, especially to girls in Kerala. Thus procedures were completed with the happy participants with awareness of positive attitudes or behaviour as noted in social learning theory. Girls could develop internal skills like self-control and stress reduction, which can support positive outward behaviour. The selection of a limited area of study and limited participants below hundred points to the scope for further expansion with more skills after the pandemic period. This module will be a small asset to society in a senior women's angle of study in a highly digital and material world of life in the planet. Including such works in the syllabus *in all classes* is also recommended to curriculum developers of India.

Conclusion

The purpose of education is generally the attainment of knowledge and the improvement of living conditions. **Any positive skill to live happily is a life skill and** creative life-skill pedagogy is seen ignored in schools in Kerala. **Internet and social media activities by children have increased several times during and after the pandemic in the world.** *Many of the adolescents cannot resist temptations from peer groups. Here, life skills like self-awareness and* **managing stress are essential for effective communication, particularly for those girls from rural India. Innovations arise when there is a problem in society. Thus to fill the gap in creative life skill training, selected adolescent rural girls were given training using a module with digital paintings on life skills by the researcher in central Kerala.** Data analysis of the results proved that the training was effective in this group. Hence, conclusions are drawn based on these findings that using creative digital painting is effective in awareness programs for life skills in these rural girls. *Social Learning theory supports this study.* Incorporating such

works in the syllabus is therefore recommended to policymakers of the nation. Words of Dr. A.P.J. Abdul Kalam are rightly applicable here, "you cannot change your future, but you can change your habits, and surely your habits will change your future".

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Career Maturity in Relation to Family Environment and Social-Economic Status among Secondary School Students

Harjinder Kaur Sandhu*

Abstract

The aim of the present study is to know the relationship of Career Maturity with Family Environment and Socio-Economic-Status among secondary school students. Descriptive survey method of research was used for collecting the data using Career Maturity Inventory (CMI-Gupta, 1989), Family Environment Scale (FES-Bhatia & Chadha; 1993) and SocioEconomic-Status Scale (SES- Upadhyaya & Saxena; 2008). A sample of 200 male and female students studying in XI grade in secondary school students studying from rural and urban areas of Amritsar District was randomly selected. Findings of the study revealed no significant difference in Career Maturity, Family Environment and Socio-Economic-Status of boy and girl students. A significant difference was found in the Career Maturity of urban and rural students as the urban students possess more Career Maturity than rural students. No significant difference was found in the Family Environment and Socio-Economic-Status of urban and rural students. A significant relationship was found between Career Maturity and Family Environment of secondary school students; but no significant relationship was found between Career Maturity and Socio-Economic-Status of secondary school students. The findings have important implications on the development of career guidance and counselling programs in schools.

Key Words: Career Maturity, Family Environment and Socio-Economic-Status

Introduction

Career maturity is the degree in which individuals are prepared to make good educational or vocational decisions. Super (1957) defined career maturity as “The readiness to make appropriate choices when engaged in planned exploration and possessing appropriate occupational knowledge, self-knowledge, and decision-making knowledge”. Readiness here

refers to the level of career development in a person. Super's career development model presumes that each level of development exhibits an appropriate career behaviour. It is important for you to have a good understanding of yourself, your personality, if you are to make intelligent career plans (Splaver, 2000). Career Maturity refers to the individual's degree of readiness to choose, to plan and to prepare for a future vocation. Zunker (2011) defined career maturity as "A continuous developmental process that presents specific identifiable characteristics and traits essential to career development". According to Themba et al., (2012) Career Maturity implies the readiness of an individual to make a career decision. Creed (2001) includes the ability of individuals to make appropriate career choices in career maturity including considerations when making reliable and consistent decisions. Besides, Career Maturity also involves various aspects of a person. Career development is a life-long journey that starts at mid adolescence and ends at retirement Bozgeyikli and Hamurcu (2009). According to Super (1963) Career Maturity is used to denote the degree of development, the place reached on the continuum of the vocational development from exploration to decline. The Career Maturity quotient may thus be conceived of as the ratio of vocational maturity to chronological age. It would indicate whether the vocational development of an individual is appropriate of this age.

The family's knowledge, beliefs, and values about work in general, and about certain professions, accompanied by the child's direct observations, are an important source of vocational knowledge. Their transmission depends on the parents' availability, the time they can afford to set aside, but also their communication skills Bryant, Zvonkovic and Reynolds (2006).

According to Seon and Kim (2008) Family, school, and community exert influence on career decision and development, and one such important context variable is the parents. Adolescents experience through their relationship with their parents plays a crucial role in their career development and decision. Parents play a role in their children's career development by means of various types of support for their children (Kim et al; 2000).

Although educational and career choices are made by an individual, but they are certainly influenced by many social and environmental factors which include socio-economic-status of the family, home and family environment, gender, age, locale and psychological factors like intelligence, personality, achievement motivation, interest, aptitude, self-concept academic

achievement etc. It is widely recognized that if learners are to maximize their potential from schooling, they will need the full support of their families. Family Environment is the quality and quantity of the cognitive, emotional, and social support that has been made available to the child within the home and connotes the psychological environment of home. It is an environment where the child learns the skills, attitudes and behaviour which could mould him into a productive and successful student.

Socio-Economic-Status (SES) is important construct which has both direct and indirect impact on career maturity of adolescents. Socio-Economic-Status (SES) is the relative position of a family or individual along a hierarchical social structure, based on access to, or control over, wealth, prestige, and power. It is used to measure social class and social status. SES is an objective assessment of income, education, and occupation to understand a person's position in society (Diemer et al; 2013). Socio-Economic-Status is evaluated as a combination of factors including income, level of education, and occupation. It is a way of looking at how individuals or families fit into society using economic and social measures that have been shown to impact individuals' health and wellbeing. Socio-Economic-Status refers to social and economic standing. Socio-Economic-Status is the perceived hierarchy resulting from the amount of possession of social resources an individual believes he/she has and controls (Han et al; 2014). The purpose of present study is to examine the effect of Socio-Economic-Status, parental support on career maturity of secondary school students.

Review of literature

Carrier maturity and family Environment

Achibe (1982) reported that males scored higher than female in Career Maturity. DeRidder (1990), however, points out that lower levels of parent education can retard adolescents' career development. Being born to parents with limited education and income reduces the likelihood of going to college or achieving a professional occupational goal and predetermines the child's vocational choice. Navin (2009) found parents are positively correlated with career exploration. Lee and Hughey (2001) found that parental attachment was positively correlated with career maturity. Sharma and Gaur (2012) revealed that there is no significant impact of various aspects

of parental modelling and vocational maturity of adolescents. Lohita and Satsangi (2013) found the family environment is significantly related to career maturity. Kalra and Gupta (2014) found no significant difference in Career Maturity of male and female adolescents. Oberai (2016) revealed that the parentally accepted Girls studying in CBSE affiliated private schools have highest Career Maturity. Alam (2016) found no meaningful relationship between family environment and career maturity of students. Aurari (2017) found a contribution of family economical level on the vocational choices of children and no contribution of parents' careers on the vocational choices of children.

Carrier maturity and Socio-Economic-Status

Arbina (2000) found that secondary school is related to future career choices and income. Bozgeyikli (2009) showed significant relationship between career maturity with Socio-Economic-Status. Osa-edoh and Alutu (2011) found significant difference in career choices of students from middle and low Socio-Economic homes. Lee et al., (2012) found that Socio-Economic-Status has a significant relationship with Career Maturity which means that low Socio-Economic-Status can act as a barrier to career consciousness and Career Maturity. Hsieh and Huang (2014) found both socio economic status and proactive personality positively associated with career decision. Cattani et.al., (2016) found that career maturity of adolescents facing a career choice is associated with other factors, and not only the Socio-Economic situation of the family. Abdinoor (2020) found there was significant gender difference on career decision-making self-efficacy and mean difference in Career Maturity. Young women reported higher level of career decision-making self-efficacy and Career Maturity than young men. Sahu et. al., (2016) revealed that urban students showed higher mean scores on total scores of Career Maturity in comparison to rural subjects. Kaur and Singh (2018) concluded that urban students are more career mature as compared to rural students; and Career Maturity has positive relationship with Socio-Economic-Status.

Objectives of the Study

1. To study the Career Maturity of secondary school students with respect to gender.
2. To study the Family Environment of secondary school students with respect to gender.

3. To study the Socio-Economic-Status of secondary school students with respect to gender.
4. To study the Career Maturity of secondary school students with respect to locale.
5. To study the Family Environment of secondary school students with respect to locale.
6. To study the Socio-Economic-Status of secondary school students with respect to locale.
7. To study the Career Maturity of secondary school students in relation Family Environment of secondary school students.
8. To study the Career Maturity of secondary school students in relation to Social- Economic- Status of the senior secondary school.

Research design: Descriptive survey method of research was employed.

Sample: A sample of 200 (male and female) students studying in XI grade secondary school students in rural and urban area of Amritsar District.

Tools:

1. Career Maturity Inventory (CMI-Gupta, 1989)
2. Family Environment Scale (FES-Bhatia and Chadha; 1993)
3. Socio-Economic-Status Scale (SES- Upadhyaya and Saxena; 2008)

Analysis and interpretation of data

Table 1: t-value on Career Maturity, Family Environment and Socio-Economic Status

Variable	Demogr aphy (Gender/ Locale)	N	Mean	S. D	SE D	t- value	Inference
Career Maturity	Boys	50	28.04	6.28	1.21	1.93	Not Significant
	Girls	50	30.36	5.76			
Family Environment	Boys	50	45.32	9.33	1.74	0.41	Not Significant
	Girls	50	46.04	7.99			
Socio	Boys	50	54.28	12.13	2.16	0.13	Not

Economic Status	Girls	50	54.00	9.25			Significant
Career Maturity	Urban	50	31.14	6.34	1.163	3.34 **	Significant
	Rural	50	27.26	5.24			
Family Environment	Urban	50	46.14	9.44	1.33	0.53	Not
	Rural	50	45.22	7.85	1.11		Significant
Socio Economic Status	Urban	50	53.68	11.921	11.921	0.43	Not
	Rural	50	54.60	9.508	9.508		Significant

*1.96 at 0.05 level **2.58 at 0.01 level

Table 1 reveals that mean scores of Career Maturity obtained by boy and girl secondary school students are 28.04 and 30.36 and S.D is 6.28 and 5.76, respectively. Further, the value of ‘t’ becomes 1.93. The obtained value of ‘t’ (1.93) is less than the table value 1.96 at 0.05 level of confidence. There exists no significant difference between boy and girl students studying in senior secondary schools on Career Maturity. Hence, Hypothesis 1 “There exists no significant difference in Career Maturity of boy and girl studying in senior secondary schools” is not rejected. The results of the study are in consonance with Salami (2008) who revealed no significant differences between the males and females in their Career Maturity; Bozgeyikli, Hamurcu (2009) found Career Maturity for the gender variable presents not a significant difference; Kalra and Gupta (2014) revealed no significant difference in Career Maturity of male and female adolescents; Sivakumar and Sridhar (2016) found XI standard boys and girls do not significantly differ in their level of Career Maturity. Manivannan & Saminathan (2018) found no significant difference between the Career Maturity scores with regard to gender; Dhull (2018) found that no significant difference between the mean score of career maturity among male and female adolescent students; Kalita (n.d) explored that no significant difference of Career Maturity in male and female higher secondary students; Shinde (2020) found no significant difference in the Career Maturity of boys’ and girls’ students in higher secondary school but the results of the study are not supported the earlier work of Achebe (1982) who revealed that males appeared more mature in job knowledge and vocational attitude than

females; Birola and Koralp (2010) explored that Career Maturity levels of females are found to be higher than those of males; Alam (2013) found significant difference between boys and girls favouring girls on Career Maturity; Mutisya and Ndeke (2016) explored that family was a major influence to students' career choice.

Table 1 reveals that mean scores of Family Environment obtained by boy and girl students are 45.32 and 46.04 and S.D is 9.33 and 7.99, respectively. Further, value of 't' becomes 0.41 The obtained value of 't' is less than the table value 1.98 at 0.05 level of confidence. There exists no significant difference between boy and girl secondary school students in a Family Environment. Hence, Hypothesis 2 "There exists no significant difference in Family Environment of boy and girl studying in secondary school students" is not rejected. The result coincides with Rani (2018) who found that there is no significant relationship between Career Maturity and home environment of adolescents; Rana and Anmol (2020) revealed no significant difference between boys and girls. The result is not in accordance to the earlier work done by Otwor, Mutisya and Ndeke (2016) who explored that family was a major influence to students' career choice; Alam (2016) found a significant difference between boys and girls on the measure of home environment; Nilsson (2017) explored the effect of attachment to parents is significantly greater for girls than for boys; Praveen and Shafeeq (2019) reveal that senior secondary school students differ significantly in Family Environment in relation to their gender.

Table 1 reveals that mean scores of Socio-Economic-Statuses obtained by boy and girl students are 54.28 and 54.00 and S.D is 12.13 and 9.25, respectively. Further, the value of 't' becomes 0.13. The obtained value of 't' is less than the table value 1.96 (at 0.05 level of confidence). Hence it is evident that there exists no significant difference between boy and girl Secondary School Students on their Socio-Economic-Status. Hence, Hypothesis 3 'There exists no significant difference in Socio-Economic-Status of boy and girl Secondary School Students' is not rejected. The result is in accordance with the earlier work done by Rani (2018) who found no significant difference in Socio-Economic-Status of boys and girls adolescent. The result is not in accordance with the earlier work done by Kumari (2020) revealed that association between boys and girls on Socio-Economic-Status was significantly related.

Table 1 reveals that mean scores of Career Maturity obtained by urban and rural students are 31.14 and 27.26 and S.D is 6.34, respectively. Further, the value of 't' becomes 3.34. The obtained value of 't' is more than the table value is 2.58 at 0.01 level of confidence. There is a significant difference between urban and rural secondary school students on Career Maturity and is in favor of urban students. Hence, Hypothesis 4 'There exists no significant difference in career choice Maturity of urban and rural secondary school students' is rejected. The result coincides with Achebe (1982) who found that urban school students are more mature than rural School students in job knowledge and vocational attitude; Alam (2013) revealed urban students have relatively more Career Maturity than rural, finding gets the support from the works of Rojewski (1994), Zeren (1999), Sahrac (2000), and Surucu (2005) found significant differences between Career Maturity and gender; Sahu (2016) revealed urban students showed higher mean scores on total scores of Career Maturity in comparison to rural subjects. The result is not in accordance with the earlier work done by Kaur and Singh (2018) found no significant difference between Career Maturity of secondary school students with respect to locale; Manivannan and Saminathan (2018) revealed no significant difference between the mean Career Maturity scores of Rural and Urban students. Singh (2020) indicates no significant difference in vocational Maturity of Undergraduates college students of Rural and Urban background; Sivakumar and Sridhar (2016) revealed rural and urban school XI standard students do not significantly differ in their level of Career Maturity; and Singh (2020) concluded there is no significant difference in vocational Maturity of Rural and Urban students.

Table 1 reveals that mean scores of Family Environment obtained by urban and rural students are 46.14 and 45.22 and S.D is 9.44 and 7.85, respectively. Further, value of 't' becomes 0.53. The obtained value of 't' is less than the table value. There is no significant difference between urban and rural secondary school students in the Family Environment. Hence, Hypothesis 5 "There exists no significant difference in Family Environment of urban and rural Secondary School Students" is not rejected. The result coincides with Pappattu and Vanitha (2017) who concluded that there exists no significant difference in the mean scores of Family Environment factors of high school students based on locale. Singh (2018) found there is no significant difference in Family Environment of rural and urban secondary school students; Singh (2020) found no significant difference between Family Environment of rural and urban undergraduate's

students on the vocational maturity. The results of the study are not supported by Alam (2016) who found a significant difference between rural and urban on the measure of home environment.

Table 1 reveals that mean scores of Socio-Economic-Statuses obtained by urban and rural students are 53.68 and 54.60 and S.D is 11.921 and 9.508, respectively. Further, value of 't' becomes 0.43. The obtained value of t' is less than the table value is 1.96 at 0.05 level of confidence. There is a significant difference between urban and rural secondary school students on Socio-Economic-Status. Hence, Hypothesis 6 "There exists no significant difference in Socio-Economic-Status of urban and rural secondary school students" is not rejected. The result coincides with Smitha and Kumar (2020) who found no significant differences seen in SES of secondary school students from different localities.

Table 2: Relationship between Career Maturity with Family Environment and Socio-Economic-Status (N=100)

Variable		'r'
Career maturity	Family Environment	0.149
	Socio-Economic- Status	0.090

* 0.138 at 0.05 level **0.180 at 0.01 level

Table 2 reveals the coefficient of correlation between Career Maturity and Family Environment of secondary school students. The coefficient of correlation between Career Maturity and Family Environment of secondary school students comes out to be 0.149. The obtained value of 'r' is more than the table value is 0.138 at 0.05 level of confidence, it becomes vivid that there is significant relationship between the two variables i.e., Career Maturity and Family Environment of secondary school students. Hence, Hypothesis 7 "There exists no significant relationship between Career Maturity and Family Environment of secondary school students", is rejected, as a significant relationship has been found between the two variables i.e., Career Maturity and Family Environment. The result revealed that positive significant correlation existed between the two variables Career Maturity and Family Environment; meaning thereby that positive family

Environment helps the students to be more mature about their career decisions. Results of the study are supported by Lee and Hudley (2001) who found parental attachment plays a more important role in Career Maturity; Kalra and Gupta (2014) found significant relationship between Career Maturity and Family Environment of adolescents. The result is not in accordance with the earlier work done by Alam (2016) found no significant relationship between Home Environment and Career Maturity of students.

Table 2 reveals the results of coefficient of correlation between career choice Career Maturity and Socio-Economic-Status of secondary school students. The coefficient of correlation between Career Maturity and Socio-Economic-Status of secondary school students comes out to be 0.090. The obtained value of 'r' is less than the table value is 0.138 at 0.05 level of confidence. It becomes vivid that there is no significant relationship between the two variables i.e., Career Maturity and Socio-Economic-Status of secondary school students. Hence, Hypothesis 8 "There exists no relationship between of Career Maturity and Socio-Economic-Status of secondary school students", is not rejected, as no significant relationship has been found between the two variables i.e., Career Maturity and Socio-Economic-Status. The result is in accordance with the earlier work done by Cattani (2016) found no significant correlation between Career Maturity and Socio-Economic-Status; Abdinoor (2020) found no significant difference of Socio-Economic-Status on career maturity. Results of the study are not supported by Bozgeyikli and Hamurcu (2009) showed significant relationship between Career Maturity with Socio-Economic-Status; Osa-edoh and Alutu (2011) explored a significant difference in educational and career choices of students from middle and low Socio-Economic homes in favour of the former; Cattani (2016) found the significant positive relationship of career maturity with Socio-Economic-Status; Kaur and Singh (2018) revealed a significant positive relationship of Career Maturity with Socio-Economic-Status.

Findings

1. No significant difference was found between boy and girl secondary school students in their Career Maturity.
2. No significant difference was found in the Family Environment of boy and girl secondary school students.

3. No significant difference was found in the Socio-Economic-Status of boy and girl secondary school students.
4. A significant difference was found in the Career Maturity of urban and rural secondary school students.
5. No significant difference was found in the Family Environment of urban and rural secondary school students.
6. No significant difference was found in the Socio-Economic-Status of urban and rural secondary school in students.
7. A significant relationship was found between Career Maturity and Family Environment of secondary school students.
8. No significant relationship was found between Career Maturity and Socio-Economic –Status of secondary school students.

Educational Implications

The present study revealed that locale influences the Career Maturity of secondary school students so rural students should be given opportunities and support in schools to develop their abilities and competences, which will strive them to work hard to achieve their career goals and take decisions accordingly. Hence, there is need to implement some strategies to improve the career maturity of rural students. Vocational guidance and counselling need to be introduced as an integrated component of the curriculum at all stages of the school curriculum with diverse objectives depending upon the needs of the students in each stage of education. This would ensure a strong foundation to make career decisions appropriately at secondary and senior secondary stages of education.

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The Role of Social Media Sites on Reading Habits among Undergraduate Students

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Abstract

The main objective of this paper is to examine the views of undergraduate students of Sonipat district in Haryana about the significance of social media usage, there are many online platforms available for language learners to improve their reading competency. This study focuses on analyzing the influence of social media on learning language, thereby enhancing their reading and cognitive skills through reading comprehension. The data were collected from 190 undergraduate students in the Sonipat district in Haryana. The finding shows that students used social media sites for communication with teachers and classmates for academics purpose but also used social media sites for connecting with family and friends. The feasibility of using online tools increases the interest among undergraduate students to enhance their critical thinking, communication, and other language skills. It also helps the students to share their knowledge and enrich their reading ability by using the E-Resources which are readily available at hand for the learners.

Keywords: Social Media Sites, Reading, E-Resources.

Introduction

In the present scenario, the number of internet users is rapidly increasing. The increased use of social media by English as a second language learners enables them to create a better opportunity to learn a language more effectively. It can engage students in improving their cognitive skills and enhancing their communication and other language skills. The usage of social media in language learning inculcates a positive approach toward their learning thereby improving their performance. Though social media has been open for several years it has come into use very recently in the language classroom. The traditional pedagogical practices can be changed and the practice of using technology in the classroom can be implemented to improve

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the cognitive skills of the students. Henceforth, social media can be used for sharing their reading resources both inside and outside the classroom. Even after getting employed the graduates face a challenge in applying their language skills in their working environment as they must do many tasks related to reading which includes reading reports, manuals, emails, meeting minutes, business agreements, and instructions related to their work, checklist, etc. Most of the graduates could identify the words and their meanings individually and find it difficult to interpret the meaning of the whole text. Al-Rahmi and Othman (2013) claimed that most of the respondents found that social media empowers academic experience. Bosch (2009) found that social media encourages students to ask questions because of the absence of face-to-face interaction. Students face discomfort in asking a question in traditional face-to-face classes. Rafiq et al. (2019) found that there is a positive effect of social media on reading habits. But distraction during exam time is also noted as an adverse effect. The study also found that WhatsApp, Facebook, Google +, and YouTube were used by the student on daily bases. Most of the student used media technology through cell phones. Hussain et al. (2021) investigated how social media sites shaped the reading habits of the degree college students in J&K, India. The study found that social media sites are used for academic as well as non-academic purposes. Most of the students used social media for exchanging reading material, communicating with classmates and teachers, reading magazines and newspapers, and for searching information. Kojo et al. (2018) statistically proved that time spend on social media significantly affects the reading habits of the students and extends their academic performance. There should be a rule and regulations that limit students from using social media sites during lectures and library sessions. Balan et al. (2019) found that students who have good reading habits can possess well academic scores and increase their reasoning skills. Besides this reading, non-academic books also increase their language skills.

Problem Statement

The main objective of this paper was to investigate the impact of social media site use on reading habits or reading habits of undergraduate students of the Sonipat district in Haryana.

Methodology

The data were collected from 190 undergraduate students by Questionnaire. Questionnaires with reading comprehension passages were given to the undergraduate students and they were asked

to respond through any one of the social media platforms such as WhatsApp, Facebook, Instagram and Google classroom and text messages. The data was collected related to the research and the observation was done.

Result And Discussion

The observation was made based on the data obtained from the questionnaire. All the respondents used social media for language learning. The analysis of the questionnaires reveals that 80 students used WhatsApp, 70 used Google classroom, 21 used Facebook, and 19 used Instagram for language learning frequently.

Table 1: Social Media Application used for Language Learning

Social Media	No of Respondents
Google Classroom	70
Facebook	21
Wats App	80
Instagram	19

The survey results were encouraging, and most of the respondents agreed that social media help improve their language skills, especially their reading skill. If students feel difficulty in understanding a few words in English they seek help for their peers or from the facilitator in the group. This helped them in enhancing their reading ability and encouraged their peers to explore new vocabulary and new concepts. As social media is interactive in nature it facilitates the students to provide feedback so that they can be aware of their strengths and weaknesses. Reading articles can help to expose them to more vocabulary. Social media has become a part of language learning and it could not be separated from using it for educational purposes. All the respondents showed interest in the use of social media to develop their reading ability with the help of online discussions.

Table: 2 Frequency of using social media by the Undergraduate Students

How frequent used social media	Frequency	Percentage
Several times	133	70%
Daily	38	20%
Weekly	9	4.7%

Occasionally	10	5.3%
Total	190	100%

The above table found that most of the students used social media several times a day i.e., 70%, followed by the students who used it daily i.e., 20% this showed that majority of students used social media for connecting with friends and for educational purposes as well.

Table:3 Academic Purpose of using social media

Academic Purpose	Frequency	Total
Interaction with teachers and students	52	27.4%
Read newspaper online	28	14.7%
Read book online	15	7.8%
To obtain/search for information	45	23.6%
To exchange reading material	50	26.4%

The response in this table showed that most of the students used social media sites for communication with teachers and classmates for academic purposes i.e., 27.4% and followed by most of the students who stated that they used social media sites for exchanging reading material i.e., 26.4%. Additionally, 23.4% stated they used social media for obtaining information. So, the result showed that most of the students used social media sites for facilitating their academic work.

On the other hand, respondents also asked about the non-academic purpose of using social media sites. 36.8% of respondents used social media for sharing content and followed by chatting with friends i.e., 28.9% and for communication i.e., 23.6%. So, the student's responses showed that the main reasons behind using social media sites for non-academic purposes are connecting with friends and family.

Table :4 Non-Academic Purpose of using social media

Non-Academics Purpose	Frequency	Total
Chatting with friends	55	28.9%
Sharing contents	70	36.8%
Making new friends	8	4.3%

Playing new games	12	6.4%
Communication	45	23.6%

Conclusion

The study investigates the impact of social media sites on the reading comprehension of undergraduate students in Haryana. The study concluded that most of the students positively used social media sites for reading purposes and for enhancing their knowledge. Social media sites are also used for the non-academic purpose of connecting with friends and family. But well designed and tailored use of social media can only help in reading habits in the students. The study also found that the benefits of social media sites for reading were not only limited to a few platforms, but learners have a lot of choice in choosing their favorite platform for learning.

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The Role of Artificial Intelligence in Fostering Mastery Learning

M Nandan*

Abstract

Artificial intelligence involves the development of machines with an adequate level of intelligence capable of performing human-like functions such as cognitive, decision-making, learning, and adapting to the environment. In recent times, Artificial Intelligence has been extensively applied in instruction and learning to improve learning outcomes. The learners' personality, i.e., whether introverted or extroverted, determines their learning preferences. For instance, Extroverted learners may prefer group activities that involve problem-solving and discussion, while introverted learners may prefer to learn through reflections from asynchronous assignments. Learning in the classroom may only be uniformly effective or proficient for some students due to differences in personality, learning preferences Etc. However, ensuring that all students have mastered key concepts, ideas, or theories is essential before moving on to subsequent topics. Simply put, it is vital to ensure that all students are on the same page. In this context, Mastery learning comes into play which is an instructional strategy that suggests that students should achieve a required level of mastery in pre-requisite and imperative knowledge before learning subsequent topics. Considering the significance of Mastery Learning, it is critical to understand how Artificial Intelligence can be leveraged to impart customized learning experiences to learners based on their learning preferences to foster mastery learning and improve the learning outcome. Considering the same, this study, through the review of relevant literature, aims to develop a conceptual model for using Artificial Intelligence to foster mastery learning among students for improved learning outcomes and identify the future scope of research.

Key words: *Artificial Intelligence, Mastery learning, Education, learning preferences, inclusive learning*

Introduction

Artificial intelligence is a discipline and the following innovations and progress that have led to computers and machines acquiring the ability to perform human-like functions indicated by cognitive abilities such as intelligence and pattern recognition, learning, adaptability, etc. Artificial intelligence has been extensively used in education (Chen et al., 2020). Similarly, according to McCarthy (2004), Artificial Intelligence is the science of instilling intelligence in machines or, more specifically, developing intelligent computer programs capable of simulating human intelligence. Therefore, Artificial Intelligence refers to computers' ability to perform actions requiring cognitive abilities.

Chen et al., (2020) found that AI and the usage of Machine Learning and adaptability have resulted in the customization of curriculum and content to provide personalized learning experiences per the learners' preferences, which has led to improved learning outcomes. Supporting the above finding, Holmes et al., (2020) imply that Artificial Intelligence in the education sector includes personalized instructional and dialogue systems, intelligent agents in game-based environments etc. Similarly, **Yufeia et al., (2020)** indicated the application of AI in education to the usage of data mining and intelligent analysis of educational data, learning analysis etc. to provide customized learning experiences. Overall, Artificial Intelligence in the context of education refers to the use of various intelligence systems, as mentioned above, to improve the learning outcome through the personalization of learning experiences and to foster independent learning.

Keefe (1979) described learning styles as the sum of cognitive, affective, and physiological factors influencing how individuals perceive, learn, and respond in a learning environment. While **Keefe (1979)** emphasized the factors that influence perception and learning, (Stewart & Felicetti, 1992) focused on the learner's preferences. They described learning styles as the educational conditions under which learners prefer to learn. Along similar lines, **Loo (2002)** describes learning styles as the persistent way a learner responds or reacts to stimuli in the learning context and further states that learning styles are determined by numerous factors, such as the learner's personality and motivations. There is no unifying definition of learning styles. However, from the above discussion, learning styles can be outlined as the preferences or ways

in which a learner would want to learn content. The learners' learning styles are influenced by intrinsic factors such as the individual's personality, motivations, Etc.

Learning styles play a crucial part in the learning process and significantly impact the overall educational environment (Moussa, 2014). In support of the above, **Hung (2012)** suggests that complementing teaching techniques to students' learning styles is critical to improving learner effectiveness. Along similar lines, Durling et al., (1996) also highlights the need for aligning teaching methods with learning styles and further indicates that failure of the same will lead to poor transmission of knowledge. The above discussion indicates the need for providing personalized learning experiences to improve learning outcomes which involves taking cognizance and consideration of the learning preferences of the learners.

Benjamin Bloom initiated the discussion on the theory of mastery learning in 1968. **Bloom (1968)** opined that most students could master the content or the subject taught to them. However, he observed a high level of variance in the grades achieved by students. Bloom emphasized the role of instruction and suggested that it is the function of instruction to find ways by which students will be able to master what they are taught. He observed inequality and opined that the educational system identifies a talented few and ignores most students concerning advanced educational opportunities. He implied that instructional strategies play a vital role in fostering mastery among students and bringing most students on the same page. Mastery learning pertains to a teaching strategy that suggests that all students must achieve a basic level of mastery in a current topic before learning the following content or topics (Bloom, 1968). Along similar lines, (Block & Burns, 1976) opined that under suitable instructional conditions, most students would be able to learn well or, more specifically, master the content or the subject taught to them. They further imply that mastery learning may result in a breakthrough regarding the improvement of student learning. Therefore, Mastery learning refers to an inclusive educational philosophy that suggests that all students must be on the same page by attaining a basic or pre-determined level of mastery in pre-requisite knowledge before learning advanced concepts or subsequent information.

The primary recommendation of mastery learning proposes that most students can master what they are taught based on the instructional strategy. **Carroll (1963)** posited that students' aptitude might not be the best predictor of academic achievements; instead, it could indicate the time

required to attain desired academic achievements. Further research (**Bloom, 1968**) found that most students reached the desired level of achievement through the optimization of time and the quality of instruction according to the learners' characteristics. Despite the effectiveness of mastery learning as discussed above, few studies (**Groff, 1974**) criticize mastery learning strategies as mechanical and inflexible. In contrast to the above, other studies (Kazu et al., 2005) opine that the mastery learning approach imparts flexibility by customizing instruction to individual learners' learning preferences and needs. Therefore, the instruction strategy has invoked a need for more consensus regarding mastery learning. Considering the advantages offered by mastery learning, The objective of this study is to suggest the effective use of Artificial Intelligence to foster mastery learning and propose a conceptual model for the same.

Review of literature:

Artificial Intelligence and Adaptive Learning

Artificial intelligence refers to the development of intelligent machines capable of simulating and performing human functions such as cognition, learning, decision-making, and adapting to the environment. Considering the capability of Artificial Intelligence to adapt to the environment, Adaptive learning has turned out to be one of the applications of Artificial Intelligence in Education.

Adaptive learning is an educational strategy that leverages computing algorithms to assemble the interactive learning experience and adapt to the learning preferences of individual students (**Kolchenko, 2018**). Similarly, Nguyen & Do (2008) opine that adaptive learning involves tailoring or customizing learning material, such as content, exercises, tests, etc., according to the learners' preferences and characteristics. Therefore, adaptive learning is an educational strategy that involves intelligently adapting instruction to the learners' preferences by customizing learning materials according to learners' characteristics.

Adaptive learning systems offer numerous benefits. Using learning analytics and real-time analysis has improved student achievement and knowledge retention. For example, Duolingo, an application used to learn spoken languages, increased learner autonomy by customizing time, finding materials, and modifying techniques to achieve better results (**Alam, 2022**). In support of the above finding, other studies Dhakshinamoorthy & Dhakshinamoorthy (2019) found that students who used a proposed adaptive learning methodology showed a performance

improvement. The study further found that the test performance of the learners in adaptive learning methodology is greater than that of learners in the traditional learning mode. (Sivakumar et al., 2015) opine that adaptive e-learning provides several benefits, such as increased knowledge retention, improved knowledge sharing, consistent delivery etc. According to Abhirami & Devi (2022), Web-based learning promotes educational equity and independent learning. Despite offering several benefits, Adaptive learning systems face certain downsides. For instance, Kolchenko (2018) points out that for adaptive learning systems to be effective, they would need high-quality and meaningful data for algorithm training, failing which the Artificial Intelligence program may perform worse. He further states that adaptive learning systems cannot replace experienced faculty as they have a firm grip on the subject and pedagogy. Furthermore, teachers have a personal connection with their students and can adapt their teaching based on experience, intuition, etc., which adaptive learning systems lack. Therefore, From the above discussion, the author argues that the teacher's or instructor's intervention is needed for adaptive learning systems to be effective and that they play a critical role in designing well-thought content, exercises, and assignments.

Mastery Learning

According to Bloom (1968), Mastery learning is a philosophy or a teaching or learning strategy according to which students should only proceed to study advanced topics or concepts after achieving a certain level of mastery in the pre-requisite knowledge or the basic concepts. It is a philosophy that encourages inclusive learning. The principal focus of this strategy is ensuring that all students have mastered key or basic concepts before studying advanced concepts. Carroll (1963) opined that scholastic aptitude is not the best predictor of academic performance. Instead, it predicts the time needed to learn a particular subject. Basis this, Bloom (1968) argued that most students could attain mastery in a subject provided that the time and instructional strategy were personalized according to the learners' characteristics. In other words, He indicated that scholastic aptitude is not indicative of academic performance.

According to Bloom (1968), The mastery learning approach involves the following steps:

1. The chapter or topics to be learned are divided into smaller units with pre-determined objectives for each unit.

2. The teaching process begins, and an identified unit is taught to the students by incorporating a suitable teaching methodology.
3. Assessment is a critical element under the mastery learning approach. Therefore, Once a unit is learned, the level of mastery attained is judged through a suitable assessment such as a quiz, written test Etc.
4. Students must showcase a certain level of mastery in the assessment. Students who do not achieve the required level of mastery are subject to remediation through additional study materials, assignments, teaching Etc., and sufficient time is allowed.
5. The learners continue the cyclical learning and assessment process until they achieve mastery and move on to the subsequent units.

Many studies have confirmed the benefits and usefulness of the mastery learning strategy. A meta-analysis study by Kulik et al., (1990) revealed that mastery learning programs positively affect the examination performance of school and college students. Similarly, Clark et al., (1983) found that students in an undergraduate program who practiced mastery learning strategy indicated higher levels of achievement. Similarly, (Hussain & Suleman, 2016) also found that mastery learning profoundly impacts student performance and knowledge retention.

Learning styles and learning preferences

The term "learning styles" indicates that learners prefer different instructional strategies or ways of learning to learn content effectively (**Pashler et al., 2008**). There are many forms of learning styles. However, one of the most accepted and popular forms of learning styles is described by the VARK modalities. Fleming & Baume (2006) opined that modal preferences affected an individual's learning behaviour and emphasized the need to match these preferences with learning strategies for improving learning outcomes. The VARK framework categorizes learners based on three modal preferences, i.e., Visual, Auditory, Kinesthetic, and a different category, i.e., Reading or Writing. The framework suggests that visual learners learn best and process information by seeing. Teaching them by showing graphs, charts, and visuals like pictures and videos would help them learn better. Auditory learners learn the finest by hearing or listening to information and conversing about it. Classroom lectures and recorded learning sessions help them better understand the content. The "R" in the acronym stands for Reading or Writing. Learners with this modal preference learn better by writing content and re-reading the same to

get a better understanding. Lastly, the "K" in the acronym stands for Kinesthetic. These learners learn best by doing things such as playing, experimenting, etc. They prefer to refine their learning through experience and practice.

Although the VARK framework is extensively used to describe learning styles, it has faced criticisms. Riener & Willingham (2010) have criticized the VARK learning styles with the argument that learners' preferences to learn content through a specific mode, i.e., Auditory, Visual, or Kinesthetic, does not have meaningful implications for their learning. In other words, they opine that the learning outcome remains the same irrespective of the preference for mode of learning. They imply that most of what we learn is stored in terms of meaning and not limited to a robust sensory mode. In support of the above argument, an experimental study on chess players' perception conducted by (Chase & Simon, 1973) revealed that what we learn is stored in terms of meaning and that we are better able to learn and recall something if the content is more meaningful to us. From the above discussion, the author argues that for learning to be effective, the content learned should be meaningful to the learner. It is critical to customize the instructional strategy according to the learner's personality rather than their modal preferences to make the content meaningful to the learner.

Considering the benefits of mastery learning, combining it with Artificial Intelligence to provide customized learning experiences to learners based on their personality and preferences is likely to improve inclusiveness and academic performance.

A conceptual model of using artificial intelligence to foster mastery learning.

Basis the above discussion, the proposed system or conceptual model will include elements from the following theories:

A. Carl Jung's Theory of personality:

Carl Jung, A Swiss Psychiatrist, was one of the first to describe the personalities of introvert and extrovert. According to him, Introverts and extroverts have the following characteristics.

1. **Introverts:** Persons with an introverted personality may focus on their inner world and rely on reflection (Jung, 1923). According to Carrigan (1960), Introverts are individuals who get energy by being in solitude and often appear shy from social activities. Introverts are not outgoing and may prefer less stimulating or social activities. They prefer to be independent, learn through reflection, spend time alone, and be self-directed.

2. **Extroverts:** Jung (1923) characterized extroverts as individuals who prefer to interact with the external world. According to (Carrigan, 1960), Extroverts are individuals who get energy from being around and interacting with others. In other words, Extroverted Individuals derive energy from stimulation. Extroverts are outgoing and may enjoy teamwork or group assignments.

B. Kolb's learning styles:

David Kolb described Kolb's learning styles in 1984. He described four learning styles as described below:

1. **Diverging:** The learners who prefer this learning style prefer to watch to obtain information and apply creativity and imagination to solve problems. They are suitable in situations that require out-of-the-box thinking, such as idea generation, Brainstorming, rapid ideation etc. Another characteristic of this type of learner is that they like to work in groups and are open-minded. They are receptive to feedback.
2. **Assimilating:** This type of learner relies on logic and emphasizes ideas and concepts more than people. They prefer to understand information and arrange them in a logical sequence or order. They prefer to be fundamentally strong and are attracted to logically sound theories. Learners with this style usually prefer reading and listening to lectures and need time to think through.
3. **Converging:** This type of learner prefers to use their knowledge and learnings to find solutions to problems or issues. They look for practical use of theories or ideas. They prefer to experiment with new ideas and are practical.
4. **Accommodating:** This learner relies on intuition rather than logic. They tend to rely on others for information. They tend to rely on others and prefer a practical and experiential approach (McLeod, 2017).

C. Preferred forms of assignments based on the discussion of Carl Jung's theory of personality and Kolb's learning styles.

Table 1:Classification of learners based on personality and learning preferences.

Personality of the learner	Preferred learning style	Preferred assignments
Introverted	Converging learning style	Assignments that will help the learner to gain critical thinking skills or the ability to think clearly, such as analysis of case lets, scenario analysis in the written form.
	Assimilating learning style	Assignments that will help the learner to conceptualize and reflect through observation such as watching videos of lectures, reading journals, identifying patterns or analogies etc.
Extroverted	Diverging learning style	Activities that facilitate out-of-the-box thinking such as brain storming, Rapid ideation etc.
	Accommodating learning style	Case studies, Group projects, group discussion etc.

As discussed in Table 1 above, Introverted learners are further classified into learners preferring a convergent learning style or an assimilative learning style. And extroverted learners are further classified into learners preferring divergent or accommodative learning styles and the preferred after class activities are mentioned for each of the combination.

A four-stage model for leveraging Artificial Intelligence to Foster Mastery learning:

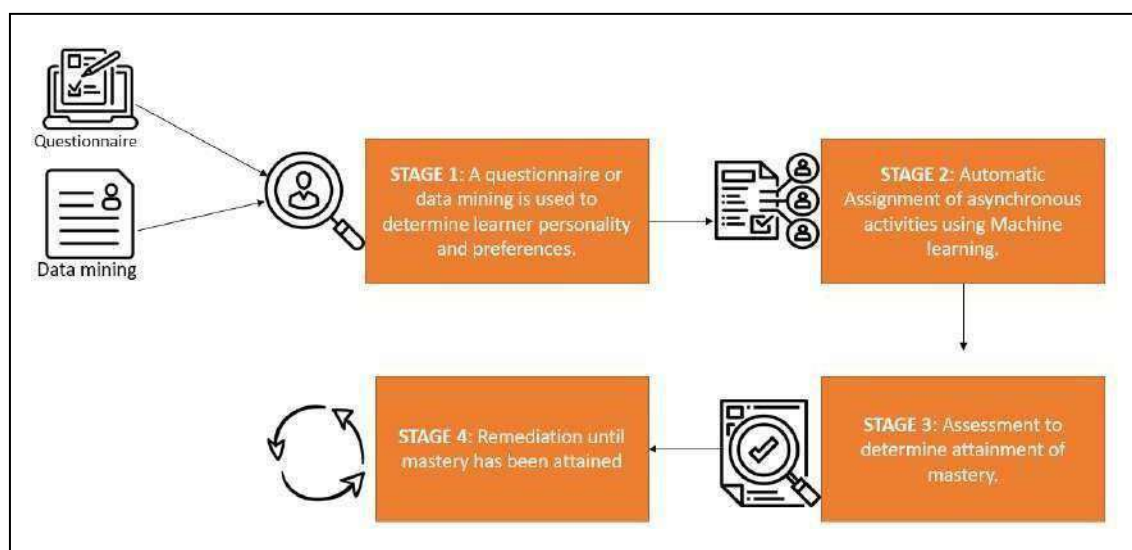


Figure 1 :- This figure illustrates a four-stage model for using Artificial Intelligence to foster Mastery learning.

Stage 1 - Determination of learner's personality: The system detects the learner's personality through a questionnaire or data mining. Data mining in the education field refers to using student data such as Age, teacher feedback and grades, etc., to determine the students' characteristics. Once the student's personality and learning style has been identified, we proceed to Stage 2.

Stage 2 - Assignment of asynchronous activities: At this stage, the learning preferences of the students have been understood. After completing one unit in class, the system automatically assigns asynchronous activities from an assignment bank or a database designed by the instructors to the learners according to their learning preferences as determined in Stage 1. The learner's personality and learning style is ascertained as discussed in Table 1. Furthermore, a fixed time is assigned to complete and submit the assignment, preferably before proceeding to the subsequent unit in class. Using Supervised machine learning at this stage is essential for automatically assigning activities based on learner characteristics. For example, Labelled data could be used to train algorithms that would be used to classify data which in turn would facilitate the automatic assignment of activities based on learner characteristics.

Stage 3 - Assessment: Assessment is done to check if the student has attained the basic or pre-determined level of mastery through a quiz or a suitable assessment method.

Stage 4 - Remediation: If the student has failed to achieve the basic level of mastery, The student is subject to remediation by providing additional resources such as materials, lectures, support Etc., The learner goes through the cyclical learning and assessment process until the desired mastery level has been obtained.

Discussion

Recognizing the need for personalizing learning experiences according to the learner's preferences to improve learning outcomes, The author suggests adopting the proposed model as an add-in to the existing Learning Management Systems of educational institutions to enhance students' learning outcomes. As indicated by **Simon and Chase**, the content learned by learners is stored in terms of meaning. Therefore, it is critical to ensure that the instructional strategy must be customized according to the learning preferences of the learners to make the content more meaningful to the learners to enhance learning outcomes.

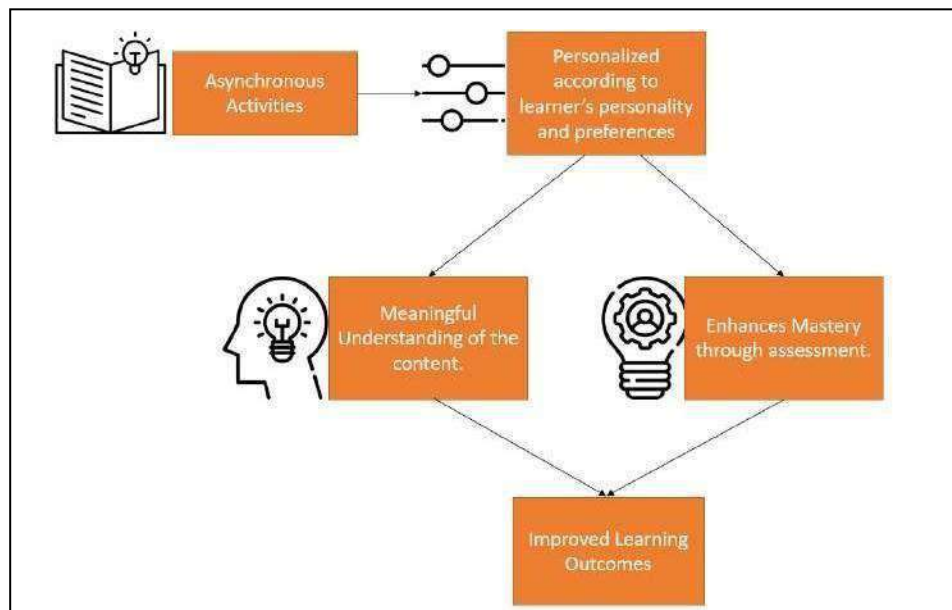


Figure 2 :- This figure illustrates the conceptualization of the proposed model for improved learning outcomes.

Considering the above argument, the proposed model suggests that learning does not occur uniformly in the classroom for several reasons. Therefore, the use of after-class hours by providing asynchronous assignments according to the learner's preferences may enhance a meaningful understanding of the content.

Therefore, As illustrated in Figure 2 above, we conceptualize the idea of the proposed model as above. Asynchronous assignments or activities, when personalized according to the learner's personality and preferences, enhance the learners' meaningful understanding of the content and fosters mastery through assessment, which would result in improved learning outcomes.

Suggestions

1. For the proposed model to be effective, the model should be integrated as an add-in or a feature of the existing learning management system of the educational institutions.
2. Teachers or instructors should design well thought asynchronous activities based on experience and expertise for the proposed model to be effective.
3. Teachers or instructors should be trained in applying artificial intelligence and machine learning to use the proposed model effectively.
4. It is suggested to capture the data of students' activities under this model to facilitate algorithmic training and to enhance the model over time by leveraging machine learning.

Conclusion and Scope for Further Research

Personalizing instructional strategy according to the learner's personality and preferences is critical to enhancing a meaningful understanding of the content and facilitating mastery learning for improved learning outcomes. The proposed conceptual model in this study suggests the use of machine learning and artificial intelligence to use after-class hours to enhance mastery learning through personalized synchronous activities. Future work or studies can focus on the practical application of the proposed model, replication, and implementation of similar models in schools and colleges in rural areas to facilitate inclusive learning. Furthermore, Future research can focus on developing practical asynchronous activities to foster critical thinking and creative thinking skills that are much needed in today's world.

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**Social Competence in Preschoolers: A Cross-Sectional Analysis of Age and Gender
Differences**

Anthea Patrao* & Triveni S.**

Abstract

Although social competence is seen as a critical component of academic and social success, it is often reduced to a set of social skills or the absence of negative behaviours. Studies that have considered social competence and its domains are scarce in the Indian preschool population. The sample of the present study consisted of 180 preschool children between 4 and 6 years old. Results showed that girls were rated significantly higher than boys on prosocial behaviors, emotion regulation, and overall social competence. For both girls and boys, gender is an important social category that impacts social competence. As far as academic skills are concerned, no gender difference was found. Additionally, significant differences were found between the three age groups in terms of social competence and its domains. In addition, our study emphasizes the development of social competence over time. Teaching, modeling, and reinforcing positive behavior with evidence-based strategies are therefore essential strategies for enhancing social competency skills. In turn, social competency skills are crucial to forming successful interactions with teachers and peers in early childhood. Limitations of the study and directions for future research have been discussed.

Keywords: prosocial behaviors, emotion regulation, academic skills, social competence, preschoolers

Introduction

In a developmental context, social competence is a multidimensional construct that takes on different meanings. Forming successful interactions with peers in early childhood is a key component to forming successful collaboration and connections in human interactions. Over the years several differing definitions of social competence have been proposed. Despite varying views on social competence, researchers agree that self-regulation, interpersonal knowledge, and

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skills are the essential components. Learning-related skills, compliance, interpersonal social skills, and group interaction comprise four key social skills dimensions (Lim et al., 2013). These skills include paying attention, working in a group, staying focused and persistent with a task, organizing one's self, exercising self-control, responding with appropriate emotions, and showing empathy. In other words, social competence is not one skill or behavior, but a combination of cognition, affect, and behavior that are coordinated in a coherent way so as to facilitate the handling of developmentally relevant social tasks. Social competence develops over time and plays a major role in determining adjustment abilities in childhood and adulthood and is also intertwined with emotional components. Having the ability to experience and express one's own emotions and to recognize others' emotions is key to successful social interactions.

Childhood development of social competence has recently attracted researchers' attention, primarily because stronger social competence is generally associated with peers' acceptance, emotional wellbeing, and school readiness as well as interpersonal relationships and social adjustment. On the other hand, there is a positive correlation between the lack of social competence, such as empathy, cooperation, and conflict resolution skills on the one hand, and children's negative behaviors and problems in their social interaction skills on the other hand (Qin & Yong, 2002).

Prior to entering preschool, a child's closest companions have been parents and caregivers. Upon transitioning to preschool, the child must learn the teacher's expectations as well as adjust to a wider peer group. A child's ability to manage emotions during the preschool period determines whether they will make successful interactions or be rejected by their peers. To be able to regulate their emotions, children need additional structure and support from their parents and caregivers.

According to studies, older children are more socially competent than younger children (Mendez et al., 2002), and girls are more socially competent than boys (Denham et al., 2003). Additionally, Diener and Kim (2004) found that boys showed more externalizing behaviour and girls showed more prosocial behaviour as children developed Social Competence. Furthermore, teachers have emphasized the idea that taking turns, following instructions, being empathetic, and not disrupting classes are as important as learning colours, shapes, letters, and numbers. Therefore, promoting these skills during the preschool period is essential.

Despite the abundance of research on the development of children's social competence from infants to pre-schoolers, the development of social competence and behavior problems during the preschool period has not been studied as much. It is during the preschool years that children interact with many peers, establish positive relationships with teachers, and develop social skills that are crucial for their development. Parents, teachers, and others in the field of development and education must understand how children develop social skills and how those skills change with development. Empirical studies on this period may shed light on the development of social competence and better contribute to the ones seeking to assist young children with low social competence, or problems associated with it. In addition, it may assist in detecting developmental difficulties in social competence and suggest ways to deal with them.

To the best of our knowledge, studies that have considered social competence and its domains are scarce in the Indian preschool population. Therefore, the main objective of this study was to examine the gender differences and the developmental pattern of social competence and its domains in Indian preschoolers.

Method

Participants

A cross-sectional design was used in this study. The sample comprised of 180 male and female preschoolers between the age range of 4-6 years. Children were recruited from semi-aided schools in the State of Goa through the method of convenience sampling. Participants were culturally and socio-economically diverse. Male and female preschoolers were equally distributed between the three age groups.

Measures

Social Competence Scale – Teacher Version

The Social Competence Scale - Teacher Version is a 25-item measure of academic skills, emotional self-regulation, and prosocial behaviours in children. It contains items adapted from Kendall and Wilcox (1979) and Gersten (1976) assessments. New items were also developed. Each of the items on the Social Competence Scale - Teacher Version describes a behaviour that children may display at school. Each statement is assessed by the teacher for its suitability to describe the child. A five-point Likert scale was used to code responses: Not at all (0), A little

(1), Moderately well (2), Well (3), and Very well (4). The higher the score, the better the social competence skills. In addition to the subscale scores, a total score for overall social competence is also provided for all 25 items. Scores are not calculated if more than half of the scale's items are missing responses. The internal consistency (alpha) for the subscales is approximately 0.96 and for the overall score it is approximately 0.98 (Corrigan, 2003).

Procedures

To obtain informed consent for the child, we explained the purpose and process of the study to parents and teachers. Parental and teacher consent was obtained before the commencement of the study. We asked parents to consent and grant permission for the child's teacher to complete the Social Competence Scale. Parents were given the opportunity to decline participation in the study before it was conducted. By doing so, they understood that participation in the study was completely voluntary and that they were free to withdraw from it at any time with no adverse consequences. Only children for whom consent was obtained from parents were included in the study. It was required that teachers have known the child they were rating for at least three months.

Statistical Analyses

The software SPSS (Statistical Package for the Social Sciences) 21.0 was used to analyse the data. Alpha of 0.05 (5%) was assigned as the statistical significance criterion. To understand the distribution of each variable, descriptive statistics were used. Based on a two-way ANOVA, we examined whether gender and age influence social competence and its domains.

Results and Discussion

A total of 180 preschoolers participated in the study. All parents supported their children's participation in the research. Male and female preschoolers were equally distributed between the three age groups. The descriptive summary is presented in Table 1 below.

As observed from the mean values in Table 1, on all domains of social competence, girls performed better than boys. There was a developmental trend with age - older preschoolers exhibited better prosocial/communication, emotion regulation, academic skills, and overall better social competence than younger preschoolers.

Table 1: Descriptive Summary of Social Competence and Its Individual Domains in Preschoolers by Gender and Age

	Gender				Age (in years)					
	Boys N=90		Girls N=90		4 N=60		5 N=60		6 N=60	
Skills	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
Prosocial	19.41	7.47	21.18	7.10	11.77	3.85	20.65	4.55	27.11	2.03
Emotion Regulation	25.17	9.38	26.79	9.36	15.93	5.45	26.40	5.15	35.60	3.69
Academic	16.77	5.80	17.28	5.33	11.65	3.79	17.87	3.87	21.55	3.67
Total Social Competence	60.44	22.14	68.31	21.27	39.35	12.57	65.00	13.03	84.28	8.16

Note. *M* and *SD* represent mean and standard deviation, respectively.

A Two-way ANOVA was performed to analyse the effects of gender and age on preschoolers' overall social competence and each of its individual domains, namely prosocial/ communication, emotion regulation, and academic skills. Summary results of the series of Two-way ANOVAs are presented in Table 2 below.

Table 2: Gender, Age, and Interaction Effects on Preschoolers' Social Competence

	Gender		Age		Gender x Age	
Skills	<i>MS</i>	<i>F</i>	<i>MS</i>	<i>F</i>	<i>MS</i>	<i>F</i>
Prosocial	63.01	3.97*	3563.54	268.79*	5.85	0.44
Emotion Regulation	118.42	5.15*	5809.69	252.85*	1.09	0.05
Academic	11.76	0.82	1502.24	104.43*	6.27	0.44
Total Social Competence	389.80	3.94*	30487.74	232.18*	23.12	0.18

Note. Gender *df* 1. Age *df* 2. Gender x Age *df* 2. * $p < 0.05$

As observed in Table 2, there was a significant gender difference in prosocial/ communication skills in favour of girls [$F(1, 174) = 3.97$; $p < .021$]. The development of social competence differs by gender, according to emerging evidence. According to Diener and Kim (2004), while prosocial behaviour ratings were significantly higher among preschool girls, externalizing behaviour ratings were significantly higher among preschool boys. Between the three age groups, significant age differences were seen in prosocial/communication behaviours as well. Tukey's HSD test for multiple comparisons found a significant difference in the mean value of prosocial/communication skills between 4-and 5-year-olds ($p = 0.000$, 95%, C.I. = [-10.455, -7.312]); 4-and 6-year-olds ($p = 0.000$, 95%, C.I. = [-16.921, -13.779]) and 5-and 6-year-olds ($p = 0.000$, 95%, C.I. = [-8.038, -4.895]). This result is consistent with a study conducted by Walker (2005), where older boys and older girls were regarded as displaying more prosocial behaviour than younger boys and younger girls. Socialization practices and societal norms that emphasize generosity and caretaking may be the cause for girls giving more and being more prosocial in nature (Gummerum et al., 2010).

A significant gender difference was found in the emotion regulation skills of preschoolers, with female preschoolers exhibiting superior emotion regulation skills as compared to male preschoolers [$F(1, 174) = 5.15$; $p .024$]. Even during the preschool years (and beyond), Saarni (1984) argued, girls are more capable than boys of regulating their emotions and conforming to social standards. This could be due to differential socialization practices, where girls are socialized to be more sensitive to others' feelings than boys, resulting in their ability to use effective strategies to cope with their emotions when interacting within a social context. As far as age differences are concerned, Cole et al., (2013) found that emotion regulation improves rapidly during the early childhood period. Across the three age groups, the present study also identified significant differences in emotion regulation skills. The Tukey's test of post-hoc analysis revealed that the mean value of emotion regulation was significantly different between 4-and 5-year-olds ($p = 0.000$, 95%, C.I. = [-12.535, -8.398]); 4-and 6-year-olds ($p = 0.000$, 95%, C.I. = [-21.735, -17.598]) and 5-and 6-year-olds ($p = 0.000$, 95%, C.I. = [-11.269, -7.131]).

Conflicting findings exist regarding gender differences in academic achievement. In this study, there were no significant differences in academic skills between male and female preschoolers. However, there was a significant age difference in academic skills across the three

age groups. After post-hoc analysis, Tukey's test revealed a significant difference in academic skills between 4-and 5-year-olds ($p = 0.000$, 95%, C.I. = $[-7.854, -4.580]$); 4-and 6-year-olds ($p = 0.000$, 95%, C.I. = $[-11.537, -8.263]$) and 5-and 6-year-olds ($p = 0.000$, 95%, C.I. = $[-5.320, -2.046]$). According to Gledhill et al., (2002), teachers perceive older children as having a higher level of academic proficiency than their younger peers, which is in line with previous data finding that younger children are referred for academic support more frequently than older students.

Our findings revealed a significant gender difference in overall social competence in favour of girls. To gain teachers' praise and avoid their criticism, girls may control and regulate their behavior more frequently than boys. As a result, girls have more opportunities to develop social competence, and their level of social competence tends to be higher than that of boys in the long run (Denham et al., 2003; Qin & Yong, 2002). We also identified significant age differences in overall social competence across the three age groups. Using Tukey's HSD test for multiple comparisons, it was found that the mean value of social competence was significantly different between 4-and 5-year-olds ($p = 0.000$, 95%, C.I. = $[-30.596, -20.704]$); 4-and 6-year-olds ($p = 0.000$, 95%, C.I. = $[-49.879, -39.988]$) and 5-and 6-year-olds ($p = 0.000$, 95%, C.I. = $[-24.229, -14.338]$). Through increasing interaction experiences, children's social competence improves continuously, resulting in older children being more socially competent than younger children (Mendez et al., 2002; Vahedi et al., 2012). The two-way ANOVA revealed that there was no statistically significant interaction between the effects of gender and age on social competence as well as its individual domains.

Conclusion, Limitations, and Scope

The current study was an attempt to examine gender differences and the developmental trends of Indian preschool children's social competence. We found gender to be an important social category that influences social competence and its domains for girls and boys. In addition, our study highlights how social competence develops over time. A potential limitation of the study is its cross-sectional nature. Studying social competence longitudinally would provide a more accurate picture of how it develops over time. Researchers have become increasingly interested in the development of social competence in childhood, due to its undeniable role in shaping adaptive abilities both in childhood and adulthood. Social-emotional development is essential for

children as much as their physical development. Regardless of how each child develops socially and navigates the social environment, there are several strategies teachers and parents can use to support them. Teachers should therefore intentionally enhance these skills by teaching, modelling, and reinforcing positive behaviours using evidence-based strategies.

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